LOCAL GOVERNMENT WEB-BASED POLICY RECOMMENDATION SYSTEM BASE ON COMMUNITY SENTIMENT: IMPLEMENTATION OF A DIGITAL REWARD SYSTEM

A Capstone Project Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In Partial Fulfillment of the Requirement for the Degree
Bachelor of Science in Information Technology
Specialized in *Service Management Program*

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JUNE 2026

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ACKNOWLEDGMENT

The researchers are deeply grateful to all those who have offered their guidance, support, and expertise in the completion of this capstone project. Without their valuable contributions, encouragement, and assistance, the successful realization of this study would not have been possible.

The University President, Dr. Mario R. Briones, and the Campus Director, Engr. Manuel Luis R. Alvarez, for their outstanding leadership.

We sincerely thank our Dean, Dr. Loyd S. Echalar, for his invaluable guidance, encouragement, and support.

To our dearest thesis adviser, Prof. Gene Marck B. Catedrilla, we extend our deepest gratitude for his invaluable guidance and unwavering support throughout the conduct of this study. His dedication, time, and encouragement have truly inspired us to pursue excellence and overcome every challenge we encountered.

We sincerely thank Prof. Jonardo R. Asor whose support and contributions made this study possible; without him, we would not have been able to learn, develop, and complete this research. We also express our appreciation to our panellists for their insightful comments and constructive suggestions, which greatly strengthened our work. Our gratitude extends to our friends for their constant support and motivation throughout this study. Above all, we thank Almighty God for His guidance, protection, and wisdom.

The Researchers

DEDICATION

This study is formally dedicated to our parents, families, and loved ones especially the Dulay, Geronimo, and Villegas families for their unwavering love, guidance, and support. Their encouragement, patience, and belief in our abilities have been a constant source of inspiration throughout the completion of this research.

This capstone project is formally dedicated to LGU, their invaluable support, guidance, and cooperation throughout the conduct of this study. The management and staff's assistance and provision of resources have significantly contributed to the successful completion of this research.

To their adviser and panellists, for their constructive guidance, expert advice, and mentorship, which enabled the researchers to think critically, refine their ideas, and develop a system of meaningful purpose.

To all their friends, classmates, and loved ones, for their encouragement, and presence throughout the completion of this study.

And above all, to Almighty God, who granted them the strength to persevere, the wisdom to gain knowledge, and the patience to continue despite challenges throughout the course of this study.

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ABSTRACT

This study, titled “Local Government Web-based Policy Recommendation System Based on Community Sentiment: Implementation of a Digital Reward System,” aims to strengthen community participation and enhance policy formulation through a data-driven, web-based platform. The system integrates machine learning–based sentiment analysis to automatically classify public comments, using a BERT model that achieved 91.4% accuracy, ensuring reliable interpretation of community feedback. A digital reward system was implemented to encourage citizen engagement by awarding points for activities such as posting comments and contributing suggestions. A total of 103 respondents and 3 IT experts evaluated the system using the ISO 25010 software quality framework and the Technology Acceptance Model (TAM). Results showed an overall ISO 25010 mean of 3.99 (Very Good), indicating strong functionality, usability, reliability, security, and portability, while the system’s acceptability received an overall TAM mean of 3.45, reflecting positive user acceptance. Sentiment analysis effectively processed community opinions, contributing to data-informed recommendations for LGU. Overall, findings demonstrate that integrating machine learning with a reward-based participation model can significantly enhance citizen engagement, improve feedback processing, and support transparent, responsive, and evidence-based local governance.

Keywords: Local Governance, Policy Recommendation System, Sentiment Analysis, BERT Model, Digital Reward System, Community Engagement, Web-Based Platform, ISO 25010, Technology Acceptance Model (TAM), Citizen Participation, Data-Driven Decision-Making

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DEFINITION OF TERMS

BERT (Bidirectional Encoder Representations from Transformers)	A deep learning model developed by Google for understanding the context of words in a sentence, used here to classify user feedback into sentiment categories.
Community Participation	The active involvement of citizens in governance processes through activities such as sharing opinions, providing feedback, and engaging in policy-related discussions via the system.
Dashboard Module	A system interface that visually displays key metrics, user activities, and sentiment analysis results to support monitoring and decision-making.
Data Analytics	The process of collecting, processing, and analyzing system data to generate insights that support evaluation, reporting, and policy formulation.
Data Privacy	The practice of safeguarding users' personal and sensitive information by ensuring secure storage, authorized access, and proper data handling.
Digital Reward System	A system mechanism that encourages user engagement by awarding points or incentives for participation activities such as commenting and submitting feedback.
E-Participation	The use of digital platforms to enable citizens to actively participate in governance, policy discussions, and decision-making processes.
ISO 25010	An international software quality standard used to evaluate systems based on characteristics such as functionality, usability, reliability, security, and portability.
Machine Learning	A branch of artificial intelligence where systems learn from data patterns to improve performance on tasks such as sentiment classification without explicit programming.
Natural Language Processing (NLP)	A field of artificial intelligence that enables computers to understand, analyze, and process human language in text form.

Policy Recommendation	The process of generating informed suggestions for local government decisions based on analyzed community sentiments and system-generated insights.
Reward Module	A system component responsible for tracking, managing, and distributing digital rewards based on user participation and engagement.
Sentiment Analysis	A technique that automatically identifies and classifies opinions expressed in text as positive, negative, or neutral.
Technology Acceptance Model (TAM)	A theoretical model used to assess user acceptance of a system based on perceived usefulness, perceived ease of use, attitude toward using, and behavioral intention.
User Engagement	The level of interaction and participation of users within the system, measured through activities such as commenting, feedback submission, and reward accumulation.

CHAPTER 1

INTRODUCTION

Project Context

Community participation plays a critical role in shaping the priorities and directions of local governance. By opening spaces for dialogue and consultation, governments gain valuable insight into the lived experiences, concerns, and aspirations of their citizens. This interaction not only strengthens the legitimacy of governance processes but also helps align public initiatives more closely with the community's needs. According to Fung (2015), deeper citizen engagement leads to more informed and equitable outcomes, while Irvin and Stansbury (2004) emphasize that collaborative approaches foster stronger civic trust and accountability. Nevertheless, maintaining sustained and meaningful participation continues to pose significant challenges, particularly in an increasingly digital and fast-paced society.

One of the central problems in local governance is the noticeable decline in community participation during policy formulation. Traditional engagement methods such as public consultations, town hall meetings, and surveys have experienced diminishing effectiveness, as citizens increasingly view them as time-consuming, bureaucratic, and ultimately ineffectual (Arnstein, 1969; Nabatchi & Leighninger, 2015). Studies by Lowndes, Pratchett, and Stoker (2001) identify structural and cultural barriers within institutions that discourage public involvement, while Michels and De Graaf (2010) note that skepticism regarding the actual impact of citizen input often leads to disengagement.

Without consistent feedback from the public, initiatives risk becoming detached from the real needs of communities. Furthermore, the absence of motivating factors or tangible rewards for participation underscores the need for innovative models that not only invite engagement but also recognize and value citizens' contributions

To address this issue, the project proposes developing a web-based system that facilitates community-driven policy recommendations, integrated with a digital rewards mechanism to incentivize participation. By incorporating Natural Language Processing (NLP) and sentiment analysis techniques, the platform will automatically classify and interpret citizen feedback based on emotional tone and thematic content. As discussed by Medaglia (2012), the strategic use of citizen-generated data can significantly improve governance responsiveness when properly analyzed and incorporated. Beyond gathering opinions, the system aims to generate actionable insights that can help local authorities craft initiatives more reflective of public sentiment. However, despite the availability of various digital participation tools, there remains a notable gap in systems that fully integrate sentiment analytics and digital incentives for participatory policymaking. Most existing platforms focus primarily on data collection without applying advanced analytics to interpret public emotions or linking engagement with meaningful rewards. Moreover, existing reward mechanisms are often tailored for commercial purposes rather than civic participation. Addressing this gap presents an opportunity to create a more dynamic and data-informed participatory framework that enhances both motivation and trust in governance processes.

The theoretical foundation of this study is anchored in Participatory Decision-Making Theory and the principles of E-Governance, both of which highlight the significance of active citizen involvement in shaping policies and governance outcomes. Arnstein's (1969) *Ladder of Citizen Participation* emphasizes that authentic engagement occurs when citizens move beyond mere consultation toward shared power and influence in decision-making. In parallel, Fung (2015) describes participatory governance as a mechanism for achieving more equitable and informed policies through inclusive and transparent collaboration between governments and citizens. The concept of E-Governance, as discussed by Medaglia (2012), extends these principles into the digital realm by leveraging technology to promote efficiency, openness, and citizen empowerment. Within this framework, integrating Artificial Intelligence (AI) and Natural Language Processing (NLP) supports data-driven policymaking by transforming unstructured feedback into measurable insights. Through techniques such as sentiment analysis, policymakers can identify public emotions, detect emerging issues, and align policy actions with the community's collective priorities. This theoretical grounding establishes the study's rationale for creating an AI-enabled participatory system that strengthens transparency, inclusivity, and accountability in local governance.

This paper focuses on the design, development, and evaluation of a digital system that leverages community sentiments to support local government policy recommendations. It will explore the technological, methodological, and participatory frameworks utilized to create a platform that meaningfully incorporates citizen input. Through this initiative, the paper aims to highlight how the strategic use of digital tools can foster more inclusive

governance, rebuild trust in public institutions, and promote a more participatory democratic process.

PROJECT OBJECTIVES

The main objective of the project is to develop an online community-based policy recommendation system through sentiment analysis. Specifically aims to:

1. Develop a web-based thread system to policy recommendations with digital reward module.
2. Utilizing a sentiment analysis model using machine learning algorithms BERT.
3. Evaluate the system's overall acceptability and software quality by applying the Technology Acceptance Model and the ISO 25010 standards, assessing key factors such as functionality, reliability, efficiency, usability, security, compatibility, maintainability, and portability.

PROJECT PURPOSE

The purpose of this project is to develop an online community-based policy recommendation system that uses Natural Language Processing (NLP) and sentiment analysis to transform unstructured public feedback into actionable policy insights. By analyzing emotions, opinions, and recurring themes in user discussions, the system identifies emerging community concerns and supports evidence-based decision-making. Through active

participation in policy dialogues, users contribute feedback that is processed by NLP models to determine sentiment trends, extract key issues, and generate recommendations that reflect collective community perspectives. A built-in reward mechanism further encourages citizen engagement, making the platform more interactive, inclusive, and representative of real policy conversations.

This project directly supports Sustainable Development Goal (SDG) 16 by promoting transparent, participatory governance; SDG 10 by empowering marginalized groups through a platform to express their needs; and SDG 9 by integrating innovative technologies to enhance civic engagement and governance systems. Ultimately, the system aims to strengthen democratic processes and produce more inclusive, data-driven, and effective policy outcomes:

Local Government Unit (LGU) and Policymakers

Local governments and policymakers gain a digital tool for efficiently analyzing community sentiment and identifying priority issues. Sentiment analysis and data visualization help administrators monitor public concerns, evaluate ongoing programs, and create more responsive and transparent policies while reducing the need for time-consuming manual consultations.

Community

Community members benefit from an accessible platform that allows them to share opinions and participate in policymaking anytime, anywhere, freely. The non-monetary reward system promotes active involvement, while transparent sentiment results help citizens understand how collective feedback

influences policy decisions, strengthening trust and collaboration with local authorities.

Future Researchers

This study provides a foundation for future innovations in digital governance and civic technology. Researchers may extend the system by improving sentiment models, adding multilingual support, or applying the framework to other governance sectors. It also serves as a reference for studies integrating AI, data analytics, and public participation to enhance e-governance platforms.

Scope and Limitations of the Study

This project focuses on developing an online community-based policy recommendation system that analyzes community feedback using sentiment analysis and Natural Language Processing (NLP). It aims to strengthen civic participation by enabling citizens to share their views and helping policymakers derive insights that reflect public concerns and sentiments. The system integrates several modules that collectively support data gathering, interpretation, and feedback dissemination.

The scope of this study is limited to a pilot implementation within the context of a local government unit (LGU). The system primarily targets community members, local officials, and stakeholders involved in local governance. It operates in a single language (English), which may restrict participation from communities with varying linguistic backgrounds. Additionally, while the system features a digital reward mechanism to motivate participation, all rewards are non-monetary, focusing on recognition and engagement rather than financial compensation. The project's analysis relies solely on data generated within the

platform and does not include integration with external data sources or predictive analytics.

Several limitations are acknowledged. The system's performance and accuracy are dependent on third-party NLP APIs, which may occasionally misinterpret local expressions, idioms, or sarcasm. Internet connectivity and device accessibility also influence User participation, potentially limiting inclusivity in areas with poor digital infrastructure. Furthermore, the evaluation of the system focuses on its usability, reliability, and effectiveness within the selected LGU pilot, and results may vary when scaled to other regions or governance contexts. Despite these constraints, the system serves as a proof of concept demonstrating how digital tools can enhance participatory policymaking through structured feedback and data-driven insights..

Dashboard Module

The Dashboard or Information Management Module functions as the system's central hub, organizing User data, feedback records, sentiment results, and reward transactions. It ensures efficient data management and reporting for Administrators. However, it relies solely on internal data and lacks advanced data validation and access control mechanisms, limiting flexibility and contextual depth.

Reward System Module

The Reward System Module promotes participation by awarding merit points to users who contribute through posts and discussions. These points correspond to non-monetary rewards such as digital badges and recognition, fostering engagement and inclusivity. Nonetheless, users cannot personalize

rewards, and the absence of tangible or financial incentives may reduce long-term motivation. The sustainability of the reward mechanism also depends on active platform usage and regular updates.

Data Analytics Module

This module visualizes community feedback into accessible charts and graphs, helping policymakers identify emerging trends and public priorities. While it provides clear, data-driven insights, its analysis is limited to internal datasets and relies on basic statistical techniques, without predictive modelling or cross-platform integration.

Sentiment Analysis Module

The Sentiment Analysis Module uses NLP APIs to classify feedback as positive, negative, or neutral, allowing policymakers to gauge public sentiment efficiently. However, its accuracy depends on external APIs that may not fully interpret local nuances or community-specific language. It also lacks the capacity to train custom models for contextual adaptation.

Notification Module

The Notification Module enhances engagement by delivering real-time alerts on new forums, updates, and earned rewards. It maintains transparency by informing users of comment statuses and interactions. Nevertheless, it only operates within the platform and does not support integration with external messaging or email systems. Users also have limited options to customize their notification preferences.

System Flow

Figure 1 illustrates the System Flow for a community-participatory system for policy recommendations. The process begins with data gathering and User input, followed by software development using Agile Scrum methodology within an Integrated Development Environment (IDE). The system includes modules for account management, forums, rewards, and notifications, with Natural Language Processing (NLP) used to analyse inputs and a reward system to foster participation. Then the system undergoes multiple levels of testing to ensure optimal performance and reliability. Finally, the system's effectiveness and quality are evaluated using different models. This framework aims to strengthen the policy formulation process by incorporating transparent, data-driven, and community-centered decision-making mechanisms.

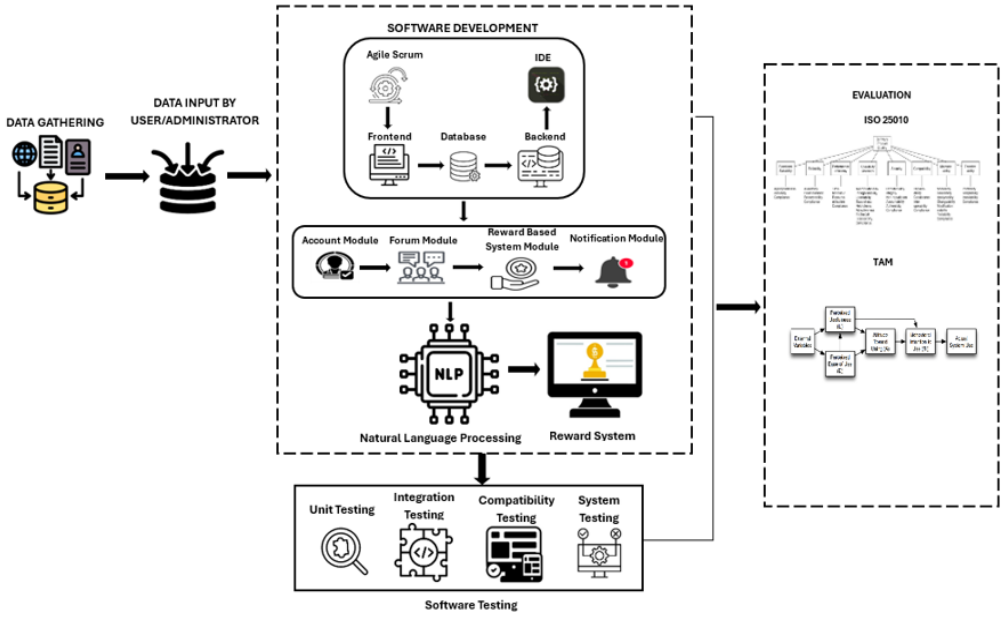


Figure 1. System Flow of Community Participatory for Policy Recommendations through Natural Processing

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter consists of a review of related literature and studies, both local and foreign, sourced from online reading resources, including news stories, posted personal reviews, blogs, and other trustworthy online publications. It focuses on additional works of literature that have a significant impact on the study's variables. It also discusses and connects relevant previous research to the study's conclusion.

Online Communities and Their Influence on Policymaking

Online communities have increasingly become critical channels for citizen engagement, providing platforms for dialogue, advocacy, and mobilization. Social media networks, public forums, and civic-oriented discussion boards serve as spaces where diverse perspectives converge, offering policymakers timely insights into public priorities and societal concerns (Purdam & Crisp, 2009; Yaqub, 2022). These digital environments do not simply aggregate opinions; they also facilitate networked participation, enabling citizens to influence one another and collectively shape perceptions of policy challenges and potential solutions.

The advent of sentiment analysis in these communities enables governments to interpret complex social signals at scale (Young et al., 2018). By examining user-generated content—such as posts, comments, and discussions—policymakers can detect trends in public reactions to proposed legislation or service delivery initiatives. This enables a form of predictive

governance, in which authorities anticipate public responses and proactively adjust policies to prevent conflicts or dissatisfaction (Yaqub, 2022). Beyond immediate feedback, online communities provide longitudinal data that reveal shifts in public sentiment over time, which is critical for evaluating the effectiveness of policy measures.

Despite these benefits, online participation is not uniformly accessible. Digital inequality, varying levels of internet literacy, and differential trust in civic institutions can limit engagement among specific populations (Boulianne, 2015; Linders, 2012). Overcoming these barriers requires inclusive platform design, mobile-first strategies, and programs that enhance digital literacy, particularly for marginalized or rural communities. International examples, such as Decide Madrid and Better Reykjavik, illustrate the potential of participatory digital ecosystems to foster transparent, accountable, and co-created governance (Susha & Grönlund, 2014). These platforms demonstrate that meaningful citizen engagement extends beyond passive commenting; it encompasses active collaboration in policy development, problem-solving, and monitoring the implementation of public initiatives.

E-Government, Digital Participation, and Public Engagement

E-government initiatives have significantly reshaped public administration by digitizing service delivery, streamlining feedback mechanisms, and promoting transparency (Sharma, 2014). Centralized platforms—ranging from online portals to mobile applications—allow citizens to access services efficiently, monitor government actions, and engage directly in governance processes. These platforms facilitate a two-way interaction

between the public and government, enhancing accountability and fostering trust in institutional processes.

Despite the potential of e-government, citizen participation is often hindered by usability challenges, low digital awareness, and inconsistent responsiveness from government agencies (Kassen, 2017). To address these issues, scholars emphasize user-centered design and co-creation approaches, in which citizens are involved throughout system development (Criado & GilGarcía, 2019; Linders, 2012). Furthermore, hybrid engagement models that integrate online and offline participation help mitigate disparities in age, socioeconomic status, and technological familiarity (Mossberger et al., 2008). These approaches ensure that digital platforms are inclusive, usable, and responsive to the needs of diverse citizen groups.

Sentiment analysis enriches digital governance by offering detailed insights into community opinions, emotional responses, and emerging service gaps (Yaqub, 2022). By monitoring real-time feedback, governments can refine policies, preemptively address conflicts, and prioritize interventions that align with public expectations (Young et al., 2018). Moreover, participatory mechanisms such as gamified engagement and open innovation frameworks encourage broader involvement, allowing citizens to co-design policies, propose solutions, and influence decision-making in meaningful ways (Janssen & Estevez, 2013). Effective implementation of these mechanisms also depends on citizen trust, particularly regarding the responsible handling of personal data and transparency in the use of feedback.

Digital Reward Systems for Civic Engagement

Digital reward systems apply gamification principles to foster engagement in civic activities, transforming participation into structured, interactive experiences (Deterding et al., 2011; Hamari et al., 2014). Features such as points, badges, leaderboards, and challenges motivate users to engage consistently, providing both tangible rewards (e.g., certificates, recognition) and intangible benefits (e.g., social validation, sense of contribution). These systems are especially effective when aligned with intrinsic motivations, such as the desire to improve community well-being, develop skills, or contribute meaningfully to policy discussions.

The integration of blockchain technology enhances transparency, accountability, and reliability within reward-based civic platforms. Smart contracts can automate the distribution of rewards, maintain tamper-proof records of participation, and reduce administrative complexity (Pop et al., 2020). Despite these advantages, ethical considerations remain crucial. Designers must ensure that reward systems avoid manipulative practices, respect user privacy, and provide equitable opportunities for participation. Properly implemented digital reward systems not only increase engagement but also generate high-quality data that can inform evidence-based policy development.

Sentiment Analysis in Community Engagement

Sentiment analysis provides a systematic way to interpret large volumes of text to uncover community attitudes, priorities, and concerns. Using natural language processing (NLP) and machine learning, sentiment analysis tools can

identify emotional patterns and categorize public feedback into positive, negative, or neutral sentiments (Liu, 2012; Cambria et al., 2017). This technique has been widely applied in contexts such as elections, disaster response, infrastructure planning, and public health monitoring (Tumasjan et al., 2010; Dewi et al., 2023; Lwin et al., 2020).

Advanced frameworks such as the Community Sentiment and Engagement Index (CSEI) further refine analysis by integrating behavioral indicators that capture levels of participation and public interest (Thakur et al., 2024). However, analyzing sentiment data presents challenges. Sarcasm, idiomatic expressions, cultural nuances, and contextual ambiguity can lead to misclassification or misinterpretation (McCabe et al., 2006). Ensuring methodological rigor, ethical safeguards, and continuous model validation is essential to maximizing the reliability and applicability of sentiment analysis for policymaking.

E-Governance and Digital Transformation in Local Government

E-governance integrates ICT into public administration to enhance efficiency, transparency, and citizen participation. Beyond digitizing existing processes, transformational e-government aims to redesign organizational workflows to become more citizen-centric (Heeks, 2006). Digital platforms support participatory decision-making, streamline service delivery, and enable data-driven governance (Misuraca & Viscusi, 2014).

However, widespread digital transformation remains hindered by issues such as unequal access to digital infrastructure, disparities in digital literacy, and concerns over data privacy (Bannister & Connolly, 2012). Addressing these

challenges requires robust infrastructure, strong institutional capacity, and comprehensive policy frameworks that protect citizens and encourage active participation.

Digital Governance in the Philippines

The Philippines has made significant progress in digital governance through national initiatives such as the eGovPH Super App, the National.

Broadband Plan, and efforts to modernize national identification systems (Andaya, 2025). At the local government level, smart city frameworks highlight the importance of ICT infrastructure, interoperability, and policy alignment to achieve sustainable and citizen-centered governance (Bithay et al., 2025).

Despite these improvements, local studies consistently show gaps in implementation. Many LGU websites provide only basic information and lack interactive features needed for meaningful citizen engagement (Lagura, 2022; Arrabaca & Base, 2017). Assessments also show wide disparities in digital readiness across urban and rural cities (Lagura, 2025). During the COVID-19 pandemic, LGUs increasingly used platforms such as Facebook to disseminate information and maintain engagement (Digital Innovation, 2023). Studies also indicate that compliance with global quality standards can improve service delivery and citizen participation (Aliling et al., 2025). However, the integration of advanced technologies such as sentiment analysis and machine learning remains underdeveloped, presenting a significant opportunity for innovation.

Artificial Intelligence and Machine Learning in Governance

AI and machine learning have enhanced governance by enabling the analysis of large-scale citizen-generated data. NLP models, including Support

Vector Machines, Naïve Bayes, LSTM networks, and BERT, significantly improve sentiment classification accuracy and enable real-time monitoring of public opinion (Liu, 2012; Young et al., 2018). AI-powered systems can automate feedback processing, detect emerging issues, and improve government responsiveness (Bertot et al., 2022; Alharbi & Papagiannidis, 2023).

However, concerns regarding algorithmic bias, data quality, and opaque decision-making highlight the need for robust AI governance frameworks (Binns, 2018). Ensuring fairness, transparency, and accountability is essential to safeguard citizen trust.

Community Empowerment and User Experience in Digital Platforms

Digital platforms can empower communities by enabling meaningful participation in decision-making processes. True empowerment occurs when citizen input influences actual policy outcomes rather than serving as symbolic or superficial consultation (Cornwall, 2008; Arnstein, 1969). Acknowledging and acting upon citizen feedback strengthens public trust and fosters sustained engagement.

User experience is a critical factor in digital participation. Systems designed using human-centered and participatory design principles are more intuitive, accessible, and aligned with community needs (Norman, 2013; Ehn & Kyng, 1992). Features such as mobile accessibility, multilingual support, and inclusive design broaden participation and ensure more representative data for policymaking.

SYNTHESIS AND RELEVANCE OF THE STUDY

These studies collectively highlight the transformative role of technology in creating participatory, data-driven, and citizen-centered governance. Online communities, e-government systems, and web-based platforms have become vital in bridging citizens and policymakers, facilitating public discourse and collective decision-making (Purdam & Crisp, 2009; Widiatmoko et al., 2023). Sentiment analysis enables governments to understand public mood, identify pressing concerns, and respond more effectively to citizen needs (Pang & Lee, 2008; Tumasjan et al., 2010), while digital reward systems, gamification, and blockchain foster sustained engagement and trust in governance (Deterding et al., 2011; Hamari et al.,

2014). Successful e-governance initiatives also require ethical foundations, User-centered design, and inclusivity to ensure equitable participation and protect citizen data (Floridi & Taddeo, 2016; Boulianne, 2015).

In the Philippine context, research by Andaya (2025), Lagura (2025), and Aliling et al. (2025) highlights both progress and challenges in digital governance. While ICT-driven platforms such as the eGovPH Super App and online feedback portals have been adopted, uneven digital readiness, limited citizen awareness, and weak sentiment integration remain.

Although prior studies demonstrate the benefits of e-participation tools, sentiment analysis, and incentive mechanisms individually, they reveal a key gap: most systems either collect citizen feedback without analyzing its emotional tone or apply engagement incentives without linking them to actionable policy outcomes. This study addresses these gaps by developing an

integrated participatory system that combines sentiment analysis, digital rewards, and policy co-creation, transforming citizen feedback into actionable insights while motivating sustained participation. By bridging these gaps, the study contributes to more inclusive, data-driven, and reward-supported civic engagement in policymaking.

Table 1. Literature Matrix

Author(s)Year	Title	Findings
Widiatmoko, B. A., Pratama, S., & Arifianto, D. (2023).	Analysis of public sentiment in e-government for policymaking. Journal of Public Administration and Policy Research	The study explores how analysing public sentiment on social media can help improve e-government policymaking. By using sentiment analysis tools, they identified how people feel about different government digital services. The findings show that understanding emotions like satisfaction, frustration, or trust allows policymakers to respond more effectively to citizens' needs, highlighting the importance of real-time feedback in creating policies that better match public expectations.
Dyachuk, D. (2023).	Sentiment Analysis in Public Health Communication. Government Information Quarterly.	Examined public health briefings in Alberta during COVID-19 using sentiment analysis. The study found that messaging remained emotionally consistent despite case fluctuations. It highlighted strategic word use and demonstrated that sentiment analysis is a valuable tool for evaluating the tone and effectiveness of public health communication.
Hutahaean, D. M., Nababan, R. S., & Muda, I. (2023).	Public participation in Indonesia's e-government transformation. International Journal of Public Policy and Administration Research.	Reveals that while Indonesia's e-government has made progress, public participation is still hindered by issues like inadequate regulations, uneven internet access, and cultural barriers such as corruption. To improve engagement, the authors recommend better regulations, expanded internet access, and addressing trust issues.
Rodrigues, M., Monteiro, V., Fernandes, B., Silva, F., Analide, C., & Santos, R. (2020).	A gamification framework for getting residents closer to public institutions. Journal of Ambient Intelligence and Humanized Computing.	Found that gamification helps bring residents closer to public institutions by making civic participation more fun, rewarding, and meaningful. Game elements like points and challenges boost engagement, build trust, and make people feel more involved in local decisions. Personalized rewards also keep citizens motivated. This supports

		your thesis idea that a digital reward system can strengthen community participation and improve how people connect with local government.
Criado, J. I., & Gil-García, J. R. (2019).	Creating public value through smart technologies and strategies. International Journal of Public Sector Management.	Explore how smart technologies can create public value in the public sector, highlighting the need for strategic implementation and collaboration among stakeholders. They discuss challenges like data privacy and digital inequality and use smart cities as an example of how these technologies can improve public services and sustainability. The authors emphasize that successful use of technology requires careful planning and inclusive strategies.
Bousquet, C. (2017).	Boosting Engagement by Gamifying Government. Data-Smart City Solutions.	They identified that gamifying government services through tools like points, badges, and challenges can significantly increase citizen engagement, especially when these systems are designed with clear incentives, transparency, and accessible digital platforms.
Doherty, S., Palmer, E., & Strater, L. (2017).	Gamification: Current Research and Applications. Proceedings of the Human Factors and Ergonomics Society Annual Meeting.	They identified that gamification effectively boosts engagement and motivation across various fields—particularly in research, industry, and education—by leveraging psychological triggers like curiosity, scarcity, and loss aversion. They highlighted successful real-world applications such as LinkedIn's use of gamified design, and noted that while gamification shows great promise, there's still a need for deeper research and more data transparency from companies to fully understand its long-term effects.

CHAPTER 3

METHODOLOGY

This chapter describes the methodology used in the design, development, testing, and evaluation of the Local Government Web-based Policy Recommendation System with a Digital Reward System. It outlines the project design, system development, and required software and hardware, as well as the testing procedures conducted to ensure system reliability and performance. The system was evaluated by community users and IT experts using the Technology Acceptance Model (TAM) and ISO 25010 to assess its quality, usability, and overall acceptability.

Project Design

Figure 2 illustrates the system architecture of the Local Government Web-based Policy Recommendation System, showing how the major modules are organized and interconnected to support the implementation of the digital reward system and provide a seamless user experience. The system is composed of several core modules, including the Report Module, Forum Module, Data Analytics Module, Notification Module, and Reward Module.

The Report Module supports data monitoring and analysis by allowing administrators to track key system activities such as user logins, forum posts, comments, and points earned. This module also helps measure forum engagement and overall system performance over time. The Forum Module enables the creation of discussion forums, viewing of posts, and commenting functionalities. The forum creation feature is accessible only to administrators, while viewing and commenting are available to both users and administrators.

The Data Analytics Module is exclusively accessible to administrators and is responsible for processing collected data, including sentiment analysis results, to generate insights that support policy recommendations. The Notification Module manages and delivers system-generated notifications to inform users and administrators of updates, activities, and rewards. Lastly, the Reward Module manages the digital reward system, allowing users to earn and view reward points, while administrators can manage the reward shop by adding, editing, or deleting rewards. Both users and administrators can view reward point balances. All system modules are connected to a centralized database that securely stores forum content, user interactions, and reward points.

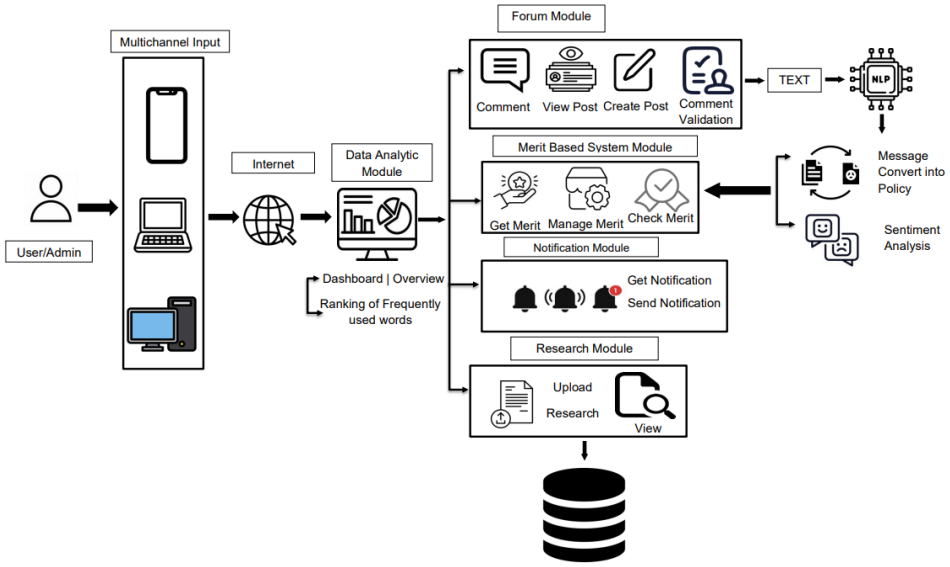


Figure 2. Top view of the proposed participatory system architecture for policy recommendations.

Figure 3 shows the Data Flow Diagram of the proposed participatory policy recommendation system. The process begins with the User or Administrator, who inputs comments, posts, and feedback through the Forum Module. These inputs serve as the system's data source (input stage). The Sentiment Analysis Module then processes this data using Natural Language Processing (NLP) and the BERT algorithm to classify sentiments as positive, neutral, or negative. The analyzed results are forwarded to the Data Analytics Module, which organizes and visualizes the data for deeper insights. Meanwhile, the Notification Module ensures users and Administrators are informed of new activities or feedback, and the Reward Module tracks User engagement and updates earned points, promoting participation. The Dashboard Module acts as the central hub, connecting all modules and providing real-time monitoring and management capabilities for Administrators. All processed data are stored in the Database, which serves as the main repository for User information, feedback, and sentiment results. Finally, the system produces a Policy Report that summarizes the analysed data into actionable insights, representing the system's final output for evidence-based policy recommendations.

Data Flow Diagram

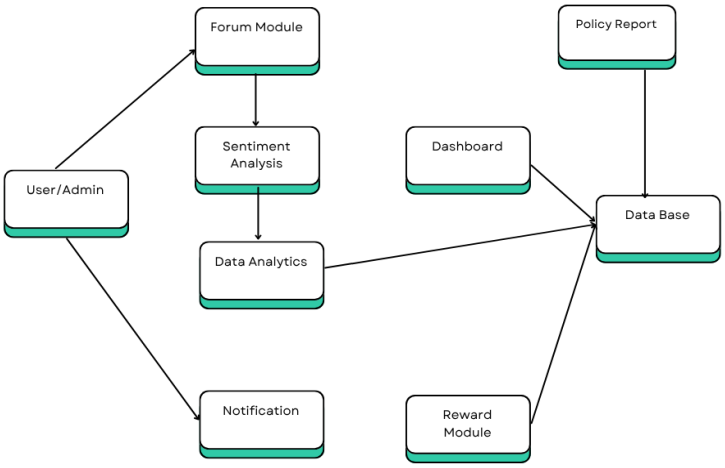


Figure 3. Data Flow Diagram

Software Development Model

Figure 4 shows the lifecycle of the system from the initial idea to the outcome, which is the implementation of a Digital Reward System for policy recommendations. By applying Agile Scrum methodologies, such as product backlog, sprint planning, daily scrum, and complete product.

By breaking down the task into smaller components and involving the entire team in the process, it becomes easier to generate innovative ideas and increase collaboration. This ultimately leads to a more effective system performance.

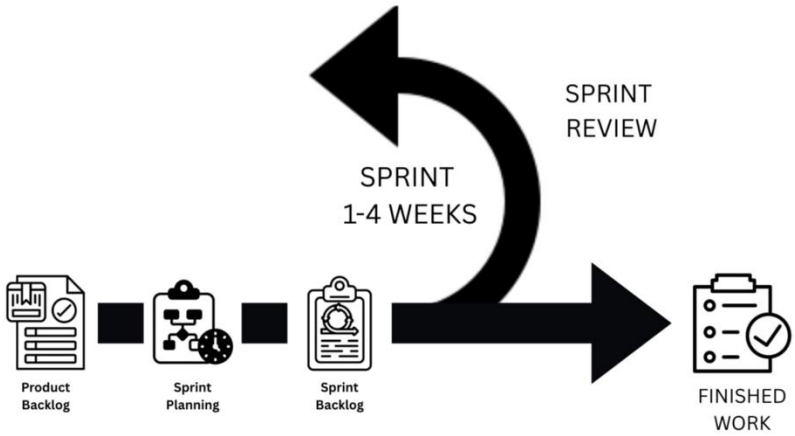


Figure 4. Agile Scrum

Product Backlog

In this phase, the researchers determine the features of the system, which contains the list that needs to be developed, and begin with the project's product backlog. This is the list of steps required for creating the software.

Table 2 presents the product backlog system for the Administrator, which prioritizes the Administrator's list that contains the estimated time for the system. This backlog serves as a guide for development planning by organizing administrative features according to priority and estimated effort to ensure efficient system implementation.

Table 2. Product Backlogs of Admin

PRIORITY	ITEMS	ESTIMATED HOURS/DAYS
1	As an Administrator, I want to access dashboard	4 Hours
2	As an Administrator, I want to manage post and announcement	2 Hours
3	As an Administrator, I want to manage users	2 Hours
4	As an Administrator, I want to manage reward shop	5 Hours
5	As an Administrator, I want to manage report	5 Hours

Table 3 shows the product backlog for the User, which outlines the prioritized features designed to support user interaction and participation within the system. These backlogged items guide the development of user functionalities that allow users to view posts, engage through comments, and earn, track, and exchange rewards while enabling the administration to monitor user activities effectively.

Table 3. Product Backlogs of User

PRIORITY	ITEMS	ESTIMATED HOURS/DAYS
USER		
1	As a User, I want to see or monitor view post	2 Hours
2	As a User, I want to comment	2 Hours
3	As a User, I want to get reward	3 Hours
4	As a User, I want to check rewards	2 Hours
5	As a User, I want to exchange reward from shop	5 Hours

Sprint Planning

Flowchart

The two figures illustrate the flowcharts for a User and an Admin in a modular web-based system, detailing how each role interacts with various system modules. For Figure 5, the process begins with either logging in or registering for an account. Once authenticated, the User can interact with four core modules, such as the Forum Module, Reward Module, and Notification Module. If the User accesses the Forum Module, users can view forums, add comments, and delete their inputs. The Reward Module allows users to check

and obtain reward points, while the Notification Module provides system updates or relevant messages.

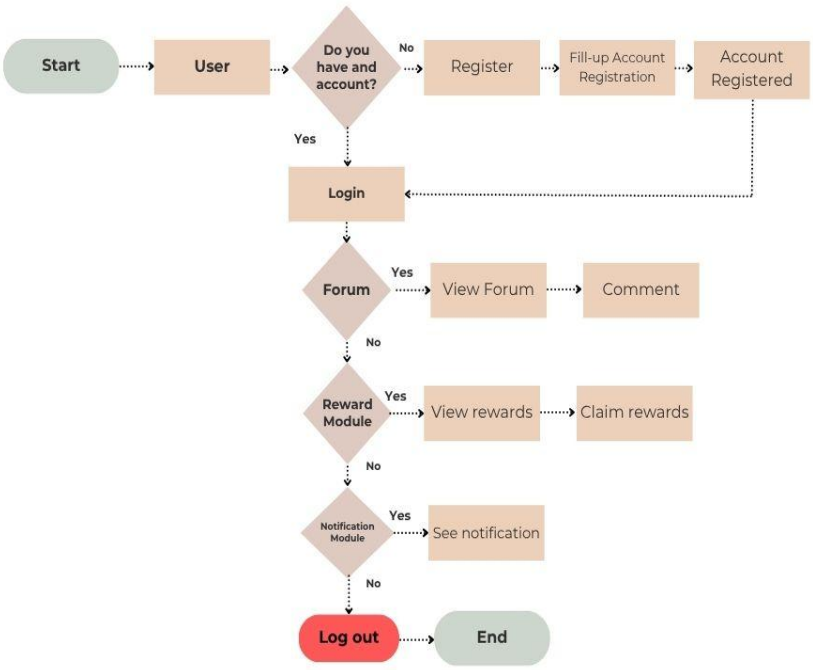


Figure 5. System Flowchart of the User

For Figure 6, the flow starts directly with login, followed by access to the same core modules but with administrative capabilities. In the Data Analytics The Admin can create activities or reports based on points. In the Forum Module, they can create new forums, view user interactions, validate comments, and assign points accordingly. The Reward Module allows them to manage the reward system, and the Notification Module lets them send and retrieve system notifications. Both users and Admins conclude their session by logging out. Overall, the system ensures a structured interaction between Users and Administrators, where users focus on participation and information

consumption, while admins manage, moderate, and enhance the system functionalities.

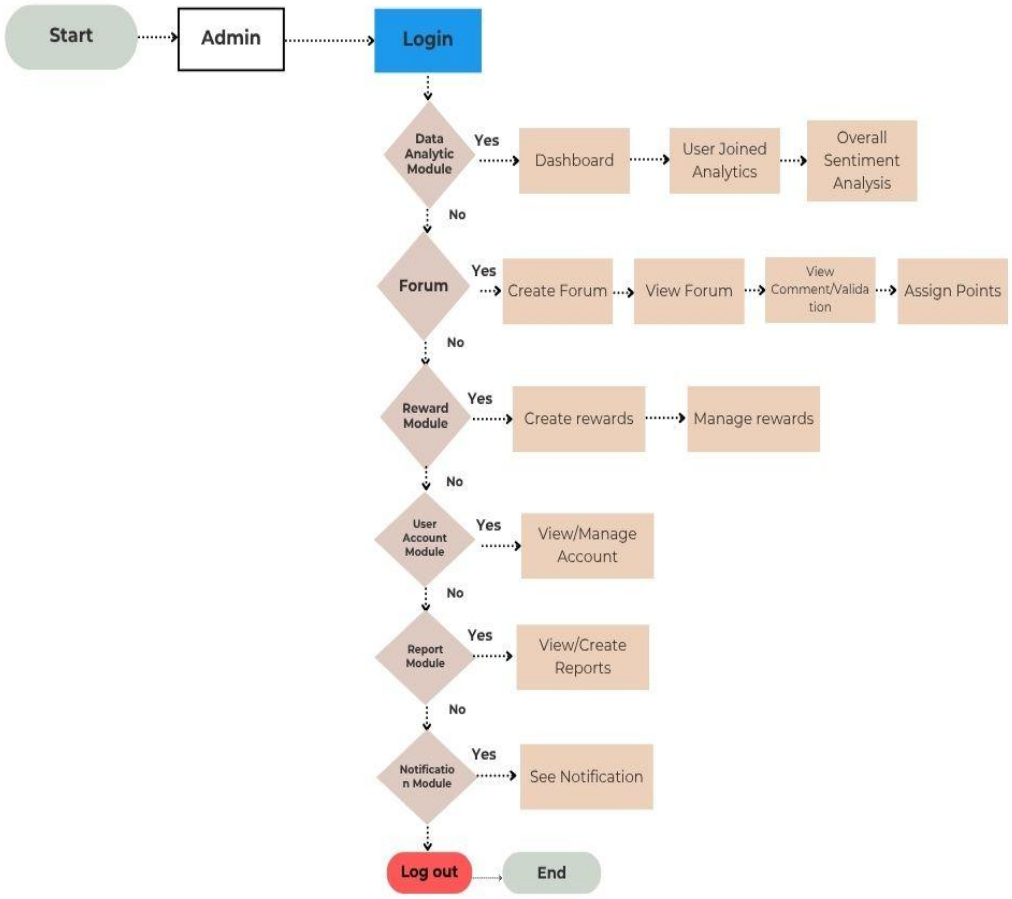


Figure 6. System Flowchart of Admin

Reward Mechanism

Figure 7, Reward Mechanism Process works by encouraging meaningful participation within the community forum, where users can share their thoughts and opinions on various topics related to governance or policymaking. After a

User posts a comment, the Administrator reviews and validates it — not based on whether it is positive or negative, but on whether it contains useful or informative content that can contribute to community understanding or decision-making. If the comment provides valuable insight, regardless of sentiment, the Admin grants the User reward points that are stored in the system's Database.

These points accumulate over time and can be redeemed for rewards through the system's reward module. This process ensures that all forms of constructive feedback — whether supportive or critical — are recognized as valuable contributions, promoting honest, thoughtful, and educational participation in the community.

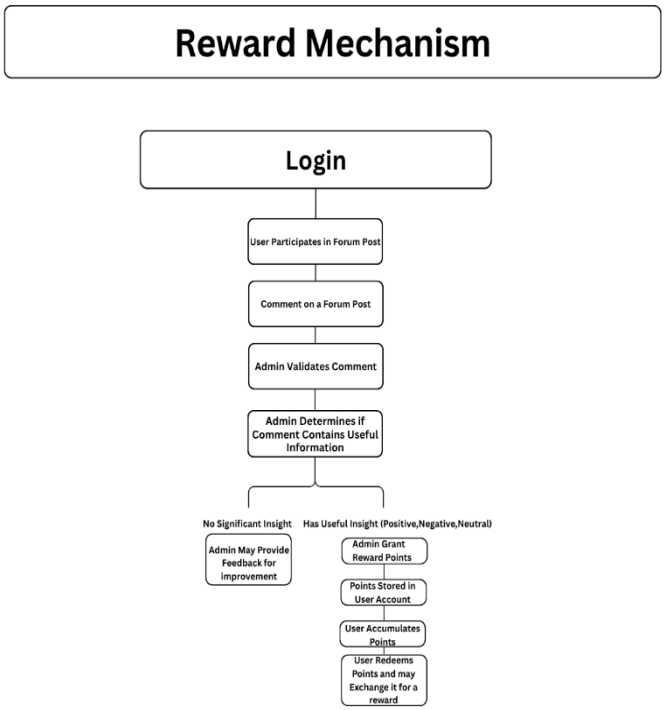


Figure 7. Reward Mechanism

Use Case Diagram

Figure 8 illustrates the modules accessible to both users and Admins. Users can register, browse posts and announcements, leave comments, earn merit points, view their accumulated points, and redeem them. On the other hand, Admins have access to a dashboard that highlights the most used word, and they can manage users, posts/announcements, research materials, and the merit points shop. Admins can also view posts, comment, and earn merit points like Users. Both roles require logging in to use the system. However, the registration process differs; users can sign up on their own to access the user panel, while only an existing Admin can register a new Admin through the Admin panel.

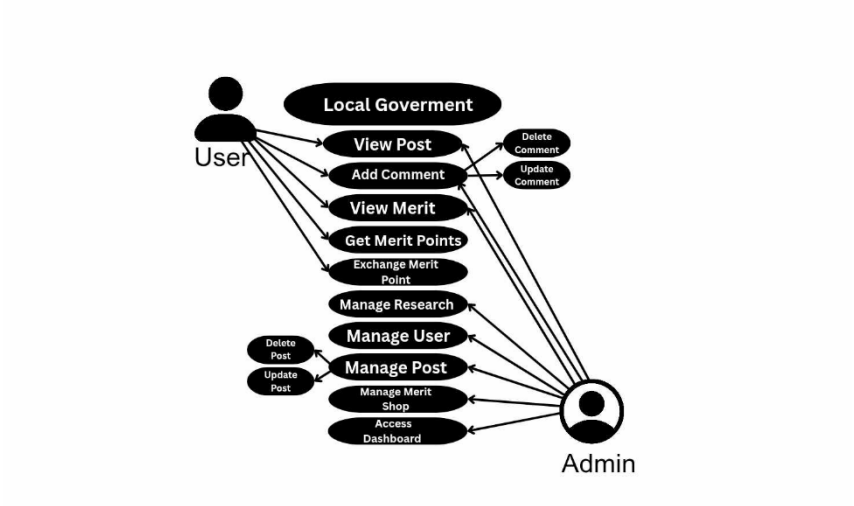


Figure 8. System Modules Use Case Diagram Overview

Materials

Software

The software tools that will be needed to develop the system are the following:

Visual Studio Code functions as the primary development workspace for writing, editing, and managing the system's code, allowing the team to build the website, connect it seamlessly with the Supabase database, and implement essential features such as community sentiment processing and digital reward mechanisms. The front end is developed using HTML, CSS, and JavaScript, with JavaScript handling API interactions, while the back end relies on Python and Flask, enhanced by tools such as pytz, Torch Transformers, and BERT for advanced sentiment analysis and natural language processing. VS Code's integrated debugging tools and extensions further streamline the development process. Supabase manages all crucial datasets—including User information, feedback, sentiment evaluations, reward points, milestones, and redemption logs—while providing real-time updates and efficient database operations.

Supporting technologies such as GitHub Desktop, GitHub, Render, CKEditor, and ApexCharts enhance the system's version control, deployment pipeline, content management, and analytics capabilities, thereby improving overall reliability and performance.

Hyper Text Markup Language HTML is responsible for building the basic structure and layout of the system's web pages, including elements like User registration forms, discussion sections, and reward displays. It organizes the content in a clear and visually appealing way, making it easier for users to navigate the platform and interact with its features. Additionally, HTML supports Administrators in managing the community by presenting data and interactions in an intuitive and organized manner, ensuring a smooth User experience across the platform.

Cascading Style Sheet (CCS) is used to control the visual appearance of the system, ensuring a neat, modern, and User-friendly experience. It defines the layouts, colour schemes, fonts, spacing, and other stylistic features that make the interface attractive and simple to navigate. This plays a crucial role in maintaining the cleanliness, modernity, and User-friendliness of the system for both users and Administrators.

JavaScript is used to add interactivity and dynamic features to the web-based system. It enhances the User experience by allowing real-time updates, form validations, interactive charts for community sentiment analysis, and responsive elements without reloading the entire page. JavaScript also helps in managing User actions, such as submitting feedback and claiming digital rewards, making the system more engaging, efficient, and User-friendly.

Python is the back-end language that powers your server logic and API endpoints (via Flask). It runs the NLP and sentiment pipelines (Torch Transformers, BERT), processes and validates feedback, and applies business rules like computing reward points and milestones. Python also interacts with Supabase to read/write data, handles timezones/scheduling (pytz), and is used for scripts, testing, and deployment automation.

Natural Language Processing helps computers understand, interpret, and generate human language. It allows systems to analyze text and extract meaning, such as identifying emotions, detecting topics, or summarizing information. NLP is also used for tasks like sentiment analysis, translation, speech recognition, chatbot responses, and text classification. In systems like yours, NLP helps evaluate community feedback by determining whether

comments are positive, negative, or neutral. Overall, NLP enables computers to process language in a way that supports better decision-making, automation, and User interaction.

Supabase serves as the system's main Database, storing all essential information, including user profiles, feedback, sentiment results, reward points, milestones, and redemption records. It provides real-time data updates, secure authentication, and fast querying, allowing the system to retrieve and save information efficiently. Supabase also integrates smoothly with Python and Flask through API calls, making data management simple and reliable.

Github is a desktop application for version control that lets developers track changes in their projects, manage branches, and collaborate efficiently. It provides a visual interface for committing updates, reviewing modifications, and syncing the project with GitHub, ensuring organized, secure code management.

Render is a cloud platform used for deploying the system, allowing the application to run online and be accessible to users. It handles server hosting, automatic updates, and continuous deployment, making it easy to publish new versions and keep the system running smoothly.

CKEditor is a web-based text editor used for content management, allowing users to create, edit, and format text easily within the system. It provides tools for rich text editing, such as styling, embedding media, and organizing content, helping maintain clean and well-formatted inputs.

Pytz Python library used for handling time zones accurately, allowing the system to convert and manage dates and times across different regions. It

ensures that timestamps for User actions, feedback, and rewards are consistent and correctly aligned with the intended timezone.

Torch Transformer are machine learning tools from the PyTorch ecosystem used for advanced sentiment analysis. They allow the system to process text using powerful pretrained models, helping it understand context, detect emotions, and generate accurate sentiment predictions.

BERT Bidirectional Encoder Representations from Transformers is a powerful machine learning model developed by Google that understands the context and meaning of text. It is used for sentiment analysis in your system, helping it accurately identify whether feedback is positive, negative, or neutral by analyzing words in context.

ApexCharts is a web-based charting library used to create interactive analytics and visualizations. It helps display data such as sentiment trends, user activity, and reward statistics through clean and dynamic charts, making system insights easier to understand.

Model Development

Figure 9, Sentiment analysis is a process in which User comments are analysed using an AI model, specifically BERT (Bidirectional Encoder Representations from Transformers). When a User posts a comment, the text is processed by BERT to analyse its content and determine its sentiment polarity. Based on this analysis, the model generates a polarity score, which is used to classify the comment as positive, negative, or neutral: A high score indicates a positive sentiment. A low score indicates a negative sentiment. A

score near the middle indicates a neutral sentiment. In the analytics phase, all comments are summarized, and the sentiment scores are averaged to produce an overall summary. This provides a general understanding of users' collective sentiment, whether positive, negative, or neutral. In short, comments are processed by the BERT model, which analyzes them and generates a polarity score that becomes the final sentiment result.

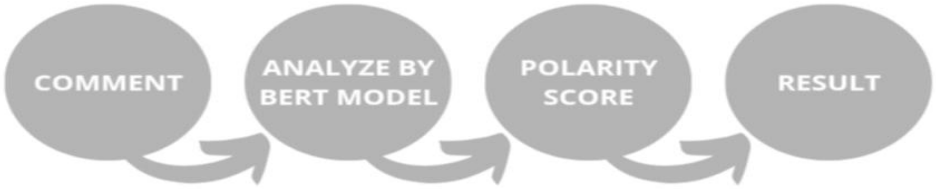


Figure 9. Process Flow of Comment Analysis Using BERT

Sentiment Analysis Model Description

To address the lack of detail regarding the sentiment analysis model, the following section explicitly defines the machine learning model, its version, and the training data used in the system.

Sentiment Analysis Model

The sentiment analysis module of the proposed participatory governance system utilizes the Bidirectional Encoder Representations from Transformers (BERT) machine learning algorithm. BERT is a pre-trained transformer-based model developed by Google AI, designed to understand the context and semantics of natural language text, making it superior to traditional models such as TextBlob or Naïve Bayes classifiers.

Initially, TextBlob was used for sentiment analysis, but it only supports English and cannot handle mixed-language comments effectively. The system switched to BERT. We used a pre-trained sentiment BERT without fine-tuning. This provides higher accuracy and contextual understanding, allowing it to classify comments in multiple languages, including English and Tagalog.

For this study, the pre-trained BERT-Base, Uncased model was implemented through a Natural Language Processing (NLP) framework. It was trained on large datasets, including the English Wikipedia and BookCorpus, enabling it to interpret complex or context-dependent sentences accurately.

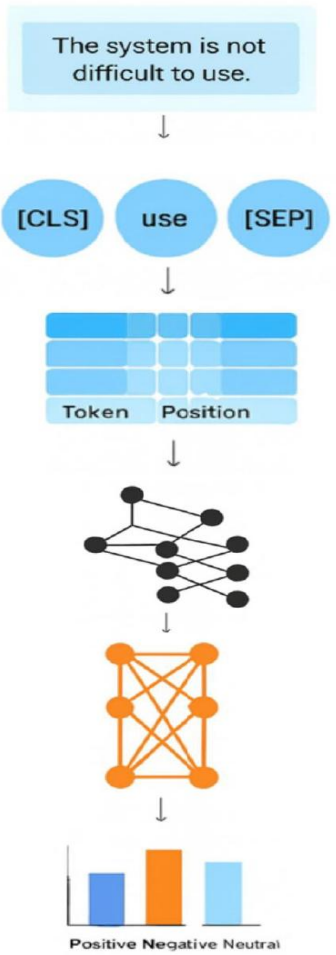
User-generated feedback is processed through BERT to classify sentiment polarity into positive, neutral, and negative categories. Predictions are displayed on the web-based dashboard, allowing administrators to view aggregated sentiment trends and topic clusters to support data-informed policy decisions.

Although highly effective, BERT has limitations with Taglish and Filipino idiomatic expressions. Future system improvements may include fine-tuning the model with locally sourced datasets to enhance cultural and linguistic accuracy.

BERT Workflow for Sentiment Analysis

Figure 10 illustrates how BERT performs sentiment analysis: it starts with the raw input text, breaks it into WordPiece tokens, and adds special markers like [CLS] and [SEP] to define the sequence. Each token is then transformed into numerical vectors using token, position, and segment embeddings before being processed by BERT's multi-layer bidirectional

A transformer, which analyzes the contextual relationships between all words in the sentence. After this deep contextual encoding, a classification layer is added during fine-tuning, allowing the model to learn from sentiment-labeled data. Once trained, the fine-tuned BERT model generates probability scores for sentiment categories—positive, negative, or neutral—and selects the class with the highest probability as the final sentiment prediction.



1 Input Text
The process serves as an initial input for BERT.

2 Tokenization
The input text undergoes WordPiece tokenization, splitting it into sub-word units (tokens). Special markers like [CLS] (start of sequence) and [SEP] (end of sequence) are added to structure the input

3 Embedding Transformation
Each token is converted into a numerical vector through BERT's embedding components: Token embeddings, Position embeddings, Segment embeddings

4 Transformer Encoding
The embeddings pass through BERT's multi-layer bidirectional Transformer system, which analyzes contextual relationships across all tokens simultaneously.

5 Fine-Tuning for Sentiment Task
A classification layer is added, and the entire BERT model becomes specialized by training on sentiment-labeled datasets.

6 Sentiment Prediction
The fine-tuned model outputs probabilities for sentiment categories (positive, negative, neutral). The class with the highest probability becomes the final prediction.

Figure 10. BERT Workflow for Sentiment Analysis

Polarity Interpretation

Table 4. Score Range Polarity Interpretation

Polarity Score	Sentiment	Interpretation
0.75 – 1.00	Positive	Very strong positive sentiment
0.50 – 0.74	Positive	Moderate positive sentiment
0.01 – 0.49	Positive	Slightly positive sentiment
0.00 – 0.80	Neutral	Balanced sentiment; no clear positive or negative tone
-0.01 – -0.49	Negative	Slightly negative sentiment
-0.50 – -0.74	Negative	Moderate negative sentiment
-0.75 – -1.00	Negative	Very strong negative sentiment

Table 4 shows that the BERT model predicts the sentiment of a comment based on the highest class probability, while the polarity score reflects the strength or intensity of that sentiment. A polarity close to +1 indicates very strong positive sentiment, while a score close to -1 indicates very strong negative sentiment, and values near zero represent a balanced tone with no dominant emotional direction. A comment is not automatically Neutral just because the polarity is low; values below 0.50 often signal that the model is detecting a mix of positive and negative cues, resulting in a balanced or uncertain classification. In some cases, BERT may still produce a high polarity value (e.g., 0.80) when it identifies strong positive words. Still, if the comment

also contains negative elements, the system may classify it as Neutral because the mixed tone lowers the model's confidence. The seven-level intensity scale is justified by examining three internal model signals: probability confidence, which indicates how certain BERT is about its predicted class; polarity absolute value, which measures how far the sentiment leans toward or away from neutral; and Softmax gaps, which show how clearly the model separates one sentiment class from the others. By combining these values, the system can categorize comments into very strong, moderate, and slight intensities for both positive and negative sentiments, while maintaining a Neutral category for balanced or contradictory messages. This approach provides a more nuanced interpretation of sentiment beyond the model's original three-class output.

Polarity Formula

where $p$pos and $p$neg are the probabilities predicted by BERT for positive and negative sentiment. The polarity indicates the strength of the dominant sentiment, while the sentiment label (Positive, Neutral, Negative) determines the direction. In our system, polarity values are shown as absolute magnitudes, and negative sentiments are labeled as "Negative" without showing a negative sign.

Polarity Score Interpretation

Table 5 shows that the polarity scores are divided into ranges to indicate both the strength and type of sentiment. High positive scores (0.75–1.00) correspond to very strong positive sentiment, e.g., *"I love the community gardening program! It's very helpful*, while moderate positive (0.50–0.74) and

slightly positive (0.01–0.49) reflect weaker positive sentiment. Neutral is assigned to polarity scores 0.00–0.80 because comments can be ambiguous, mixed, or low-confidence, such as *"The program is okay, some parts are good, some are not"*; this captures situations where BERT cannot clearly distinguish positive or negative sentiment. Negative polarity is shown in the same ranges as positive but labeled as "Negative" and indicates the magnitude of negative sentiment: slightly negative (0.01–0.49), moderate negative (0.50–0.74), and very strong negative (0.75–1.00), e.g., *"The project is terrible, it wastes time."* This system allows us to represent both the intensity (polarity magnitude) and direction (sentiment label), while explaining why Neutral can sometimes have high polarity values, depending on the balance of positive and negative cues in a comment.

Table 5. Polarity Score Interpretation with Examples

Polarity Score	Sentiment	Example Comment	Interpretation
0.75 – 1.00	Positive	I love the community gardening program! It's very helpful.	Very strong positive sentiment: polarity and label align
0.50 – 0.74	Positive	The project is useful, but could be better organized	Moderate positive sentiment
0.01 – 0.49	Positive	The program is okay, some things are good	Slightly positive sentiment
0.00 – 0.80	Neutral	The program is okay, some parts are good, some are not.	Weak, ambiguous, or mixed sentiment; includes comments where BERT has low confidence or polarity is small.
0.01 – 0.49	Negative	Some aspects are bad, but others are okay	Slightly negative sentiment; magnitude only, system hides negative sign.
0.50 – 0.74	Negative	The project is poorly organized and confusing	Moderate negative sentiment; magnitude only

Algorithm Analysis

Table 6 shows that BERT achieves exceptionally high performance in benchmark evaluations, particularly on the SQuAD question-answering task. The results indicate that both the BERT ensemble and single-model versions consistently outperform other models, achieving Exact Match (EM) scores of 85.1%-87.4% and F1 scores of 91.8%-93.2%. These high EM values demonstrate BERT's ability to correctly identify precise answers, while the strong F1 scores reflect its balanced precision and recall, enabling it to interpret context accurately even in complex or ambiguous text. Compared to competing architectures such as nlnet, BERT leads in both EM and F1 metrics, confirming its reliability and effectiveness for tasks that require deep language understanding. Such strong performance metrics make BERT a suitable model for analyzing User sentiments, as it can accurately detect tone, intent, and context in community feedback.

Table 6. BERT Performance on SQuAD Benchmark Tasks (Exact Match and F1 Scores)

PRIORITY	Test EM	Test F1
1st Place Ensemble - BERT	87.4	93.2
2nd Place Ensemble - nlnet	86.0	91.7
1st Place Single Model - BERT	85.1	91.8
2nd Place Single Model - nlnet	83.5	90.1

Table 7 highlights BERT's performance on several Natural Language Inference (NLI) tasks, demonstrating its strong capability to understand relationships between sentences. BERT achieves accuracy scores of 86.7% on MultiNLI, 91.1% on QuestionNLI, and 86.3% on SWAG, surpassing previous state-of-the-art models such as OpenAI GPT, which scored lower across all

datasets (82.2%, 88.1%, and 75.0%, respectively). These results show that BERT excels not only in extracting factual answers but also in reasoning about language, making correct judgments about entailment, contradiction, or neutrality. Its performance across multiple NLI benchmarks demonstrates robustness and generalization, reinforcing its suitability for sentiment analysis, where interpreting subtle emotional cues and contextual relationships is essential.

Table 7. BERT Accuracy Across Natural Language Inference (NLI) Tasks

System	MultiNLI	QuestionNLI	SWAG
BERT	86.7	91.1	86.3
OPENAI GPT (Prev. SOTA)	82.2	88.1	75.0

Hardware

Table 8 presents the hardware specifications of the computer used in developing the system. The 16GB RAM and Intel Core Ultra 5 125U processor provide sufficient processing power and multitasking capability to support system development, testing, and debugging processes. Additionally, running the system on Windows 11 Pro ensures compatibility with modern development tools and a stable environment for software implementation.

Table 8. Hardware Requirements

Components	Specification
Random Access Memory (RAM)	16GB
CPU	Intel Core Ultra 5 125U processor, Graphics 3.60 GHz and P-cores with a max speed of 4.30 GHz.
Operating System (OS)	Windows 11 Pro

Sprint backlog

The researchers share their priorities and discuss the sprint goals. It will be done through collaboration by the team: The researchers will provide a sprint backlog and pick the tasks that need to be done in developing the project.

Table 9 shows that the researchers have outlined the functionalities that will be executed in the coming sprint, which is a list of functionalities that need to be built.

Table 9. Sprint Backlog

Priority	Items	Task	Estimated Hours/ Days
Admin			
1	As an Administrator, I want to access dashboard to see the most frequently used word in the comment by the User	Design Dashboard with ranking of frequently used word	2 Hours
		Develop Dashboard	3 Hours
2		Test Dashboard	2 Hours
	As an Administrator, I want to manage post/ announcement by creating post and announcement	Design forum page for post and announcement	2 Hours
		Develop forum page	1 Hours
3	As an Administrator, I want to manage users by creating, monitoring, and deleting.	Test forum page	1 Hours
		Design manage User page	1 Hours
		Develop manage User page	1 Hours
		Tests manage User page	1 Hours
4	As an Administrator, I want to manage reward shop by adding, editing, and deleting enlisted incentives.	Design manages reward shop page	2 Hours
		Develop manage reward shop page	2 Hours
		Tests manage reward shop page	1 Hours
5	As an Administrator, I want to manage report module to post, update, and delete.	Design manage report page	2 Hours
		Develop manage report page	1 Hours
		Tests manage report page	1 Hours

Project Testing and Evaluation

Project Testing and Evaluation were conducted to ensure that the system functions properly and meets the project objectives. Different testing methods were applied to verify system functionality, reliability, and performance. The system was also evaluated by users and IT experts to assess its quality and acceptability.

Integration Testing

In this study, integration testing ensured that the Forum, Sentiment Analysis, Reward, Notification, and Data Analytics modules functioned cohesively by verifying correct data exchange, synchronized reward point updates, and accurate sentiment results across the centralized Database during real user interactions.

Compatibility Testing

In this study, integration testing ensured that the Forum, Sentiment Analysis, Reward, Notification, and Data Analytics modules functioned cohesively by verifying correct data exchange, synchronized reward point updates, and accurate sentiment results across the centralized Database during real user interactions.

System Testing

In this study, system testing validated that all modules, including user authentication, forum discussions, sentiment analysis, reward allocation, notifications, and administrative dashboards, worked correctly as an integrated

whole and met the defined project objectives, performance requirements, and reliability standards.

Population of the Study

The end-user respondents were purposively selected individuals who were knowledgeable in using digital systems and applications. They were guided by the researcher during system testing to ensure proper operation and understanding of the features. Data were gathered through a Google Form survey to evaluate the system's usability, functionality, and overall user experience.

The IT experts were professionals with strong academic and practical backgrounds in information technology, including master's degree professors from Laguna State Polytechnic University (LSPU) and practitioners working in various IT-related fields such as system development, networking, and information security. They used hard copy evaluation instruments to assess the system's security, authentication, and technical reliability.

Data Collection and Instrument

All the data collected for this study were obtained from first-hand sources through face-to-face surveys and Google Forms. The researcher created a survey questionnaire and distributed it to the participants, consisting of both Users and IT Experts. After testing the system, they answered the survey instrument designed to determine whether the developed system is effective and meets User requirements. The questionnaire comprised criteria based on the Technology Acceptance Model (TAM) and ISO 25010 software quality standards. For the TAM evaluation, the researcher used a 4-point Likert scale

with the following equivalents: 1–Poor, 2–Fair, 3–Good, and 4–Excellent, while the ISO 25010 technical evaluation used a 5-point Likert scale interpreted as 1–Poor, 2–Fair, 3–Good, 4–Very Good, and 5–Excellent.

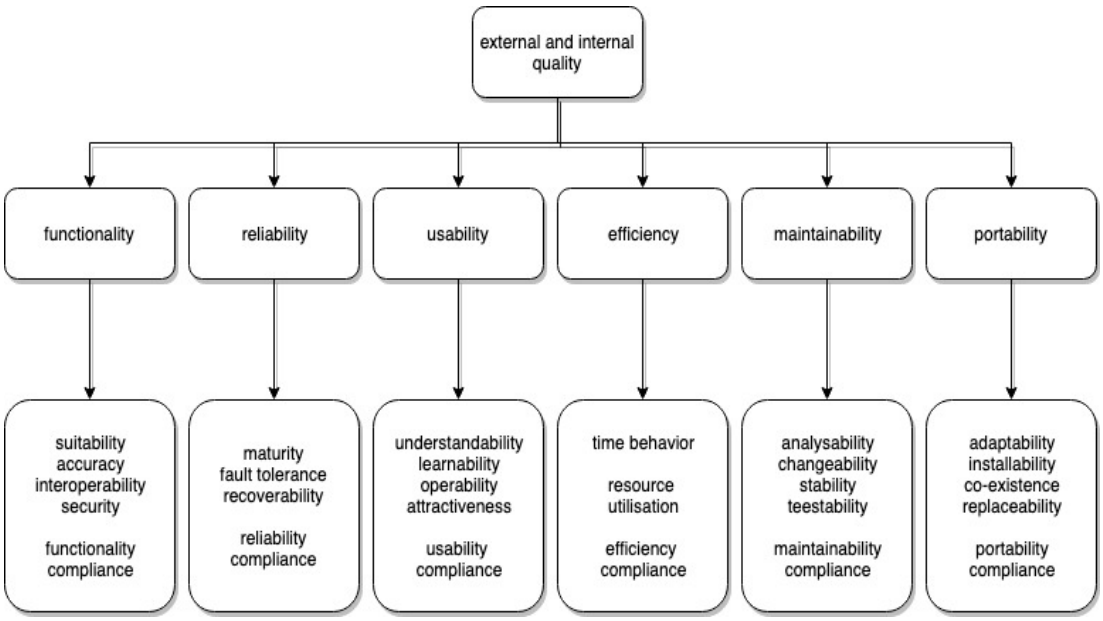


Figure 11. Quality Characteristic OF ISO 25010

Technology Acceptance Model

To evaluate user acceptance and adoption of the system, surveys were developed using the Technology Acceptance Model (TAM) and administered to end users. The survey focused on key constructs adapted from Diaz et al., and the questionnaire's reliability was verified using Cronbach's alpha to ensure consistency before data analysis. As shown in Table 10, the results indicate good to excellent internal consistency across all constructs, with Perceived Usefulness (α = 0.900), Behavioral Intention to Actual Use (α = 0.912), and Attitude Toward Using the System (α = 0.938), demonstrating excellent reliability, while Perceived Ease of Use (α = 0.841) showed good reliability. User evaluations were collected through Google Forms after system testing, while IT

experts completed hard-copy questionnaires to allow a detailed technical assessment. Overall, the Cronbach's alpha values confirm that the survey items were reliable, clear, and appropriate for assessing system acceptability.

Table 10. Reliability test result for Technology Acceptance Model

Item	Number of Items	Cronbach's	Interpretation
Perceived Usefulness (PU)	5	0.900	Excellent
Perceived Ease of Use (PEOU)	5	0.841	Good
Behavioral Intention to Actual Use	5	0.912	Excellent
Attitude Toward Using the System	5	0.938	Excellent

Legend above 90 = excellent, 0.80-0.89 = Good, 0.70-79 = Acceptable 0.60-0.69 = Questionable, 0.50-0.59 = Poor Below 0.50 = Unacceptable

Sampling Design

The purposive sampling technique was employed in this study. This approach involves deliberately selecting participants based on specific criteria relevant to the system's testing and evaluation. A total of 103 endusers were chosen from the community to participate in the review through a Google Form survey, as they were considered capable of providing meaningful feedback based on their actual use of the application. In addition, three (3) IT experts were purposively chosen to conduct a technical evaluation of the system using hard copy assessment instruments. These IT professionals examined the application to verify that it is functionally correct, secure, and acceptable according to established software quality standards.

System Performance Evaluation

To validate the performance of the proposed system, metrics such as accuracy, precision, recall, and execution time were used to evaluate its effectiveness in sentiment classification and overall reliability.

Statistical Treatment of Data

The researcher used purposive sampling to select 103 community respondents and 3 IT experts who were directly involved in evaluating the Local Government Web-based Policy Recommendation System. This approach ensured that the participants were appropriate and capable of providing relevant assessments of the system's sentiment analysis features, usability, functionality, and digital reward mechanism. After testing the system, respondents evaluated their experience using a Technology Acceptance Model (TAM)–based survey, which applied a 4-point Likert scale to measure perceived Usefulness, perceived ease of use, attitude toward using, and behavioral intention. This combined methodological approach allowed the researcher to gather focused, meaningful feedback from actual users while obtaining expert validation on system quality, resulting in a comprehensive evaluation of the system's acceptability and effectiveness.

Table 11 shows that the ISO 25010 evaluation was conducted by three (3)

IT experts assessed the system based on eight software quality characteristics. The researcher uses a 5-point Likert scale to evaluate the experts' technical judgment. Table 10 presents the total data along with the computed mean and standard deviation. The ISO data display graphically the

survey questionnaire deviation from 1 (Poor), 2 (Fair), 3 (Good), 4 (Very Good), and 5 (Excellent).

Table 11. Likert Scale Technique for ISO

RATING	INTERPRETATION	RANGE
5	Excellent	4.21 – 5.00
4	Very Good	3.41 – 4.20
3	Good	2.61 – 3.40
2	Fair	1.81 – 2.60
1	Poor	1.00 – 1.80

Table 12 presents the total data, along with the computed mean and standard deviation. The data display the survey questionnaire deviation from 1 (Poor), 2 (Fair), 3 (Good), and 4 (Excellent) graphically).

Table 12. Likert Scale Technique for TAM

RATING	INTERPRETATION	RANGE
4	Excellent	3.25 – 4.00
3	Good	2.50 – 3.24
2	Fair	1.75 – 2.49
1	Poor	1.0 1.74

Ethical Considerations

The study adhered to strict ethical standards to protect the rights and privacy of all participants. Informed consent was obtained before data collection, with respondents clearly informed of the study's purpose, voluntary participation, and their right to withdraw at any time without consequence.

Personal information was kept confidential and used solely for research objectives, and all participants were treated with fairness, respect, and dignity throughout the process. To ensure privacy and compliance with the Data Privacy Act of 2012, all collected citizen data, including system-generated feedback, underwent data anonymization and encryption, with personal identifiers removed before analysis or reporting.

User comments used for sentiment analysis were processed only in aggregated or anonymized form, and access to stored information was limited to authorized system Administrators. All research procedures were conducted transparently, and findings were presented accurately to maintain integrity and uphold ethical data-handling standards.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter shows the results by objectives, evaluation of the survey answers, and project testing. It also discusses how well the system works based on user feedback and testing results.

Result by Objectives

Developed a web-based thread system to policy recommendations with digital reward module.

Figure 12 (a) shows the User's Landing page and mobile views. On the homepage, Users can see the existing post and join different forums.

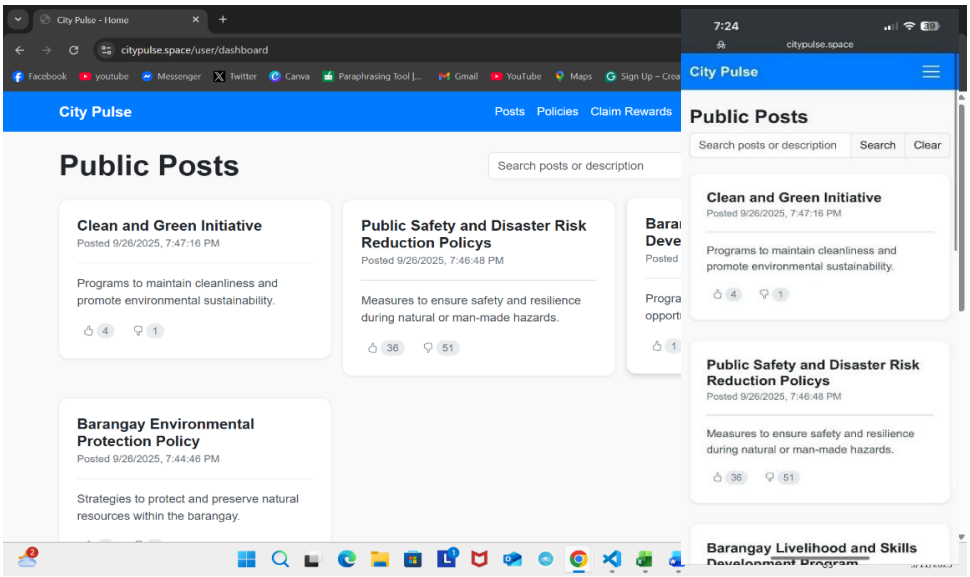


Figure 12. (a) User's Landing page on Web and Mobile

Figure 12 (b) shows the Administrator Landing Page, which serves as the main control panel for managing the City Pulse platform. From this page, administrators can view key system statistics such as the number of active and inactive users, monitor user growth through the users' Joined Analytics graph, and review community engagement via the Overall Sentiment Analysis for

different policy posts. The page also displays visual summaries of sentiment distribution and the most claimed rewards, allowing administrators to assess public feedback and platform activity quickly. This landing page centralizes essential information, enabling efficient oversight and decision-making within the system.

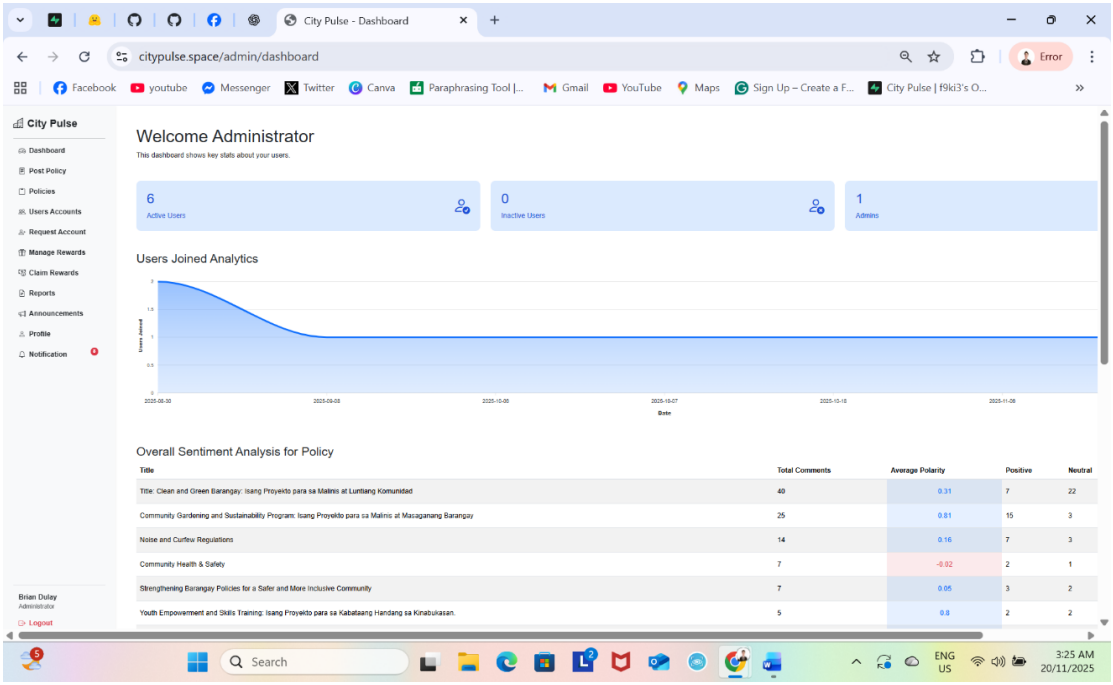


Figure 12. (b) Admin Landing Page

Figure 13 shows the Admin interface of City Pulse, specifically the Create Post page, where the Administrator can compose and publish new posts by entering the title, description, and full content. It represents the post-creation module of the Admin dashboard, which allows Administrators to write, manage, and prepare announcements or policy updates by providing all required details before publishing.

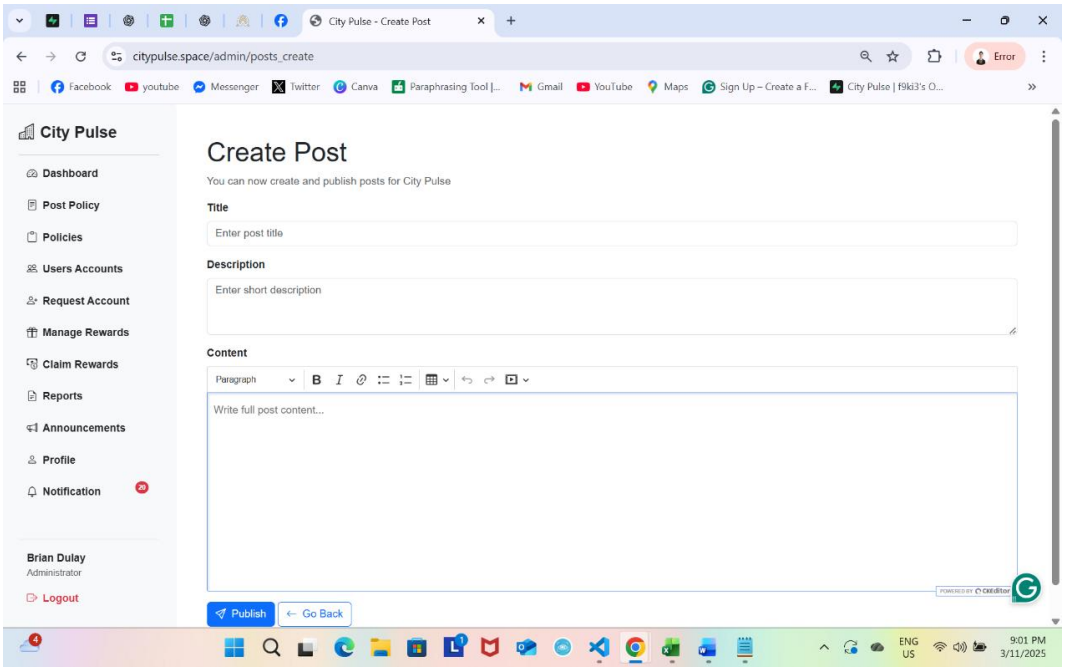


Figure 13. Admin Dashboard – Create Post Interface

Figure 14 shows the Admin Manage Posts page, where administrators can create, edit, view, and delete forum policy posts. This interface allows efficient management of announcements and policy updates for the community.

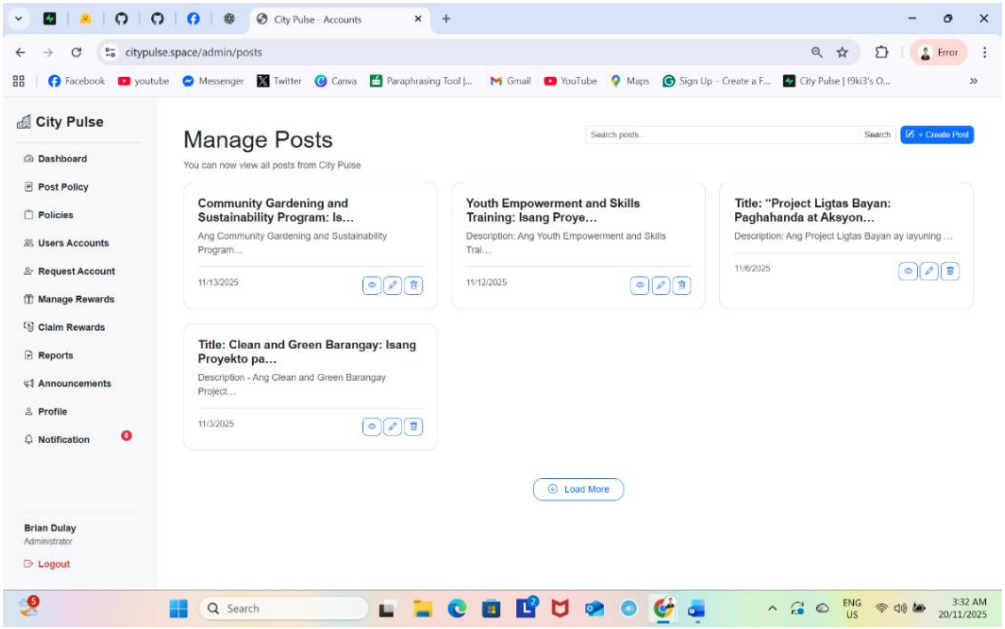


Figure 14. Admin Manage Posts Page

Figure 15 shows the web and mobile views of the forum policies page, where users can join discussions by exchanging ideas and sharing their insights with the community. And highlights the responsive design of the forum policies page, demonstrating consistent functionality and usability across desktop and mobile devices.

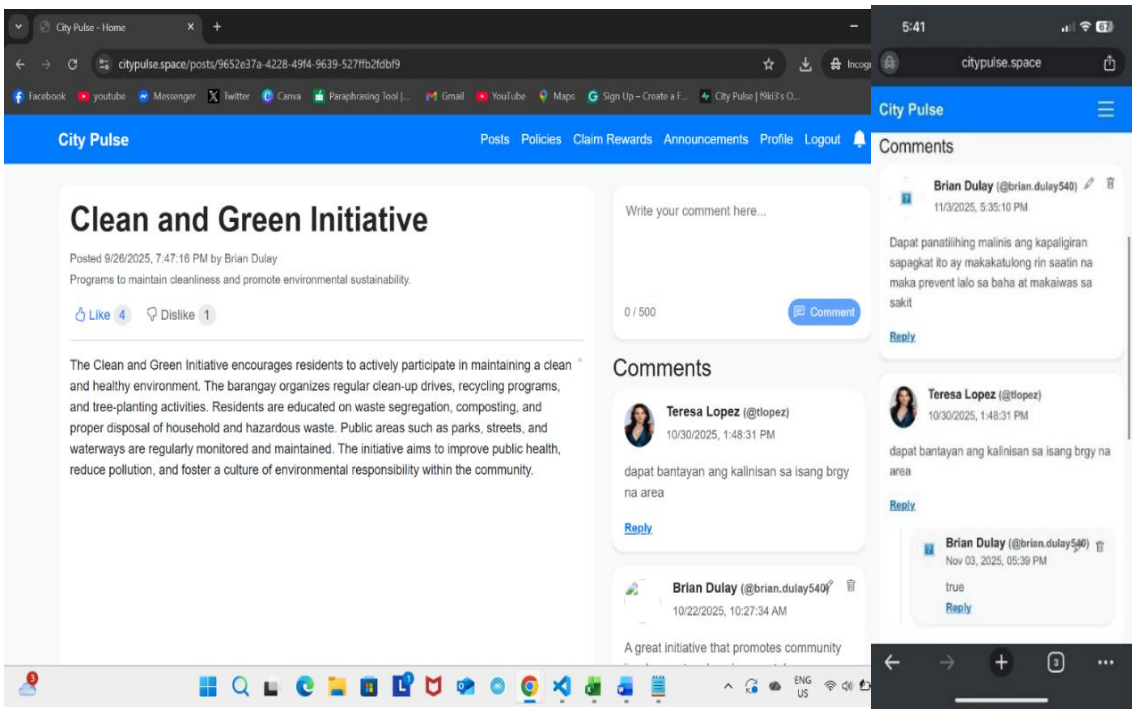


Figure 15. Forum Page on Web and Mobile

Figure 16 illustrates the system's sentiment-linked point-granting interface, where Administrators can evaluate User comments and assign points based on sentiment classification. By enabling the awarding of points directly from sentiment feedback, the system promotes constructive participation while simultaneously giving policymakers real-time visibility into how citizens emotionally respond to policy discussions. This enhances policymakers' responsiveness by allowing them to identify which topics generate positive engagement or critical sentiment immediately.

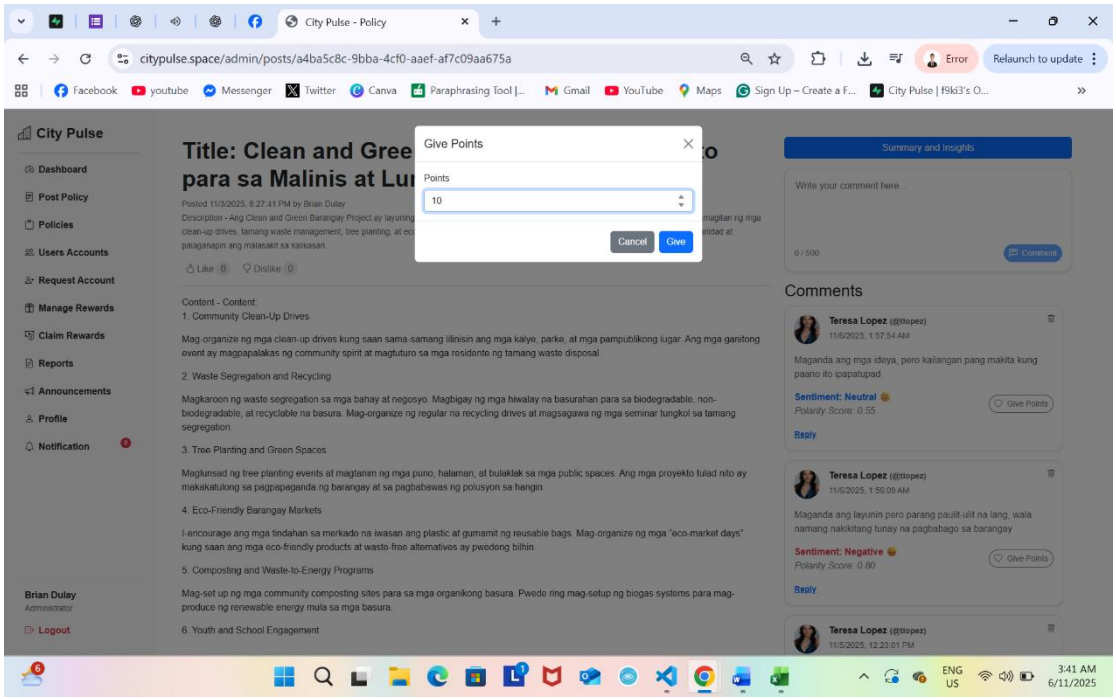


Figure 16. Granting Points for Sentiment-Based Feedback

Figure 17 presents the User dashboard showing accumulated points and available reward options, which are directly influenced by sentiment-validated interactions. This visualization not only motivates citizens to share meaningful feedback but also provides policymakers with an overview of participation levels driven by different policy issues. High user activity and point accumulation serve as indicators of public interest, enabling policymakers to prioritize high-impact concerns and respond promptly.

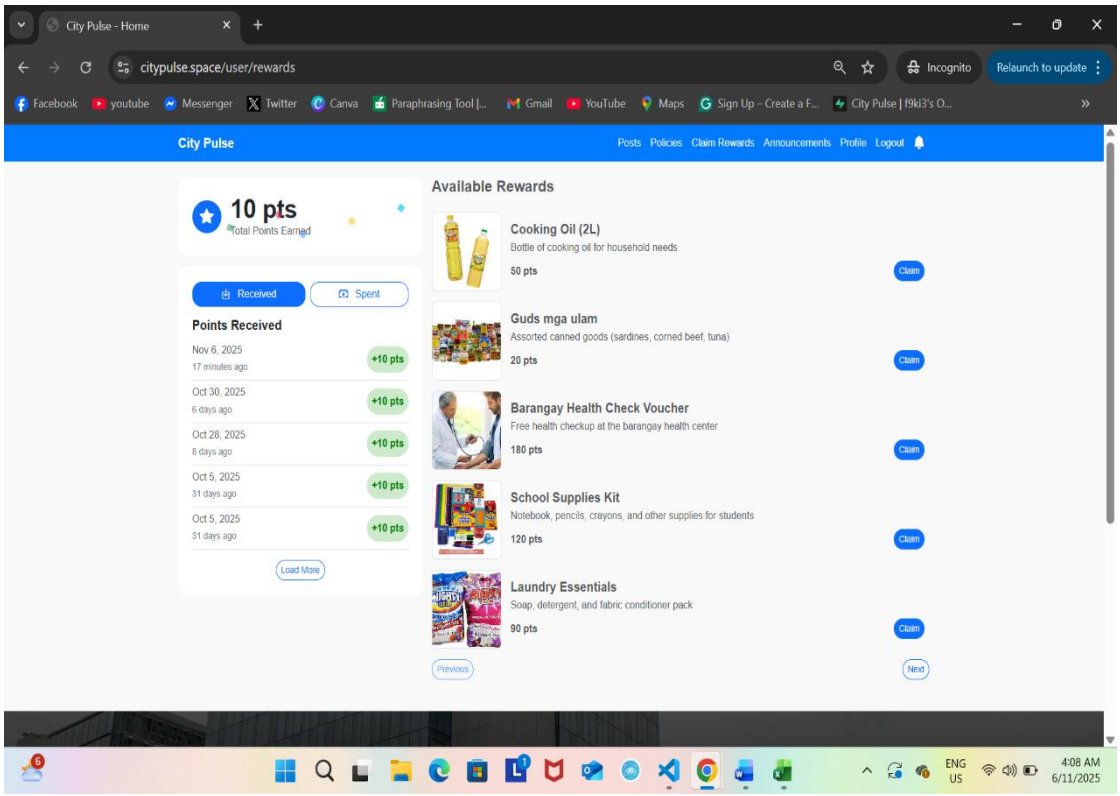


Figure 17. User Rewards Points Redemption Page

Figure 18 shows the reward management interface that allows Administrators to configure incentives for sentiment-based participation. By monitoring which rewards are frequently redeemed, policymakers can indirectly measure user satisfaction and engagement levels. This visualization supports quick decision-making by revealing which policy discussions attract the most active and emotionally engaged users, enabling more targeted and timely policy responses.

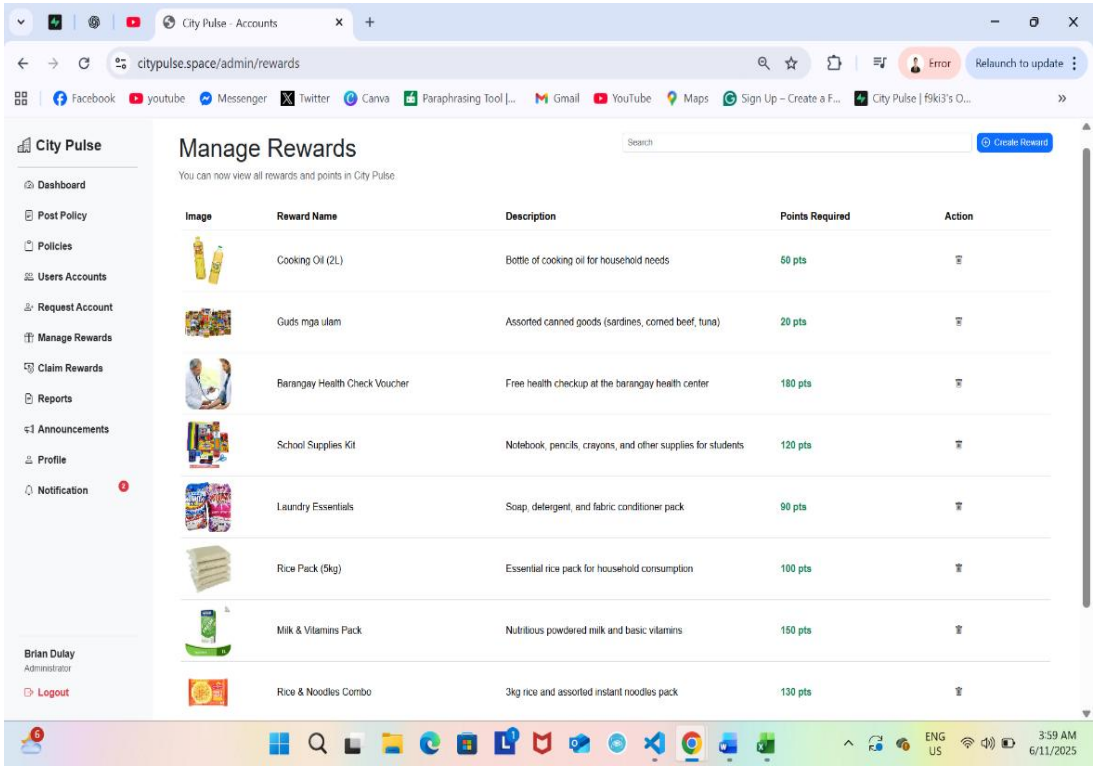


Figure 18. Rewards Management Interface

Utilizing sentiment analysis model using machine learning algorithms.

Figure 19 shows the City Pulse Policy Post Interface, where residents submit comments about community policies. The system's sentiment analysis model processes each comment and assigns a sentiment label—positive, negative, or neutral—along with a corresponding polarity score. This automated evaluation provides policymakers with an immediate understanding of how users perceive posted policies.

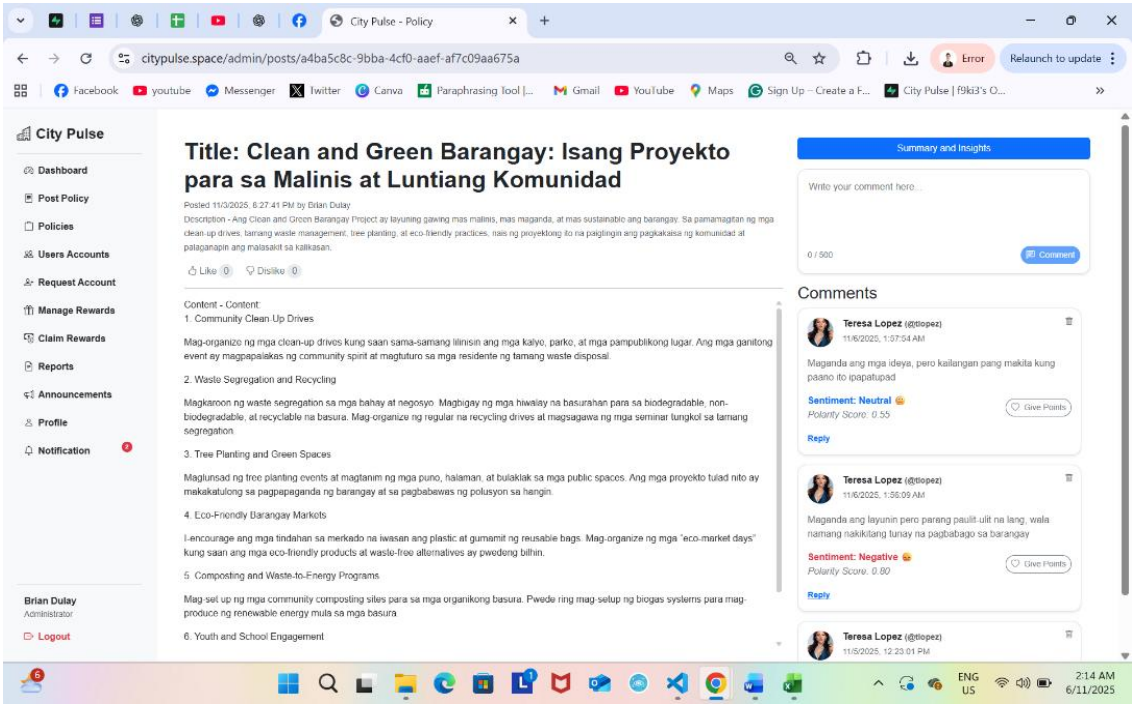


Figure 19. Sentiment-Based Policy Feedback

Figure 20 further demonstrates how machine learning improves feedback interpretation by displaying a sentiment breakdown of 39 analyzed comments. In this sample, seven comments were classified as positive, 11 as negative, and 21 as neutral, yielding a polarity score of 0.3, indicating a slightly negative overall sentiment. The visualization helps decision-makers quickly identify concerns, patterns, and areas needing improvement. Through these analytics, policymakers can respond more efficiently and refine future policy decisions based on actual community reaction.

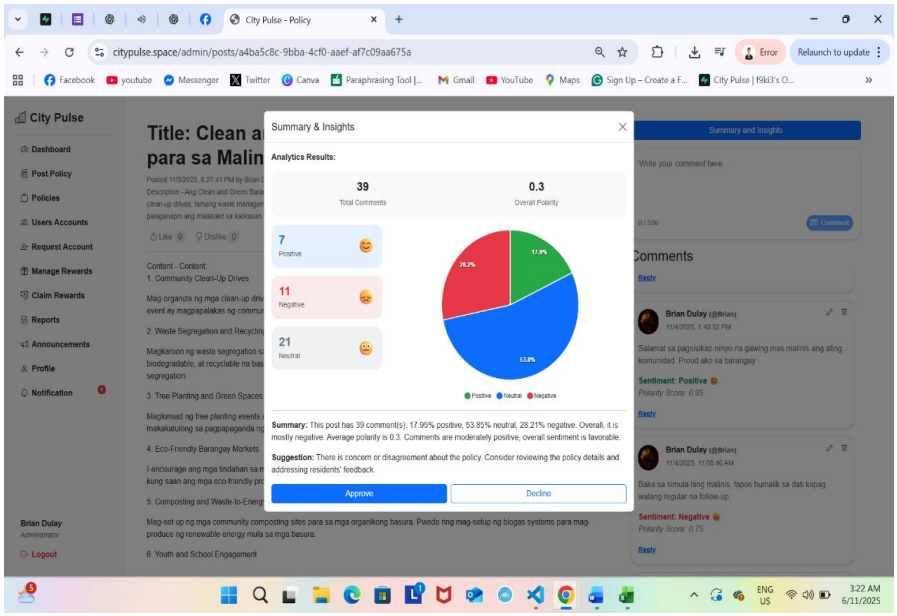


Figure 20. Sentiment Analysis of Community Feedback on Policy Posts

Figure 21 shows the Admin Dashboard, where aggregated sentiment results are visualized in real time. The dashboard highlights the distribution of positive, neutral, and negative sentiments across policy posts and identifies which policies are most discussed or controversial. This level of visualization allows administrators to monitor community sentiment trends and evaluate policy performance based on evidence rather than assumption. By transforming public comments into actionable insights, the sentiment analysis feature supports more informed, data-driven policymaking and improves the overall responsiveness of LGU officials.

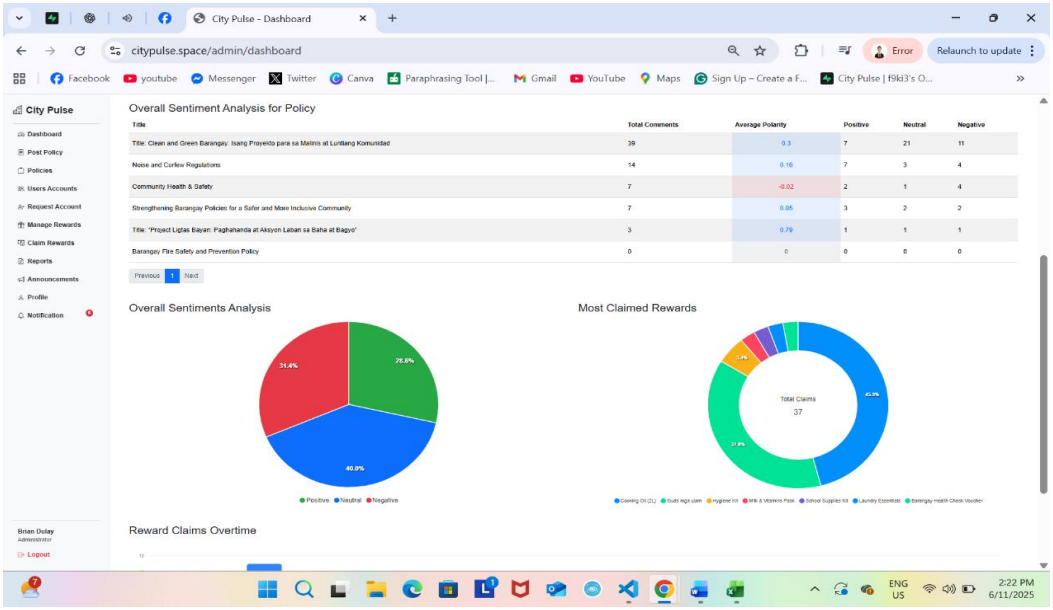


Figure 21. Analyzing Policy Feedback and Claim Trends on the City Pulse Dashboard

Evaluate the system's software quality based on the ISO 25010 standards to assess its performance in terms of functionality, reliability, usability, efficiency, security, compatibility, maintainability, and portability.

Table 13 presents findings indicating that the system demonstrates a high level of functional suitability for its intended use in a local government environment. Essential features are delivered effectively, and core functions operate reliably, demonstrating consistent compliance with operational requirements. Key strengths include the system's capability to perform primary tasks that support policy-related processes and user engagement. At the same time, the results also highlight areas for improvement in the quality, accuracy, and analytical depth of generated outputs. Overall, the system satisfies the functional requirements defined by the ISO 25010 standard, with additional refinements recommended to enhance reporting accuracy and decision-support effectiveness.

Table 13. Functional Suitability

Item	Mean	Standard Deviation	Interpretation
The system provides all required policy.	3.67	0.58	Very Good
Outputs (reports, data) are accurate.	3.33	0.58	Good
Reward system performs as intended.	4.33	0.58	Excellent
User feedback and sentiment.	3.33	0.58	Good
System supports intended government.	3.67	0.58	Very Good
Overall Mean	3.67	0	Very Good

Table 14 indicates that the system demonstrates high reliability, providing stable, consistent performance that meets user expectations. Core functions operate dependably under both normal and demanding conditions, ensuring uninterrupted access and smooth operation. Strengths of the system include its stability and continuous availability, which support user confidence and seamless workflow. At the same time, the findings highlight opportunities for improvement in areas such as data handling and performance under heavy use, suggesting that optimizing these aspects could further enhance overall system reliability. Overall, the system demonstrates reliable operation aligned with its intended purpose.

Table 14. Reliability

Item	Mean	Standard Deviation	Interpretation
The system runs without crashes or errors.	4.67	0.58	Excellent
Data is saved and retrieved correctly.	4.00	1.00	Very Good
System can recover after errors.	4.00	0.00	Very Good
Uptime availability.	4.33	1.15	Excellent
Consistent operation during heavy use.	4.00	1.00	Very Good
Overall Mean	4.2	0.47	Very Good

Table 15 indicates that the system demonstrates high performance efficiency, maintaining fast, reliable operation across various tasks. Core processes operate smoothly, allowing multiple users to interact with the system simultaneously while reward calculations and responses remain timely. The system's strengths lie in its overall responsiveness and ability to handle concurrent activity effectively, ensuring a seamless user experience. At the same time, the evaluation suggests opportunities to optimize resource utilization and further enhance stability during peak usage. Overall, the system is efficient and scalable, supporting real-world demands while maintaining consistent performance.

Table 15. Performance Efficiency

Item	Mean	Standard Deviation	Interpretation
System loads quickly	4.00	1.00	Very Good
Handles multiple Users efficiently	4.33	0.58	Excellent
Reward calculations are processed quickly	4.00	1.00	Very Good
Efficient use of server resources	3.67	0.58	Very Good
Overall responsiveness	4.33	0.58	Excellent
Overall Mean	4.07	0.23	Very Good

Table 16 indicates that the system demonstrates high usability, offering an intuitive, user-friendly experience that meets its audience's needs. Users can easily navigate and interact with the system, thanks to clear guidance and accessible features, which contribute to overall satisfaction. Strengths include the system’s engaging user experience and inclusive design, which support efficient task completion and positive interactions. While usability is generally strong, minor variations in perception suggest opportunities to provide additional guidance for certain users. Overall, the system is well-designed, easy to use, and fosters consistent user satisfaction.

Table 16. Usability

Item	Mean	Standard Deviation	Interpretation
Interface design is intuitive.	4.00	0.00	Very Good
Instructions and features are easy to understand.	4.00	1.00	Very Good
Feedback and forms are User-friendly.	4.00	1.00	Very Good
User experience is satisfying.	4.33	0.58	Excellent
Accessibility for all Users.	4.33	0.58	Excellent
Overall Mean	4.13	0.41	Very Good

Table 17 indicates that the system demonstrates a strong level of security, providing reliable protection of user data and controlled access to its features. Key measures ensure that sensitive information is safeguarded and errors are managed without compromising data integrity. The system's strengths lie in maintaining overall data protection and preventing unauthorized access, which supports user confidence in its security. At the same time, minor areas for improvement include enhancing authentication and encryption processes to strengthen data transmission and login security further. Overall, the system is perceived as secure and dependable, with only slight refinements needed to optimize protection mechanisms.

Table 17. Security

Item	Mean	Standard Deviation	Interpretation
User data is protected and private.	4.00	1.00	Very Good
Login and authentication are secure.	3.67	0.58	Very Good
Prevents unauthorized access.	4.00	1.00	Very Good
Data transmission is encrypted.	3.67	1.53	Very Good
Handles errors without exposing data.	4.00	1.00	Very Good
Overall Mean	3.87	0.34	Very Good

Table 18 indicates that the system demonstrates a high level of compatibility, performing reliably across multiple devices, browsers, and external integrations. Its strengths include seamless operation on different platforms and effective synchronization of user-related processes, ensuring smooth interaction and data handling. Minor areas for improvement involve interoperability with existing government databases, where some users experienced challenges. Overall, the system is perceived as highly compatible, supporting efficient operation across diverse technical environments while offering further opportunities to enhance integration with legacy systems.

Table 18. Compatibility

Item	Mean	Standard Deviation	Interpretation
Works on different browsers.	4.33	0.58	Excellent
Works across various devices (PC, tablet, mobile).	4.67	0.58	Excellent
Integrates with external data or APIs.	4.00	1.00	Very Good
Reward system syncs with User data.	4.67	0.58	Excellent
Compatible with existing government databases or systems.	3.33	1.15	Good
Overall Mean	4.20	0.28	Very Good

Table 19 indicates that the system demonstrates a high level of maintainability, enabling efficient maintenance and future development with minimal disruption. Its strengths lie in its modular design and the ability to perform updates with minimal downtime, which support consistent system upkeep. Areas for potential improvement include code readability, update procedures, and the effectiveness of debugging tools, which further streamline maintenance activities. Overall, the system is generally maintainable, providing a stable foundation for ongoing enhancements and support.

Table 19. Maintainability

Item	Mean	Standard Deviation	Interpretation
Code is easy to modify and update.	3.33	0.58	Good
System updates can be applied easily.	3.33	0.58	Good
Error logs and debugging tools are effective.	3.33	0.58	Good
Modules are reusable for future development.	3.67	0.58	Very Good
Maintenance requires minimal downtime.	3.67	0.58	Very Good
Overall Mean	3.47	0	Very Good

Table 20 indicates that the system demonstrates excellent portability, allowing it to operate seamlessly across different platforms and environments. Users found the system easy to install, deploy, and migrate data, while its adaptability to environmental changes highlights outstanding flexibility. The only minor area for improvement involves simplifying configuration processes to enhance user convenience further. Overall, the system excels in portability, supporting efficient deployment and smooth transitions across varying technical contexts.

Table 20. Portability

Item	Mean	Standard Deviation	Interpretation
Can be deployed on different platforms.	4.33	0.58	Excellent
Installation process is simple.	4.33	0.58	Excellent
Configuration is straightforward.	4.00	1.00	Very Good
Data migration between environments is smooth.	4.33	0.58	Excellent
Adapts easily to environmental changes.	4.67	0.58	Excellent
Overall Mean	4.33	0.19	Excellent

Table 21 shows that the system maintains a strong overall level of software quality, performing consistently well across all ISO 25010 categories. Key strengths include stability, usability, portability, and performance efficiency, which contribute to a reliable and user-friendly experience. Areas such as functional accuracy, maintainability, and certain security features present minor opportunities for enhancement, particularly in optimizing reporting precision, code readability, and data protection processes. Overall, the system is well-designed, robust, and capable of supporting its intended functions effectively, with only slight refinements needed to achieve peak quality across all dimensions.

Table 21. Software Quality Evaluation Summary

Category	Overall Mean	Overall SD	Interpretation
Functional Suitability	3.67	0	Very Good
Reliability	4.2	0.47	Very Good
Performance Efficiency	4.07	0.23	Very Good
Usability	4.13	0.41	Very Good
Security	3.87	0.34	Very Good
Compatibility	4.20	0.28	Very Good
Maintainability	3.47	0	Very Good
Portability	4.33	0.19	Excellent
Overall	3.99	0.17	Very Good

Test and evaluate the acceptability of the system using Technology Acceptance Model.

Table 22 indicates that users perceive the system as highly useful in supporting decision-making and community engagement. The platform effectively facilitates understanding of community sentiments, improves local government processes, and encourages active participation. These results suggest that the system can meaningfully enhance operational efficiency and policymaking outcomes, reinforcing its value to both administrators and the community. Additionally, the high perceived usefulness may increase overall user trust in the system and encourage more consistent utilization of its features.

Table 22. Results of Perceived Usefulness

Item	Mean	Standard Deviation	Interpretation
The system helps understand community sentiments on policies.	3.55	0.50	Excellent
Improves the quality of LGU decision-making.	3.45	0.54	Excellent
Motivates community participation through rewards.	3.46	0.52	Excellent
Makes policy recommendation tasks faster.	3.37	0.56	Excellent
Enhances overall effectiveness of community engagement	3.53	0.52	Excellent
Composite Mean	3.47	0.53	Excellent

Table 23 shows that users find the system intuitive and user-friendly, with smooth navigation and straightforward processes. Its design allows users to operate it with minimal assistance, promoting accessibility and efficiency. This ease of use indicates that the system can be readily adopted by users with varying levels of technical proficiency, supporting widespread engagement. Moreover, the user-friendly design may reduce training requirements and facilitate quicker onboarding for new users.

Table 23. Result Perceived Ease of Use

Item	Mean	Standard Deviation	Interpretation
The interface is easy to navigate.	3.33	0.51	Excellent
System can be used without technical help.	3.16	0.66	Good
Feedback submission process is straightforward.	3.48	0.52	Excellent
Reward system features are easy to understand.	3.41	0.51	Excellent
Learning the system is quick and effortless.	3.40	0.53	Excellent
Overall Mean	3.36	0.55	Excellent

Table 24 demonstrates strong user intention to continue engaging with the system, recommending it to others, and utilizing its features regularly. The digital reward mechanism further supports participation, reinforcing consistent use. These findings imply that the system is likely to achieve sustained adoption and active involvement, contributing to long-term community engagement. Furthermore, a strong behavioral intention suggests that the system can become an integral tool in supporting local government policy initiatives.

Table 24. Results of Behavioral Intention to Actual Use

Item	Mean	Standard Deviation	Interpretation
Intention to continue using the system.	3.46	0.52	Excellent
Recommending to others.	3.53	0.54	Excellent
Frequency of use for feedback or policy checking.	3.41	0.53	Excellent
Using digital rewards to engage.	3.42	0.52	Excellent
Preference over traditional feedback methods	3.42	0.57	Excellent
Overall Mean	3.45	0.54	Excellent

Table 25 indicates a very positive user attitude toward the system, reflecting enjoyment, confidence, and belief in its benefits for public participation. Users generally feel comfortable navigating the platform and express willingness to recommend it. The overall perception highlights the system's success in fostering user satisfaction and trust, which are crucial for continued usage and acceptance. In addition, the positive attitude can help drive advocacy among users, encouraging wider adoption within the community

Table 25. Attitude Toward Using (ATU)

Item	Mean	Standard Deviation	Interpretation
Enjoyment while using the system.	3.51	0.52	Excellent
Recommending to others.	3.49	0.54	Excellent
Belief that it improves public participation.	3.49	0.54	Excellent
Confidence and comfort in using the system.	3.45	0.52	Excellent
Overall positive perception of the system.	3.61	0.53	Excellent
Overall Mean	3.51	0.53	Excellent

Table 26 shows that the system is highly accepted across all Technology Acceptance Model dimensions, including usefulness, ease of use, behavioral intention, and attitude. Users consistently report satisfaction and positive experiences, confirming the system's effectiveness in achieving the objectives of usability, engagement, and acceptability. This overall positive reception suggests strong potential for long-term adoption and integration into local government processes. It also indicates that the system is well-aligned with user needs, which is critical for its sustained success and impact.

Table 26. Overall Evaluation results of TAM survey answers per category

Item	Mean	Standard Deviation	Interpretation
Perceived Usefulness (PU)	3.47	0.53	Excellent
Perceived Ease of Use (PEOU)	3.36	0.55	Excellent
Behavioral Intention to Actual Use (BI)	3.45	0.54	Excellent
Attitude Toward Using (ATU)	3.51	0.53	Excellent
Overall Mean	3.45	0.54	Excellent

Legend: 3.26 – 4.00 = Excellent, 2.51 – 3.25 = Good, 1.76 – 2.50 = Fair, 1.00 – 1.75 = Poor

System Validation Summary

The overall validation of the Local Government Web-based Policy the Recommendation System was established by integrating findings from both the Technology Acceptance Model (TAM) and the ISO 25010 software quality evaluation. Results from TAM indicate that users rated the system highly in Perceived Usefulness, Perceived Ease of Use, and Behavioral Intention to Actual Use, demonstrating that it is effective, user-friendly, and engaging. These outcomes highlight users' strong willingness to continue utilizing the system and recommend it for fostering community participation.

Concurrently, the ISO 25010 evaluation confirmed that the system Concurrently, the ISO 25010 evaluation confirmed that the system performs at a Very Good to Excellent level across key software quality attributes, including functionality, reliability, usability, and maintainability. This demonstrates that the platform meets established standards for software performance, stability, and

efficiency, ensuring dependable and consistent use in local governance operations.

Overall, the combined results from TAM and ISO 25010 validate that the system is both technically robust and highly acceptable to users. Its reliability, usability, and strong user acceptance support its suitability as an effective tool for promoting participatory governance, facilitating informed policy making, and enhancing community engagement.

Thematic Analysis of User Feedback

In addition to quantitative evaluation, a qualitative thematic analysis was conducted to identify recurring themes from open-ended responses gathered from Users of the City Pulse community participatory platform. This analysis provided deeper insights into User experiences, satisfaction, and suggestions for improvement, complementing the numerical results obtained through statistical evaluation.

Table 27 shows that users highly value the system’s usability and responsiveness, praising the intuitive interface and quick system feedback. While the platform is successful in fostering community engagement, some users highlighted the need for more transparent reward tracking. These insights suggest that improving reward visibility could further motivate participation and sustain user interest. Additionally, the qualitative feedback reinforces quantitative findings, confirming that City Pulse effectively supports participatory governance while providing actionable directions for future enhancements.

Table 27. Summary of User Feedback Themes

Theme	Description	User Statement
Usability	Users found the system intuitive, clean, and easy to navigate on both desktop and mobile.	“I like how clear and User-friendly the system forum looks.”
Response Speed	Users appreciated the fast validation of their comments and forum activity.	“The system responds quickly whenever I send feedback or join a discussion.”
Reward Transparency	Participants suggested clearer tracking of earned points and rewards.	“It would be great to see how close I am to earning a reward or badge.”
Community Engagement	Users valued being able to express opinions that influence local discussions.	“It feels great knowing my comment can contribute to real community decisions.”

Table 28 indicates that the system demonstrates high reliability in sentiment classification, achieving an accuracy rate of 91.4% along with strong precision and recall scores. Efficient processing ensures that results are available nearly in real-time, supporting timely analysis of user feedback. These results suggest that the sentiment analysis model can be trusted to provide accurate insights into community opinions. Furthermore, the use of a manually annotated test dataset validates the academic rigor of the evaluation, enhancing confidence in the system’s performance metrics.

Table 28. System Performance Metrics

Metric	Result (%)	Description
Accuracy	91.4	Model correctly classified 91.4% of User sentiments.
Precision	89.7	Among positive predictions, 89.7% were correct.
Recall	90.2	The model successfully retrieved 90.2% of relevant sentiments.
Execution Time	1.2s per request	Average time taken to process and classify sentiment.

Table 29 shows that sentiment results vary across policy topics, with environmental initiatives and youth programs generating the most positive feedback. Neutral comments dominate most discussions, reflecting users' preference for balanced or factual input, while negative comments provide constructive criticism on policy implementation. These findings suggest that the system successfully captures a wide range of perspectives, offering valuable insights for policymakers. Additionally, the balance of positive, neutral, and negative sentiments highlights City Pulse's ability to facilitate honest and constructive engagement, supporting informed decision-making and participatory policymaking.

Table 29. Overall Sentiment Analysis for Policy Discussions

Title	Total Comments	Average Polarity	Positive	Neutral	Negative
Clean and Green Barangay: Isang Proyekto para sa Malinis at Luntiang Komunidad	40	0.31	7	22	11
Noise and Curfew Regulations	14	0.16	7	3	4
Community Health & Safety	7	-0.02	2	1	4
Strengthening Barangay Policies for a Safer and More Inclusive Community	7	0.05	3	2	2
Youth Empowerment and Skills Training: Isang Proyekto para sa Kabataang Handang sa Kinabukasan	5	0.80	2	2	1
Project Ligtas Bayan: Paghahanda at Aksyon Laban sa Baha at Bagyo	3	0.79	1	1	1
Barangay Fire Safety and Prevention Policy	0	0	0	0	0

Acceptance Comparison

After presenting and discussing the Technology Acceptance Model and ISO 25010 evaluation results, insert the following paragraph to strengthen your comparative discussion:

This acceptance rate aligns with findings from comparable digital governance and e-participation platforms. For instance, Aliling, Santa Ana, and Soberano (2025) reported similarly high User satisfaction and system quality ratings in their *digital governance initiative in Paracale*, which achieved excellent usability and reliability scores based on ISO 25010:2011 standards.

Likewise, Kim and Park (2020) observed that incorporating gamification and reward systems in e-government platforms significantly enhanced User engagement and perceived usefulness. These parallels suggest that the Local Government Web-based Policy Recommendation System demonstrates a level of acceptance consistent with successful digital governance models, reinforcing its potential to promote participatory and transparent policymaking. *(Cite as: Aliling, Santa Ana, & Soberano, 2025; Kim & Park, 2020.)*

Result of Software Testing

The following results were obtained via unit testing, integration system, and compatibility testing.

Unit Testing Result

Table 30 demonstrates the Forum Post and Reward System. Modules are fully functional, with all tested cases passing successfully. Initial failures in comment and reply handling were identified and resolved, confirming the module's reliability. These results highlight that the module can support user interaction and reward validation effectively in real-time scenarios. Additionally, the variation in response time due to internet speed or file size suggests the system performs optimally under typical usage conditions. The successful testing confirms that the module meets functional requirements and enhances user engagement within the platform.

Table 30. Unit Testing for Forum Post and Reward System

TESTED CASE	EXPECTED OUTPUT	ACTUALOUTPUT/ RESPONSE TIME	REMARKS
Create Forum Post	The Admin can create a post. While the Admin is waiting for the comment.	The system successfully created the post and showed the content and description. (1-10 seconds)	PASSED
Comment/Reply in Forum	User can comment/reply in the forum	The system shows that the User successfully commented/replied on the forum (1-3 seconds)	PASSED
Comment Validation	The comment of the User is validated by the Admin to get points.	The system successfully validates the comment of the User to get points	PASSED
Delete Comment	The User can delete their comment	The system shows that the User deleted their own comment only (1 second)	PASSED

Table 31 confirms that all data analytics functions operate correctly, providing accurate outputs for sentiment analysis and user participation. The admin-only access ensures secure handling of sensitive data, maintaining system integrity. These results indicate that the module is reliable for monitoring community engagement and supporting administrative decisions. Moreover, the consistent 1-second response time demonstrates the module's efficiency in delivering analytics promptly. The unit testing ensures that administrators can trust the accuracy and accessibility of data for informed decision-making.

Table 31. Unit Testing for Data Analytics Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/ RESPONSE TIME	REMARKS
Admin only access	The Admin is only one who has access	The system successfully shows the data analytic module on Admin can access (1 second)	PASSED
Sentiment Analytics	The overall sentiment analysis is shown	The system successfully shows the data analytics for the overall sentiment analysis (1 second)	PASSED
User Joined Analytics	The summary of all Users who have joined is shown.	The system successfully shows the data analytics for the overall User participation (1 second)	PASSED

Table 32 shows that the Notification module successfully informs users of new comments and replies. Initial issues with notification redirection were debugged and resolved, confirming correct system behavior. The results indicate that the system can maintain timely user engagement and improve responsiveness to forum activity. The module's reliable notifications also contribute to keeping users actively involved in discussions. Overall, this ensures that the system effectively supports user participation and the platform interactivity.

Table 32. Unit Testing for Notification Module

TESTED CASE	EXPECTED OUTPUT	ACTUALOUTPUT/ RESPONSE TIME	REMARKS
Get notification	The User is notified if the new comments or reply appear in the system.	The system successfully shows the User get a notification if there are new comments and replies in forum (3seconds)	PASSED

Table 33 confirms that the Sentiment Analysis module accurately classifies comments, generates polarity scores, and handles mixed-language inputs. All tested functions passed, demonstrating the system's efficiency and accuracy. These results support reliable real-time sentiment monitoring to inform participatory governance decisions. The successful handling of mixed-language inputs highlights the system's adaptability to diverse user comments. In addition, the dashboard display of sentiment analytics ensures that administrators can quickly assess overall community opinion.

Table 33. Unit Testing for Sentiment Analysis Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT/ RESPONSE TIME	REMARKS
Analyse Comment Sentiment	The system should classify the comment as positive, negative, or neutral.	The system successfully analyses the comment and displays the correct sentiment label (2 seconds).	PASSED
Generate Polarity Score	The system should generate a polarity score based on the BERT model.	The system successfully generates an accurate polarity score for the input comment (2 seconds).	PASSED
Handle Mixed Language Input	The system should correctly analyse comments containing English and Tagalog.	The system successfully processes and analyses mixed-language comments with correct sentiment output (3 seconds).	PASSED
Display Sentiment Analytics Dashboard	The system should show the overall sentiment analytics.	The system successfully displays the sentiment analytics in the dashboard (2 seconds).	PASSED

Compatibility Testing

Table 34 shows that the system performs best on Microsoft Edge while maintaining good accessibility, best practices, and SEO compliance across all tested browsers. Performance variations on Chrome and Opera GX suggest minor optimization opportunities. Overall, these results confirm the system's broad compatibility for end users. This ensures that users can reliably access the platform regardless of their preferred browser or device. The findings also indicate the system's readiness for deployment in varied technical environments.

Table 34. Compatibility Testing of the System

Browser	Web Page	Performance	Accessibility	Best Practices	SEO
Chrome	Login Page	50	90	61	91
Chrome	Admin Create Forum Page	62	100	100	90
Chrome	Forum Page	59	79	93	82
Chrome	Admin Forum Page	61	94	100	91
Chrome	User Reward Page	55	92	100	91
Chrome	Admin Reward Page	61	95	100	91
Chrome	Admin Data Analytic Page	55	96	100	90
Microsoft Edge	Login Page	92	90	78	91
Microsoft Edge	Admin Create Forum Page	90	100	100	90
Microsoft Edge	Admin Reward Page	89	94	100	91
Microsoft Edge	User Reward Page	82	92	100	91
Microsoft Edge	Admin Data Analytic Page	54	95	100	90
Opera Gx	Login Page	63	90	79	91
Opera Gx	Admin Create Forum Page	52	100	100	90
Opera Gx	User Reward Page	55	92	100	91
Opera Gx	Admin Reward Page	62	94	100	91
Opera Gx	Admin Data Analytic Page	60	95	100	90
Opera Gx	Sentiment Analytics of Forum Page	60	93	100	91

Integration Testing

Table 35 illustrates that the system's modules operate cohesively, successfully passing integration tests after initial debugging. Smooth data exchange between modules ensures reliable end-to-end performance and confirms the system's readiness for real-world deployment.

Table 35. Integration Testing of the System

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
Forum Module and Reward System	The system should display all forums created by the Admin on the User interface. The Admin is also responsible for validating comments and replies to grant User points.	The system successfully displayed all created forums on the User interface. Comments and replies were validated by the Admin, and points were correctly awarded to Users (1–10 seconds).	PASSED
Data Analytics Module	The system should display data analytics on the Admin page and retrieve all sentiment data from forum posts.	The system successfully displayed data analytics when accessed by the Admin, showing the overall sentiment distribution of forum discussions.	PASSED
Sentiment Analysis Module	The system should display the overall sentiment of each forum topic by category.	The system successfully analysed and displayed the sentiment of forum posts by category (3 seconds).	PASSED
Notification Module	The system should notify Users when new comments or replies are posted.	The system successfully sent notifications to Users regarding new comments and replies.	PASSED
Dashboard Module	The system should function as the central hub, managing all User information, feedback, sentiment analysis results, and reward transactions for analysis and reporting.	The system efficiently organized and displayed all collected data in a centralized dashboard, ensuring easy access and management for administrators (4–6 seconds).	PASSED

CHAPTER 5

SUMMARY, CONCLUSION, AMD RECOMMENDATION

This chapter summarizes the study's main findings. It also presents the significance of the research and relates findings to the objectives and problems written in the introduction part of the study. Recommendations must be made in this chapter. This part usually directs the reader to conduct further research on some specific areas related to the study.

SUMMARY

The City Pulse System is a web-based community policy recommendation platform designed to support data-driven governance by combining sentiment analysis with a digital reward mechanism. The system allows citizens to provide feedback on local policies through a forum interface, while NLP-powered sentiment analysis, implemented with a pre-trained BERT model, transforms unstructured text into insights that help policymakers detect emerging issues and gauge public opinion. The integrated reward module motivates users to participate more actively, leading to higher engagement and sustained civic involvement.

Developed through the Agile Scrum methodology, the system includes four major modules: forum discussions, Admin post management, sentiment analysis, and the digital reward module. Technical implementation using HTML, CSS, JavaScript, Supabase, and Python ensured accessibility across devices, while modular development enabled continuous refinement. System evaluation showed strong performance: sentiment classification was accurate, the reward system effectively increased participation, and the forum and Admin modules

functioned smoothly. User testing using the Technology Acceptance Model (TAM) yielded an overall Excellent rating, indicating high usefulness, ease of use, and an intention to continue using the platform.

Beyond technical development, the project highlights how digital Beyond technical development, the project highlights how digital platforms can enhance participatory governance. By combining AI-driven insights with community engagement, the system promotes transparency, inclusiveness, and evidence-based policy formulation. Ethical considerations— such as anonymized data handling and fairness in reward distribution—were incorporated to ensure responsible governance. Ultimately, the findings show that City Pulse serves as a practical and sustainable model for integrating sentiment analytics into policymaking, reinforcing citizen trust and collaboration with local authorities.

CONCLUSION

In conclusion, the City Pulse System successfully achieves the following objectives:

Developing a Web-based Thread System for Policy Recommendations with a Digital Reward Module

The City Pulse System successfully created an accessible thread-based platform that enables structured policy discussions and real-time community participation. The digital reward module proved effective in motivating citizens to share feedback by converting participation into redeemable points. This offers a novel approach to civic engagement, combining discussion, feedback analytics, and gamified rewards in a single system. A key lesson learned is that

simple incentives significantly increase participation, especially in local governance platforms.

Utilizing a Sentiment Analysis Model Using Machine Learning Algorithms

The project implemented a machine learning–based sentiment classifier that classifies public reactions as positive, neutral, or negative. By automatically summarizing sentiment trends, the system provides policymakers with data-driven insights instead of relying on manual interpretation. The contribution lies in integrating BERT-based sentiment analysis into a local policy feedback system, which is still uncommon in LGU-level platforms. A major lesson learned is that even without full fine-tuning, pre-trained NLP models can provide meaningful sentiment insights that support policy evaluation.

Testing and Evaluating the Acceptability of the System Using the Technology Acceptance Model (TAM)

TAM results showed an overall Excellent rating, confirming the system's usefulness, ease of use, and strong user acceptance. This demonstrates that the system not only functions well technically but is also highly acceptable to real users, contributing evidence that AI-assisted civic platforms can be adopted by communities. The key lesson learned is that intuitive design and clear features directly increase user trust and consistent usage.

Evaluate the system's software quality based on the ISO 25010 standards to assess its performance in terms of functionality, reliability, efficiency, usability, security, compatibility, maintainability, and portability.

The ISO 25010 evaluation resulted in a very good confirmation that the system meets high software quality standards. Its strong performance in functionality, usability, and portability validates the system as a stable and efficient model for digital governance tools. The lesson learned is that continuous testing and modular development greatly improve system quality and maintainability, even under limited resources.

Overall, the City Pulse System successfully met its objectives by providing an accessible, interactive platform for public participation, integrating sentiment analysis to generate data-driven insights, and demonstrating strong User satisfaction and acceptance under the Technology Acceptance Model. It stands as an innovative and effective digital governance tool that promotes transparency, engagement, and evidence-based policymaking within local government settings.

RECOMMENDATION

The evaluation of the system demonstrates its strong performance in usability, reliability, and portability, indicating that it is already a valuable tool for community-driven policymaking. To further enhance its effectiveness and maximize its impact on citizen engagement and policy development, the following recommendations are proposed:

1. Develop a live analytics dashboard that provides insights into sentiment trends, participation rates, and feedback patterns. This will support policymakers in making faster, data-driven decisions and identifying emerging community concern.
2. Conduct pilot implementations in various Local Government Units to evaluate scalability, adapt features to different local governance contexts, and gather user feedback to refine the system further.
3. Introduce multi-factor authentication, CAPTCHA verification, and encrypted communication channels to safeguard user information, while maintaining a seamless experience for participants.
4. Link the system with LGU dashboards and national open data portals to enhance transparency, facilitate policy tracking, and encourage interdepartmental collaboration.
5. Collaborate with civic groups, schools, and local organizations to raise awareness about the platform, educate users on its benefits, and encourage active, sustained participation in policy discussions.
6. Future research can incorporate AI-based techniques such as predictive topic modelling, trend forecasting, or bias detection to enhance the system’s analytical depth and provide even more.

7. The system should enhance security and data privacy by Implementing stronger authentication, encrypted data handling, and regular security checks to ensure user information remains safe and protected.

Future Work

Future work may focus on improving the system's contextual understanding and localization rather than expanding language support, since BERT already handles over 100 languages. To enhance the accuracy of government-related feedback, future researchers may fine-tune BERT using Filipino, Taglish, and local dialect datasets to capture better culturally specific expressions, indirect sentiments, and sarcasm. The system may also be expanded to include emotion detection, topic modelling, and toxicity filtering to provide deeper insights into community concerns and ensure safer public interactions. Additionally, integrating predictive analytics and real-time sentiment monitoring can help local government units anticipate emerging issues and make more proactive decisions, thereby fostering a more responsive, data-driven governance ecosystem.

To support scalability and policy transparency, the system may also be integrated with Local Government Unit (LGU) dashboards **or** Open Data systems**.** Such integration would enable real-time policy monitoring, data sharing, and cross-departmental collaboration, strengthening the platform's role as a participatory governance tool.

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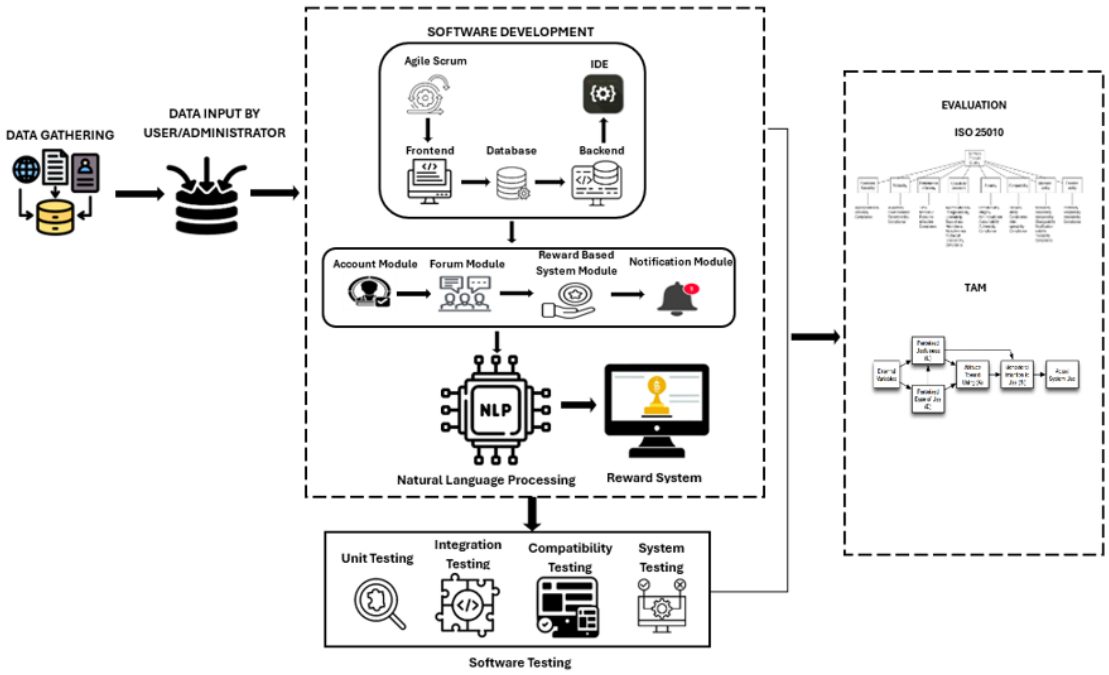
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APPENDIX

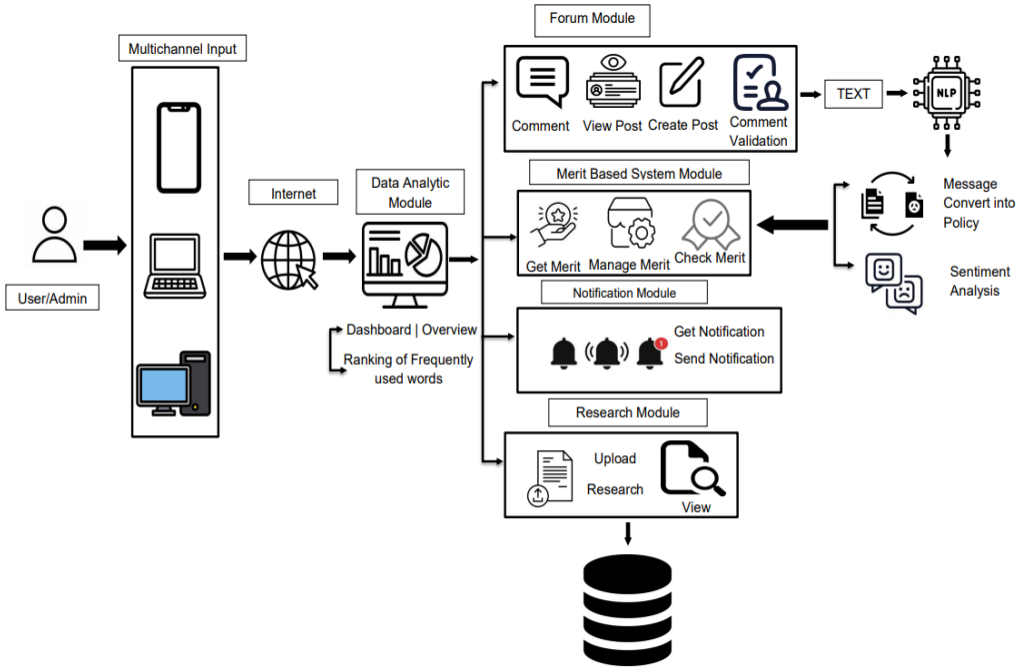
APPENDIX A

Figures

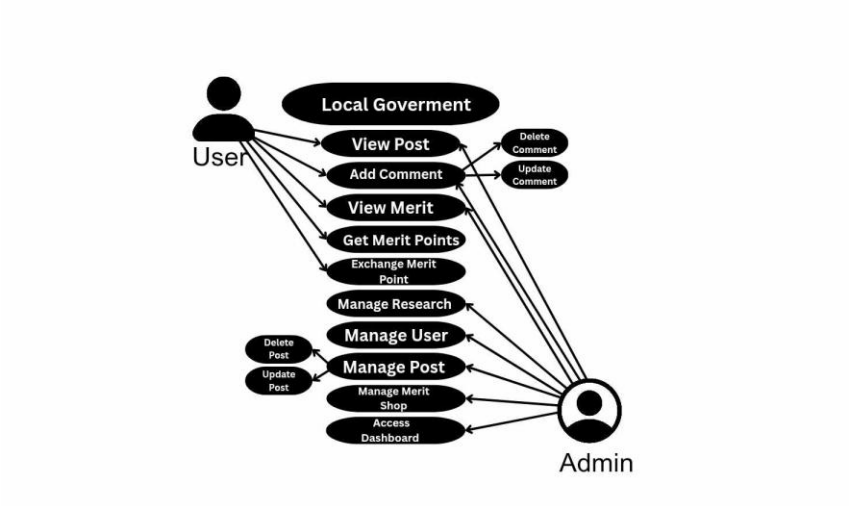
System Flow



System Architecture

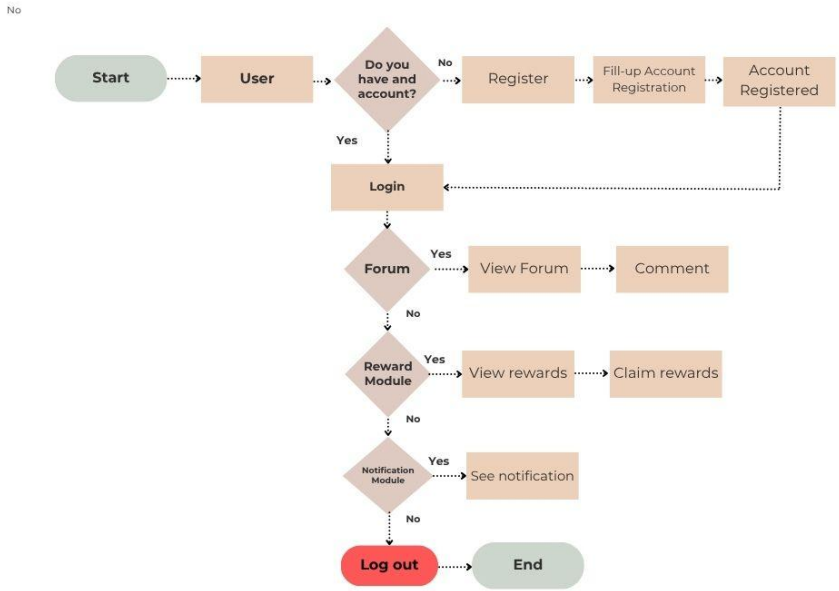


Use Case Diagram

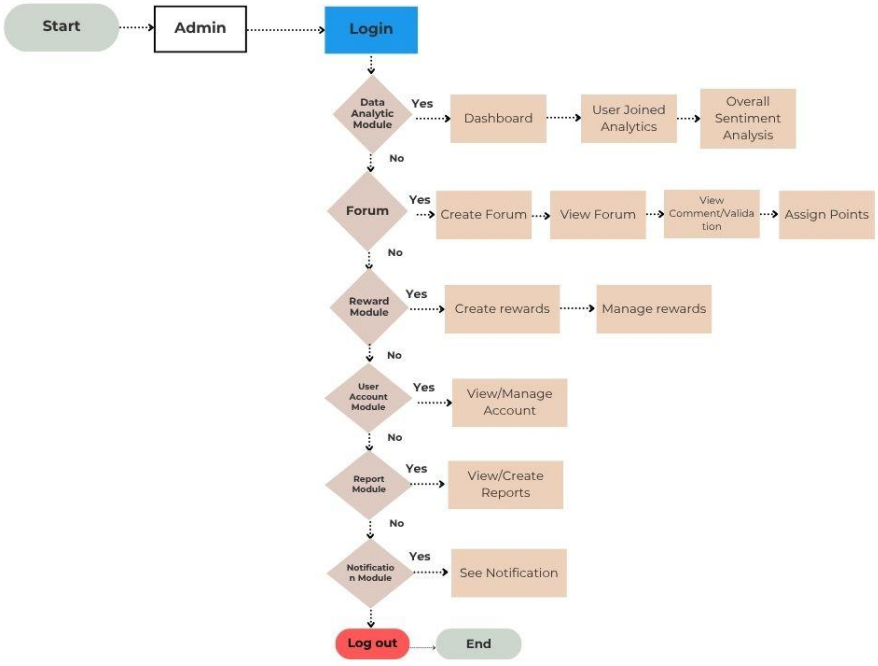


Flowchart

System Flowchart of the User



System Flowchart of the Admin



APPENDIX B

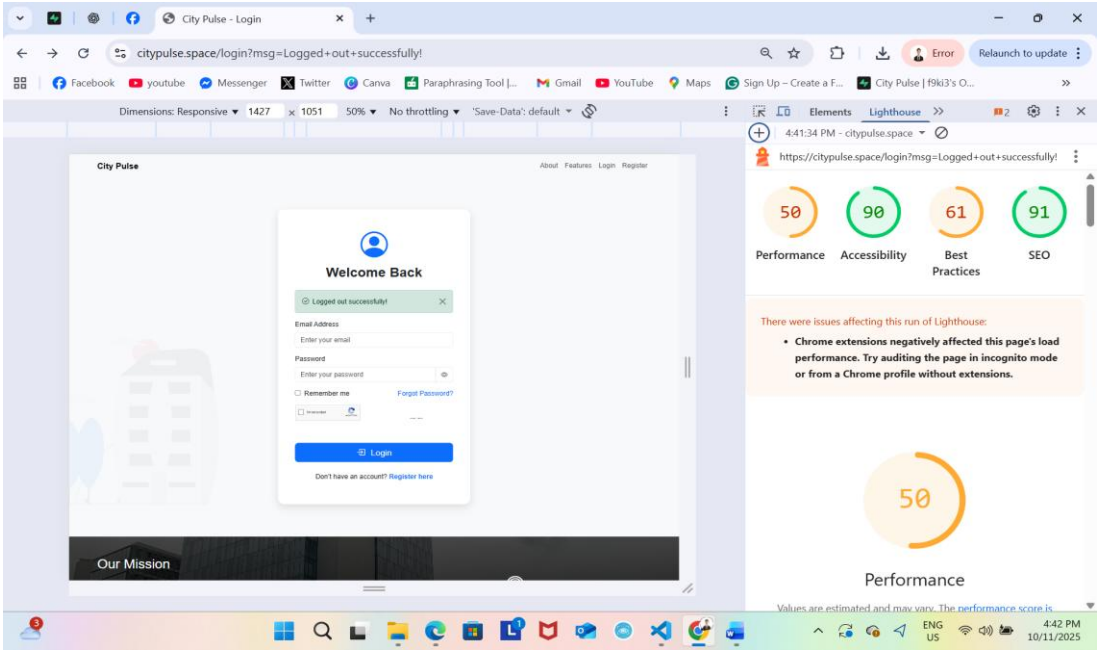
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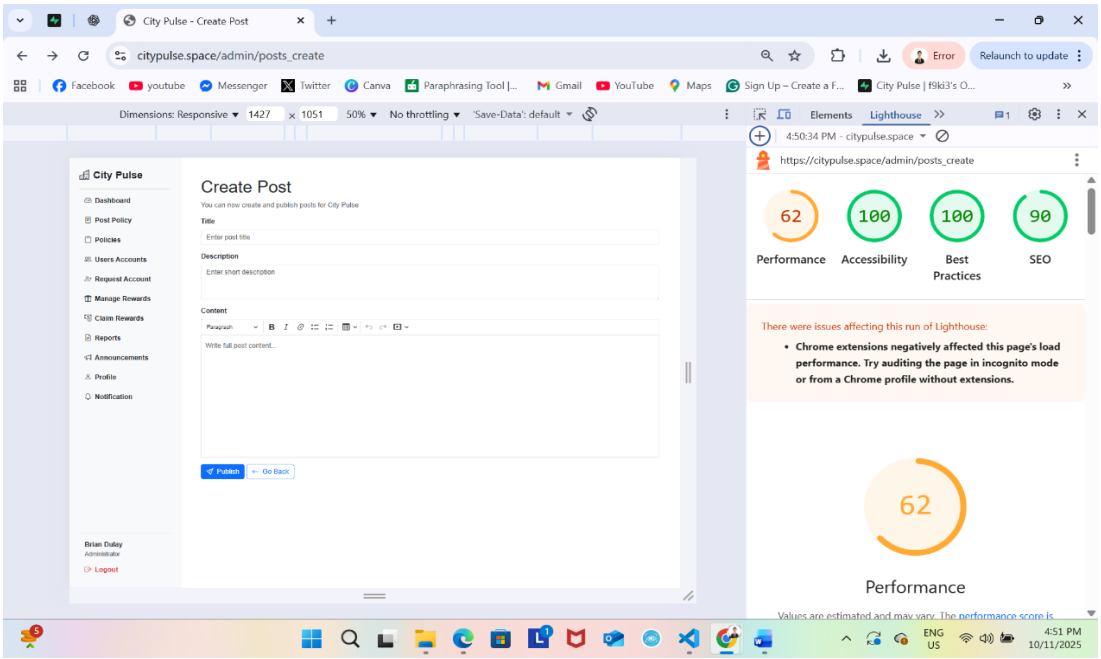
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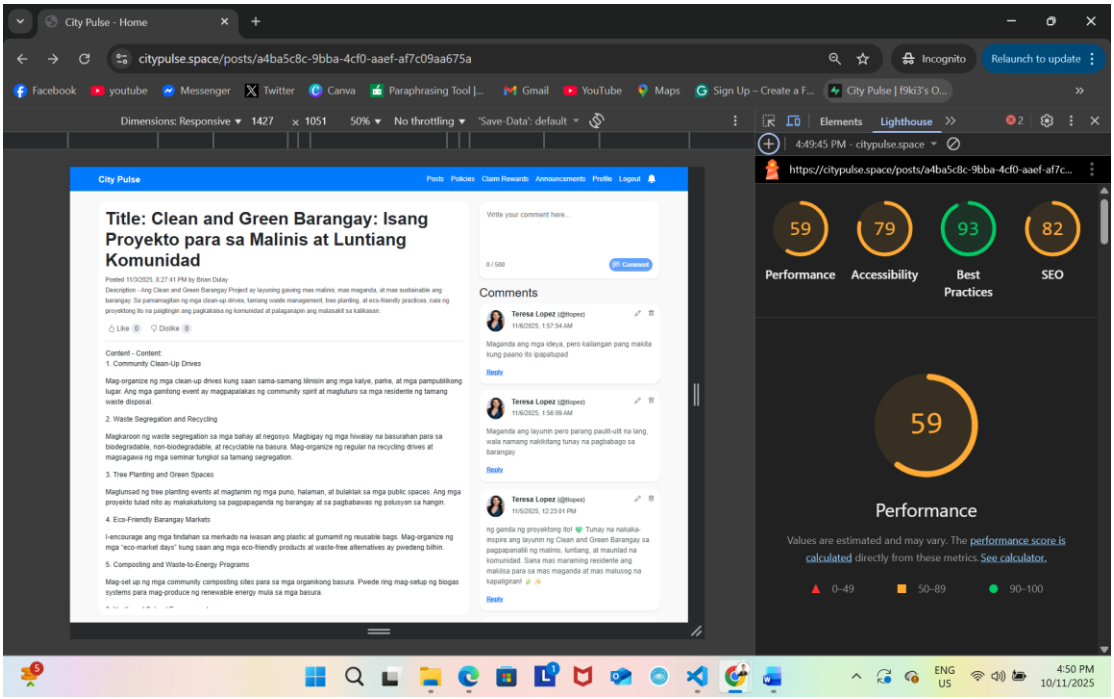
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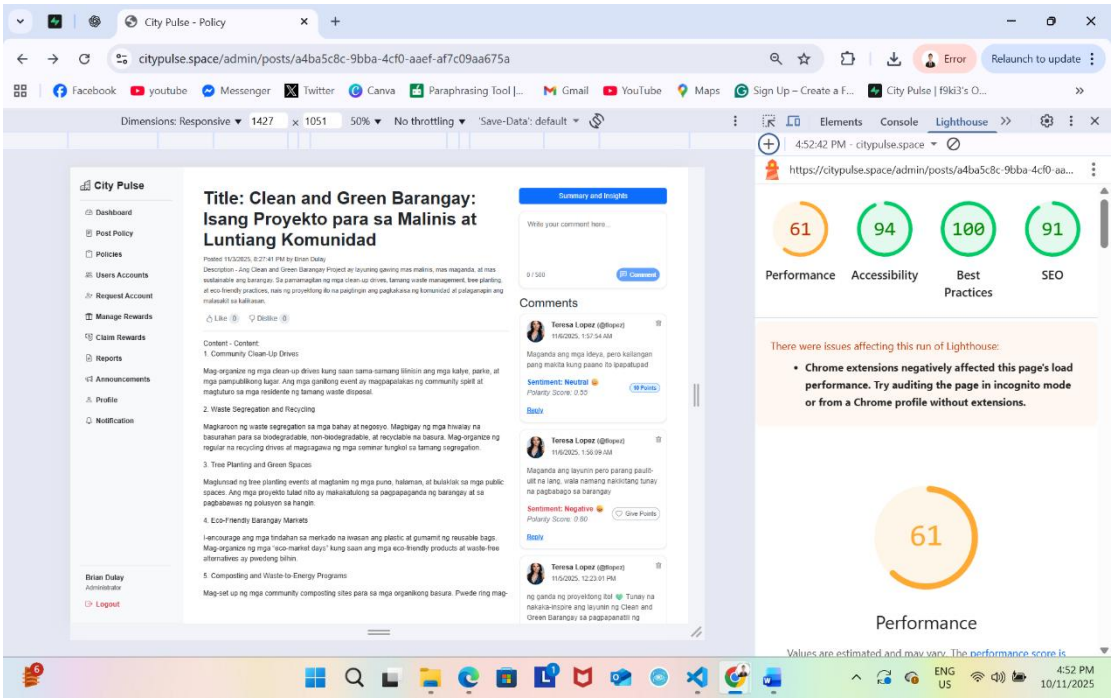
Develop a web-based thread system to policy recommendation with digital reward module.

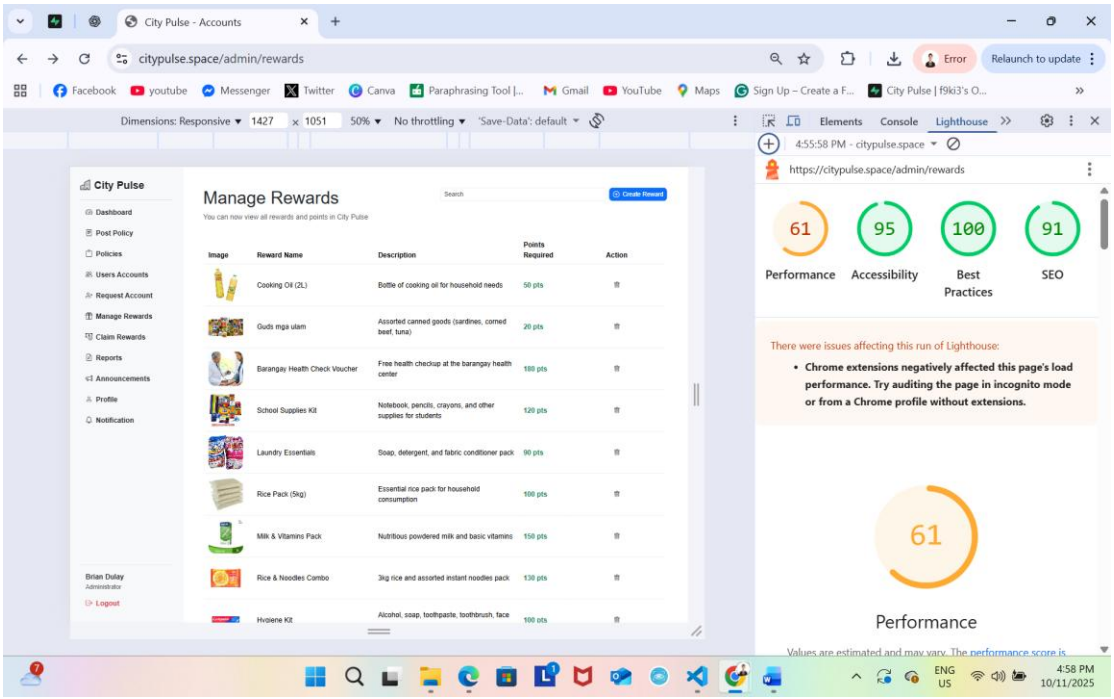
Chrome Lighthouse

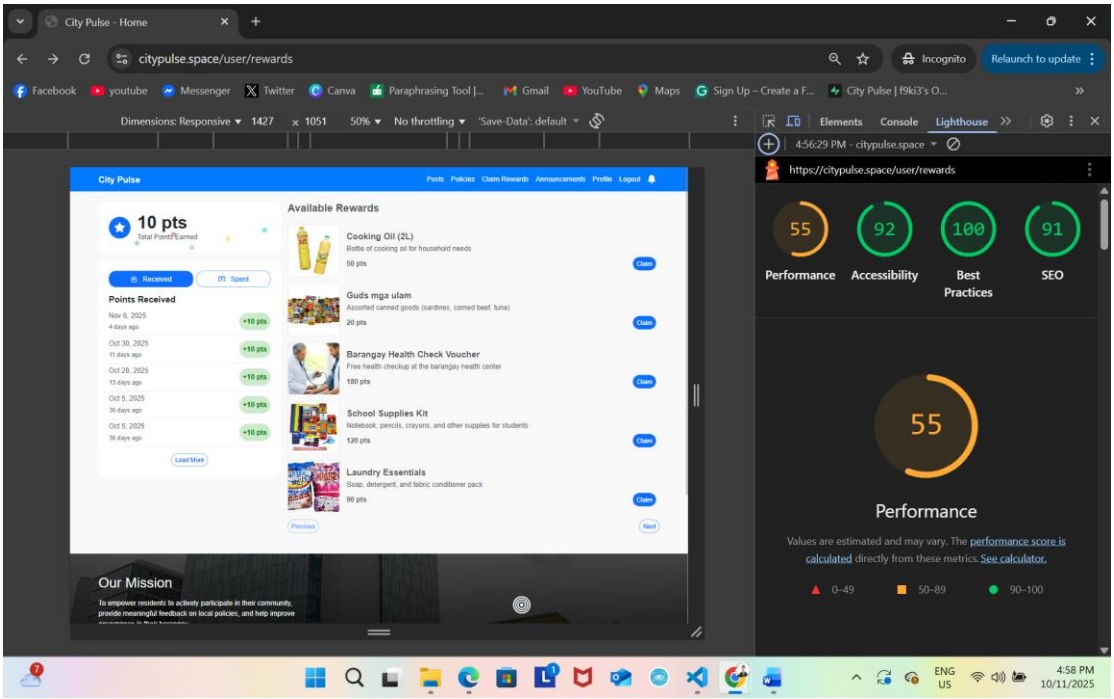




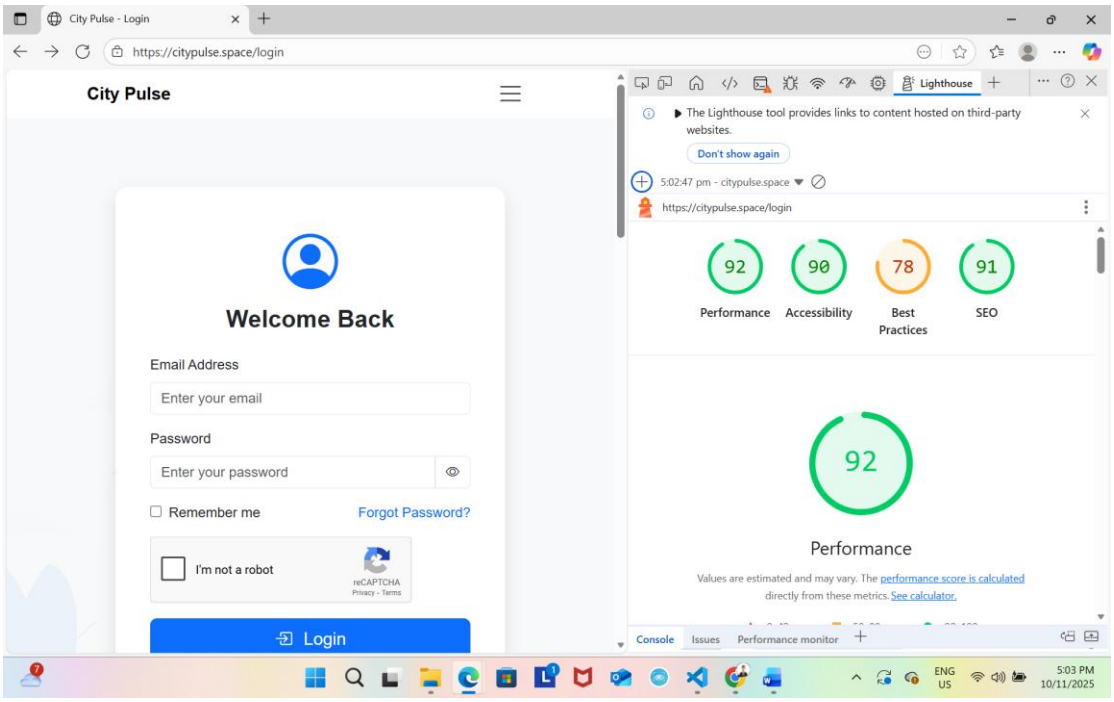

City Pulse - Home
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City Pulse
Posts Policies Claim Rewards Announcements Profile Logout
Title: Clean and Green Barangay: Isang Proyekto para sa Malinis at Luntiang Komunidad
Comments
Teresa Lopez
Performance 59
Accessibility 79
Best Practices 93
SEO 82
59
Performance
Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.
0–49 50–89 90–100
ENG US
4:50 PM
10/11/2025

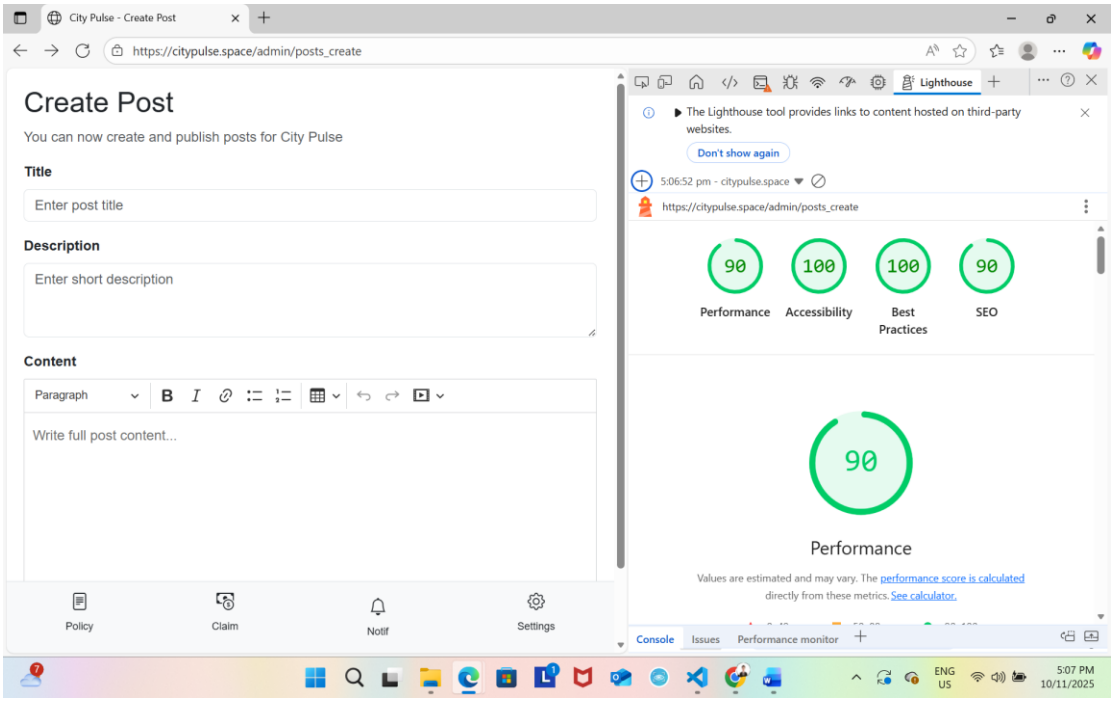

City Pulse - Policy
citypulse.space/admin/posts/a4ba5c8c-9bba-4cf0-aaef-af7c09aa675a
Error
Relaunch to update
Dimensions: Responsive 1427 x 1051 50% No throttling 'Save-Data': default
Elements Console Lighthouse
4:52:42 PM - citypulse.space
https://citypulse.space/admin/posts/a4ba5c8c-9bba-4cf0-aa...
City Pulse
Dashboard
Post Policy
Policies
Users Accounts
Request Account
Manage Rewards
Claim Rewards
Reports
Announcements
Profile
Notification
Brian Dulay
Administrator
Logout
Title: Clean and Green Barangay: Isang Proyekto para sa Malinis at Luntiang Komunidad
Summary and Insights
Comments
Performance 61
Accessibility 94
Best Practices 100
SEO 91
There were issues affecting this run of Lighthouse:
Chrome extensions negatively affected this page's load performance. Try auditing the page in incognito mode or from a Chrome profile without extensions.
61
Performance
ENG US
4:52 PM
10/11/2025

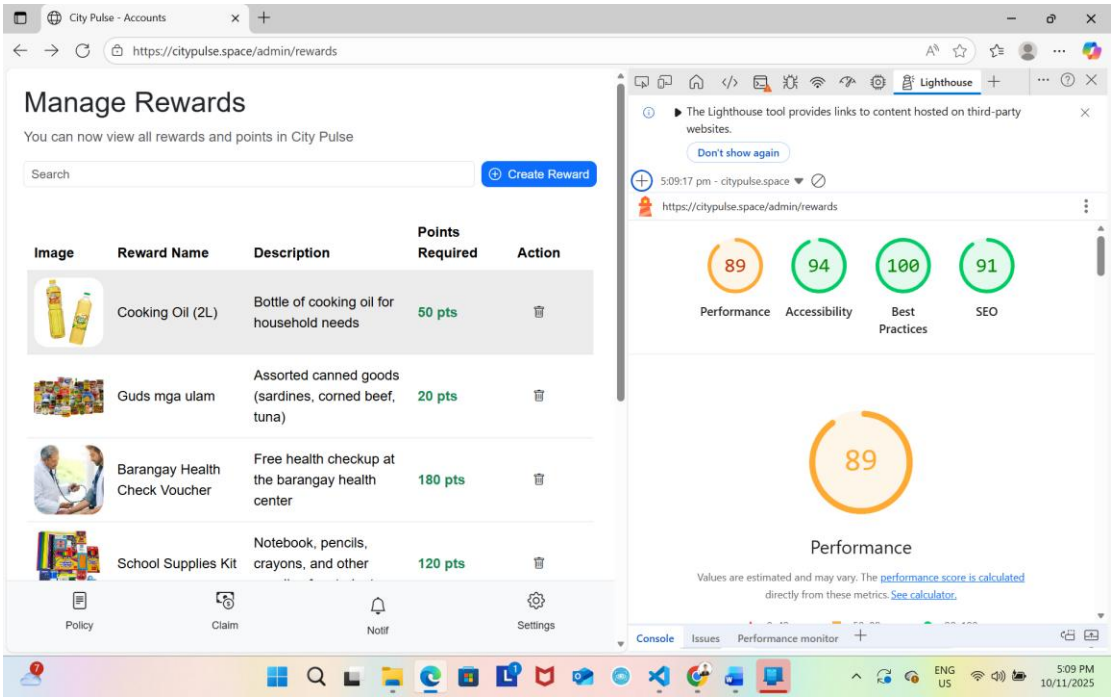

City Pulse - Accounts
citypulse.space/admin/rewards
Manage Rewards
61
95
100
91
Performance
Accessibility
Best Practices
SEO
There were issues affecting this run of Lighthouse:
Chrome extensions negatively affected this page's load performance. Try auditing the page in incognito mode or from a Chrome profile without extensions.
61
Performance

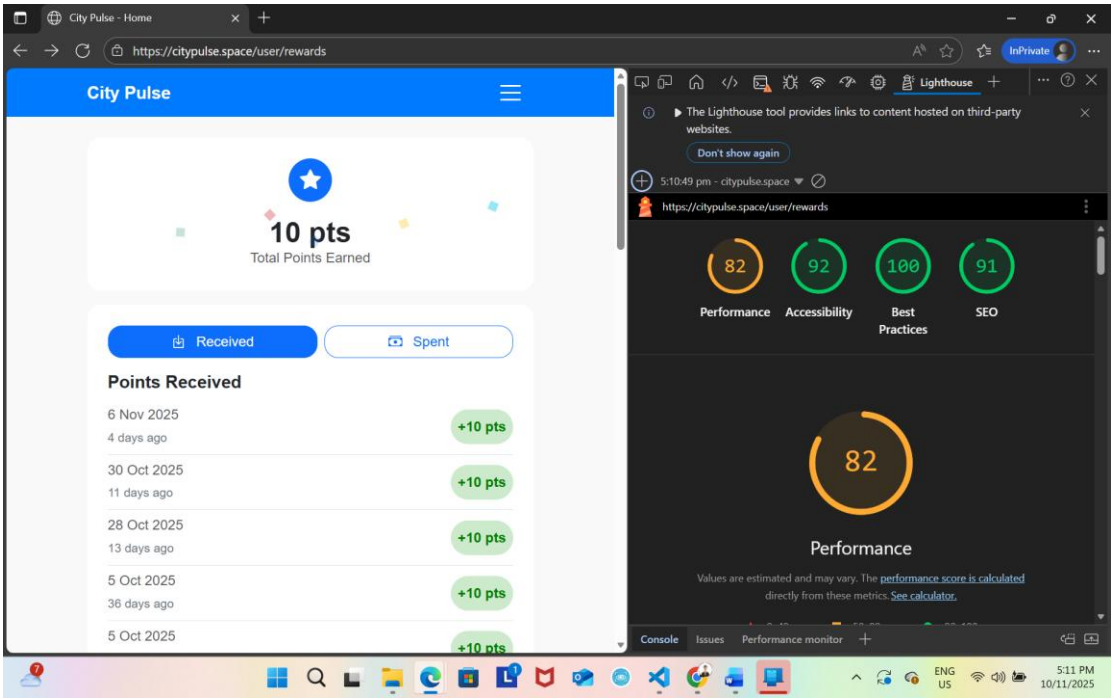

City Pulse - Home
citypulse.space/user/rewards
10 pts
Available Rewards
Cooking Oil (2L)
Guds mga ulam
Barangay Health Check Voucher
School Supplies Kit
Laundry Essentials
Our Mission
55
92
100
91
Performance
Accessibility
Best Practices
SEO
55
Performance

Microsoft Edge Lighthouse

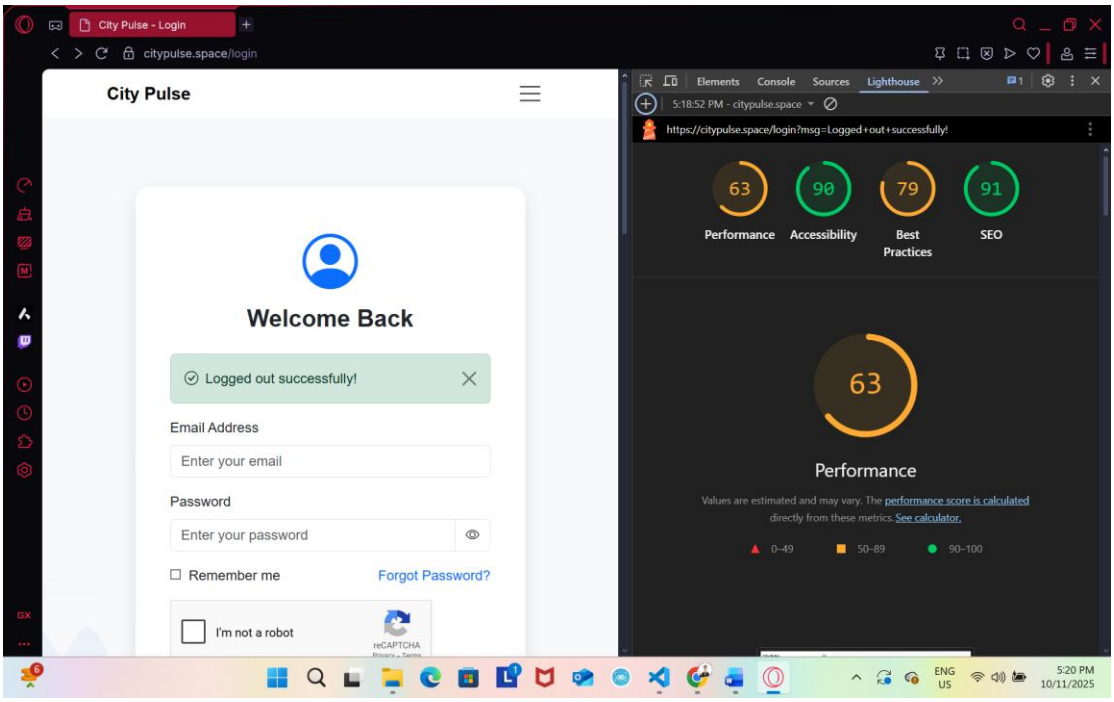


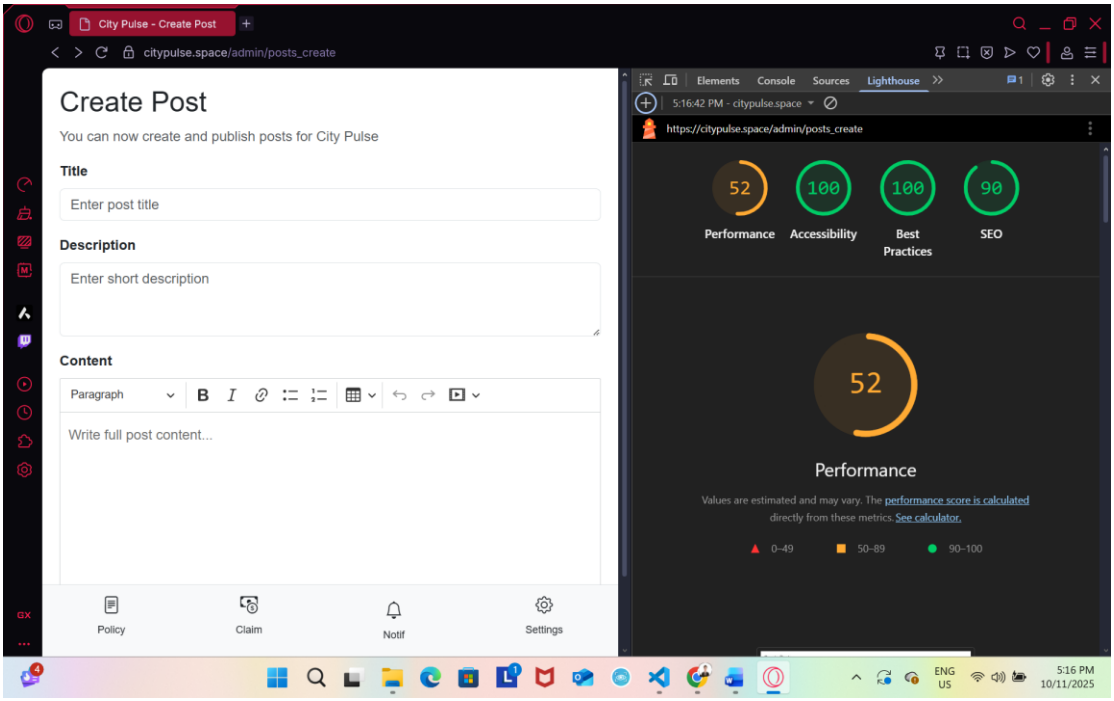


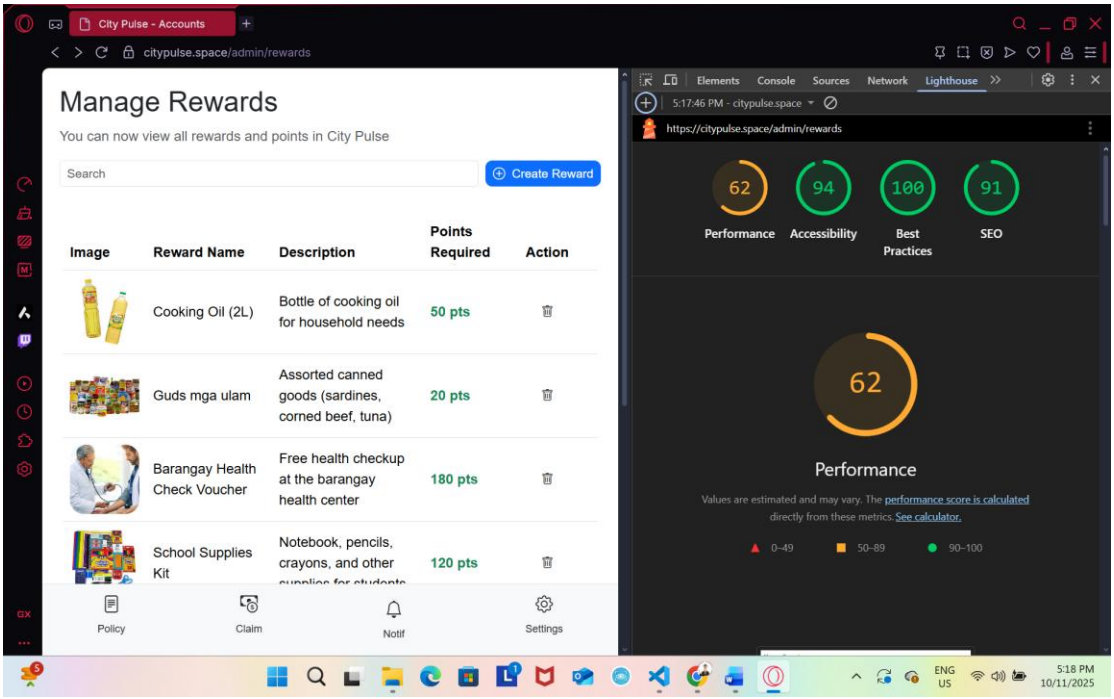

City Pulse - Accounts
https://citypulse.space/admin/rewards
Manage Rewards
You can now view all rewards and points in City Pulse
Search
Create Reward
Image
Reward Name
Description
Points Required
Action
Cooking Oil (2L)
Bottle of cooking oil for household needs
50 pts
Guds mga ulam
Assorted canned goods (sardines, corned beef, tuna)
20 pts
Barangay Health Check Voucher
Free health checkup at the barangay health center
180 pts
School Supplies Kit
Notebook, pencils, crayons, and other
120 pts
Policy
Claim
Notif
Settings
Lighthouse
The Lighthouse tool provides links to content hosted on third-party websites.
Don't show again
5:09:17 pm - citypulse.space
https://citypulse.space/admin/rewards
89
94
100
91
Performance
Accessibility
Best Practices
SEO
89
Performance
Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.
Console
Issues
Performance monitor
ENG US
5:09 PM
10/11/2025

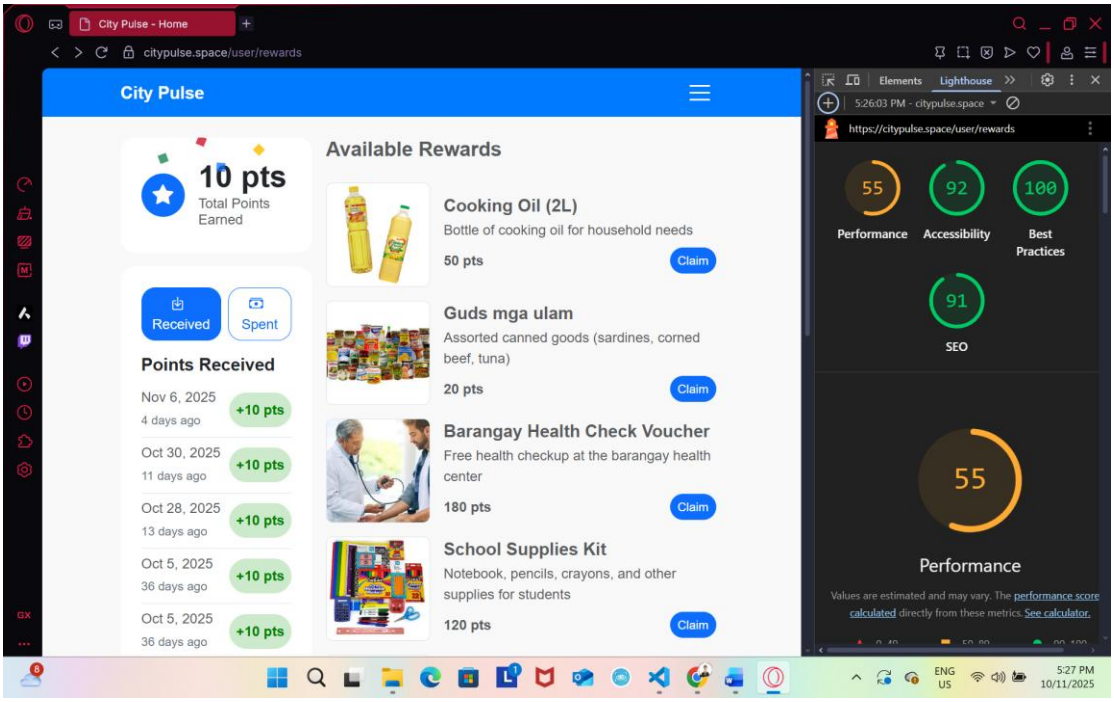

City Pulse - Home
https://citypulse.space/user/rewards
InPrivate
City Pulse
10 pts
Total Points Earned
Received
Spent
Points Received
6 Nov 2025
4 days ago
+10 pts
30 Oct 2025
11 days ago
+10 pts
28 Oct 2025
13 days ago
+10 pts
5 Oct 2025
36 days ago
+10 pts
5 Oct 2025
+10 pts
Lighthouse
The Lighthouse tool provides links to content hosted on third-party websites.
Don't show again
5:10:49 pm - citypulse.space
https://citypulse.space/user/rewards
82
92
100
91
Performance
Accessibility
Best Practices
SEO
82
Performance
Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.
Console
Issues
Performance monitor
ENG US
5:11 PM
10/11/2025

Opera Gx


City Pulse - Login
citypulse.space/login
City Pulse
Welcome Back
Logged out successfully!
Email Address
Enter your email
Password
Enter your password
Remember me
Forgot Password?
I'm not a robot
Elements
Console
Sources
Lighthouse
https://citypulse.space/login?msg=Logged+out+successfully!
63
90
79
91
Performance
Accessibility
Best Practices
SEO
63
Performance
Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.
0–49
50–89
90–100
ENG US
5:20 PM
10/11/2025

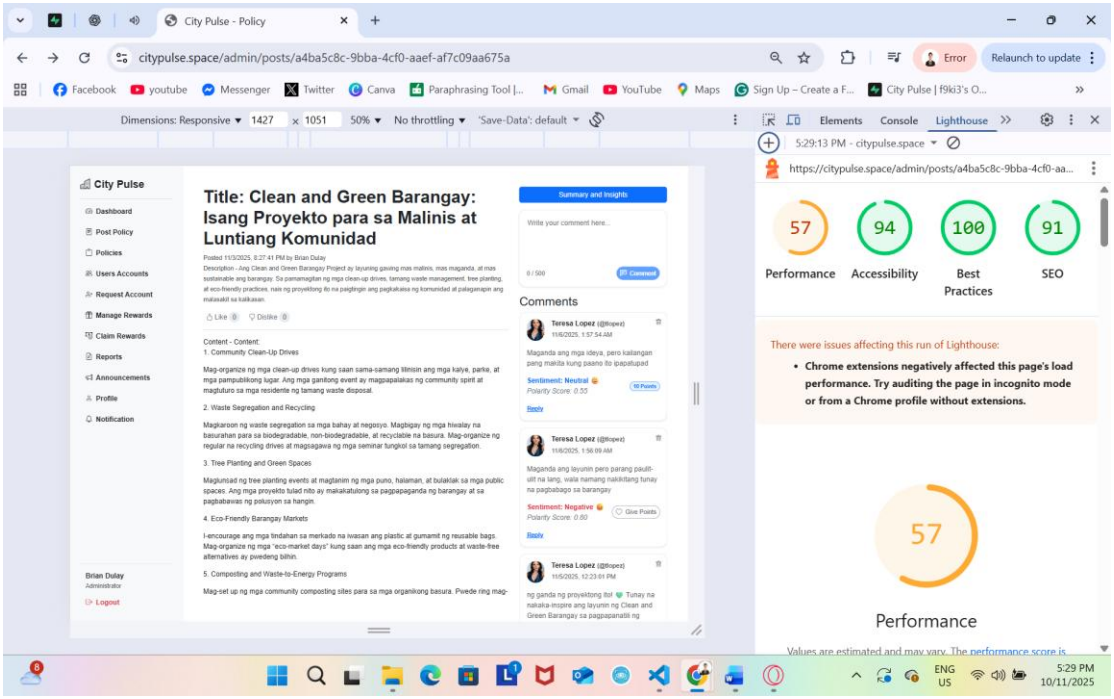

City Pulse - Create Post
citypulse.space/admin/posts_create
Create Post
You can now create and publish posts for City Pulse
Title
Enter post title
Description
Enter short description
Content
Paragraph
Write full post content...
Policy
Claim
Notif
Settings
Elements
Console
Sources
Lighthouse
https://citypulse.space/admin/posts_create
52
100
100
90
Performance
Accessibility
Best Practices
SEO
52
Performance
Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.
0–49
50–89
90–100
ENG US
5:16 PM
10/11/2025

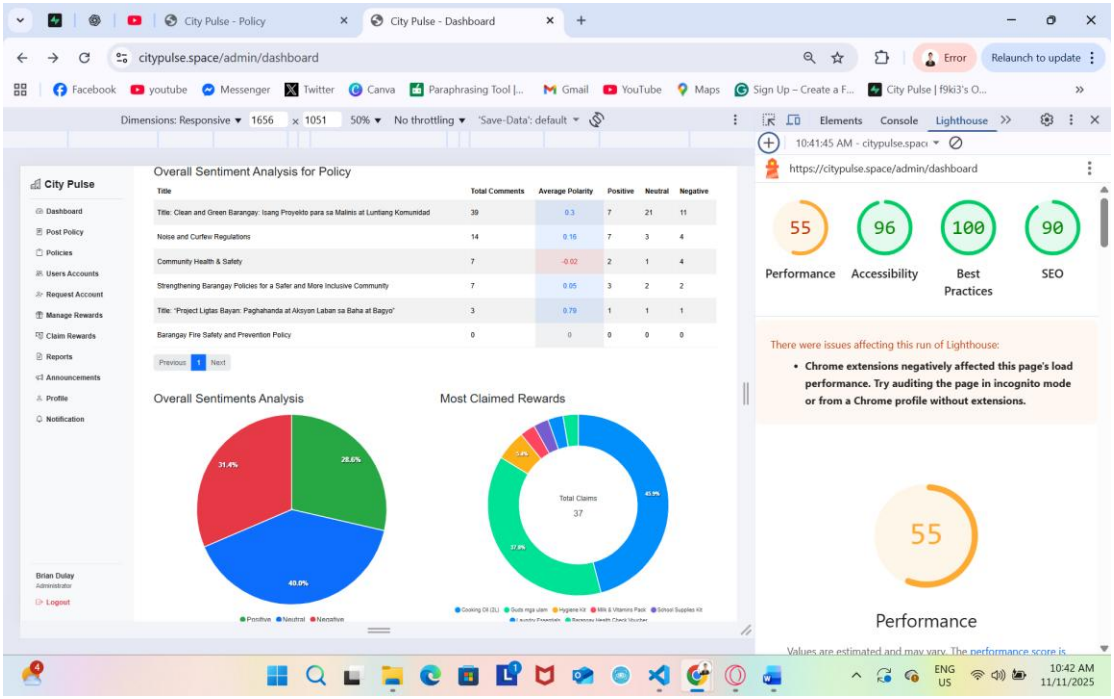

City Pulse - Accounts
citypulse.space/admin/rewards
Manage Rewards
You can now view all rewards and points in City Pulse
Search
Create Reward
Image
Reward Name
Description
Points Required
Action
Cooking Oil (2L)
Bottle of cooking oil for household needs
50 pts
Guds mga ulam
Assorted canned goods (sardines, corned beef, tuna)
20 pts
Barangay Health Check Voucher
Free health checkup at the barangay health center
180 pts
School Supplies Kit
Notebook, pencils, crayons, and other
120 pts
Policy
Claim
Notif
Settings
Elements
Console
Sources
Network
Lighthouse
5:17:46 PM - citypulse.space
https://citypulse.space/admin/rewards
62
94
100
91
Performance
Accessibility
Best Practices
SEO
62
Performance
Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.
0–49
50–89
90–100
ENG US
5:18 PM
10/11/2025


City Pulse - Home
citypulse.space/user/rewards
City Pulse
10 pts
Total Points Earned
Received
Spent
Points Received
Nov 6, 2025
4 days ago
+10 pts
Oct 30, 2025
11 days ago
+10 pts
Oct 28, 2025
13 days ago
+10 pts
Oct 5, 2025
36 days ago
+10 pts
Oct 5, 2025
36 days ago
+10 pts
Available Rewards
Cooking Oil (2L)
Bottle of cooking oil for household needs
50 pts
Claim
Guds mga ulam
Assorted canned goods (sardines, corned beef, tuna)
20 pts
Claim
Barangay Health Check Voucher
Free health checkup at the barangay health center
180 pts
Claim
School Supplies Kit
Notebook, pencils, crayons, and other supplies for students
120 pts
Claim
Elements
Lighthouse
5:26:03 PM - citypulse.space
https://citypulse.space/user/rewards
55
92
100
91
Performance
Accessibility
Best Practices
SEO
55
Performance
Values are estimated and may vary. The performance score calculated directly from these metrics. See calculator.
ENG US
5:27 PM
10/11/2025

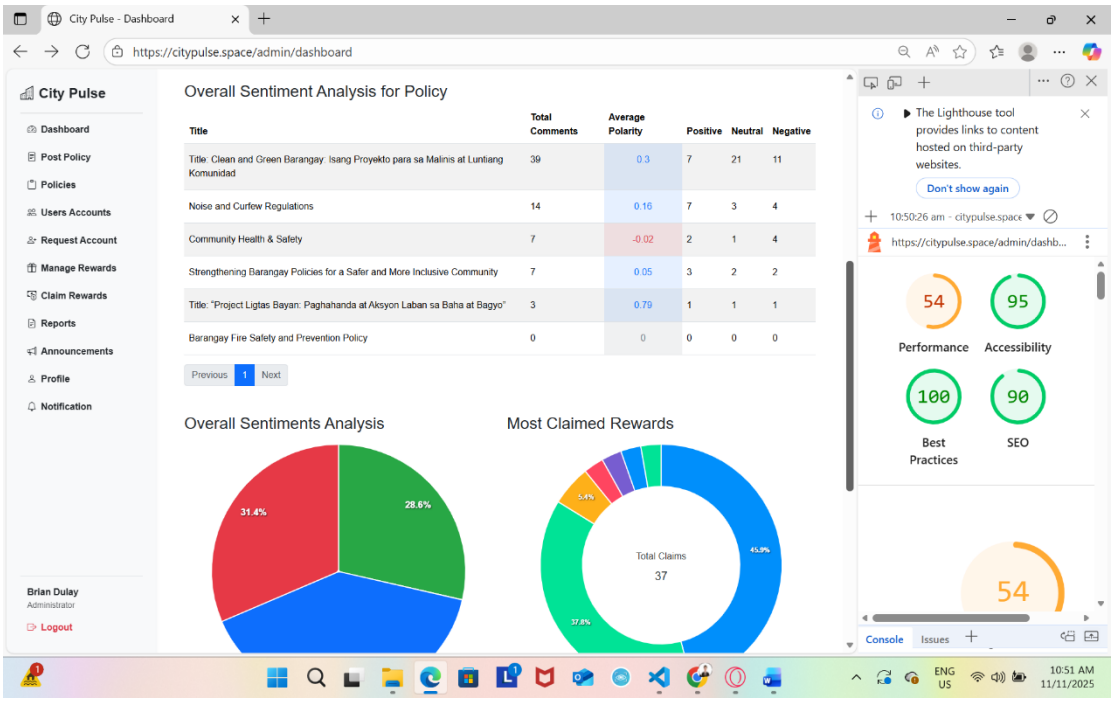
Utilizing a sentiment analysis model using machine learning algorithms.

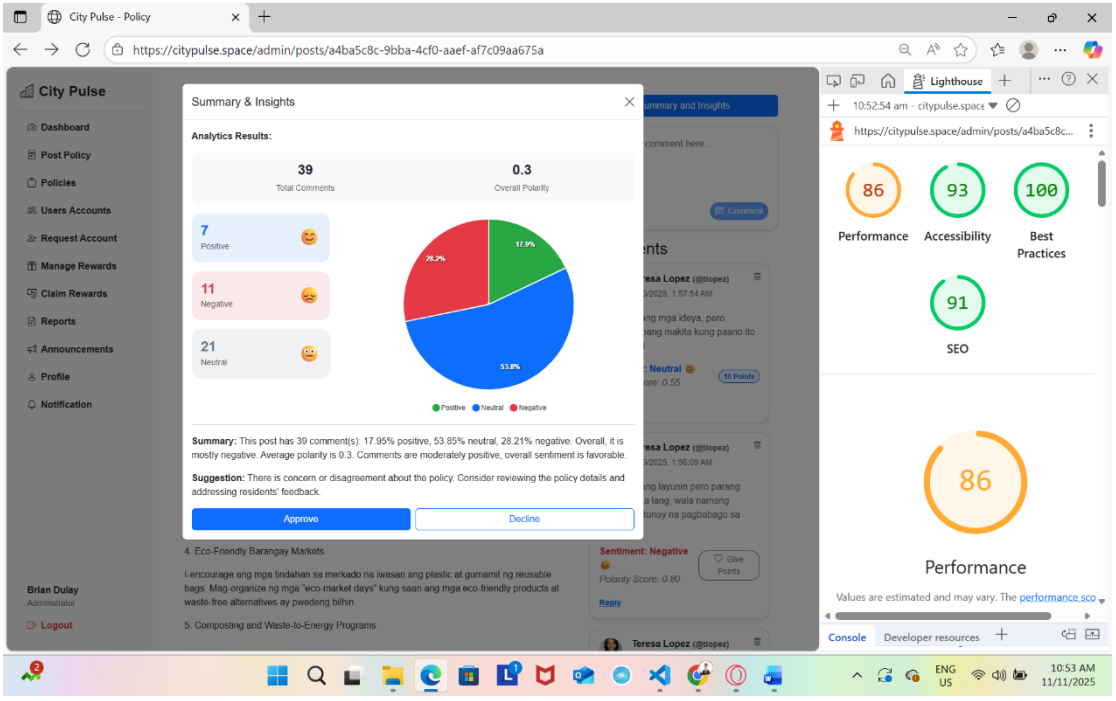
Chrome Lighthouse



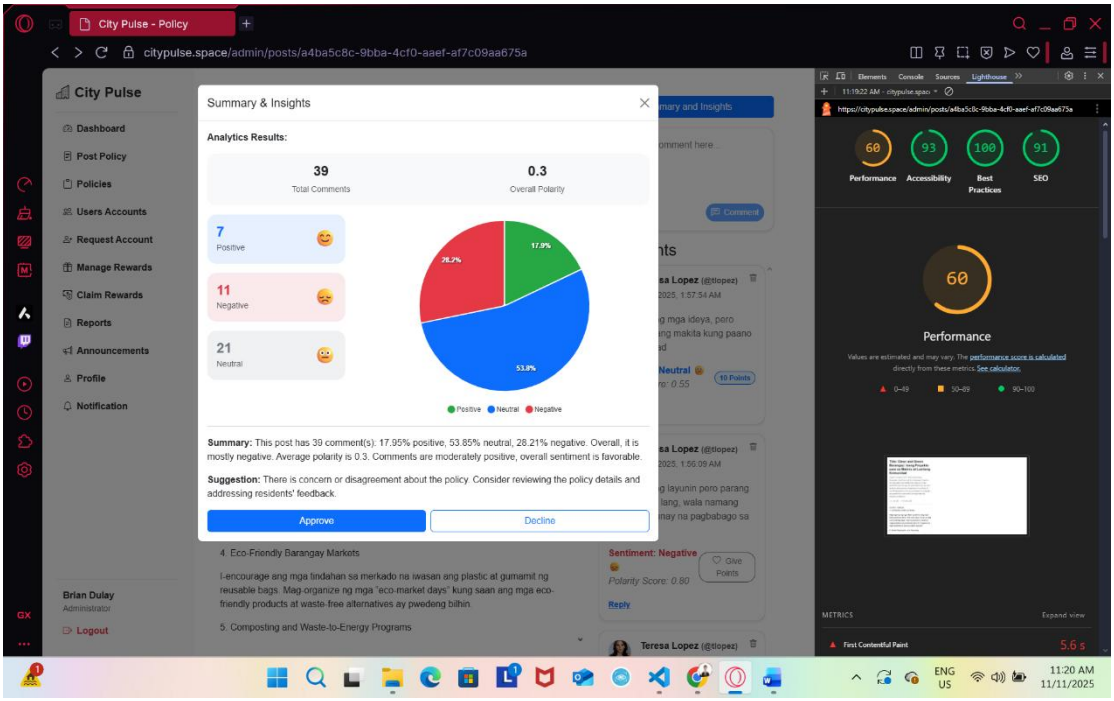


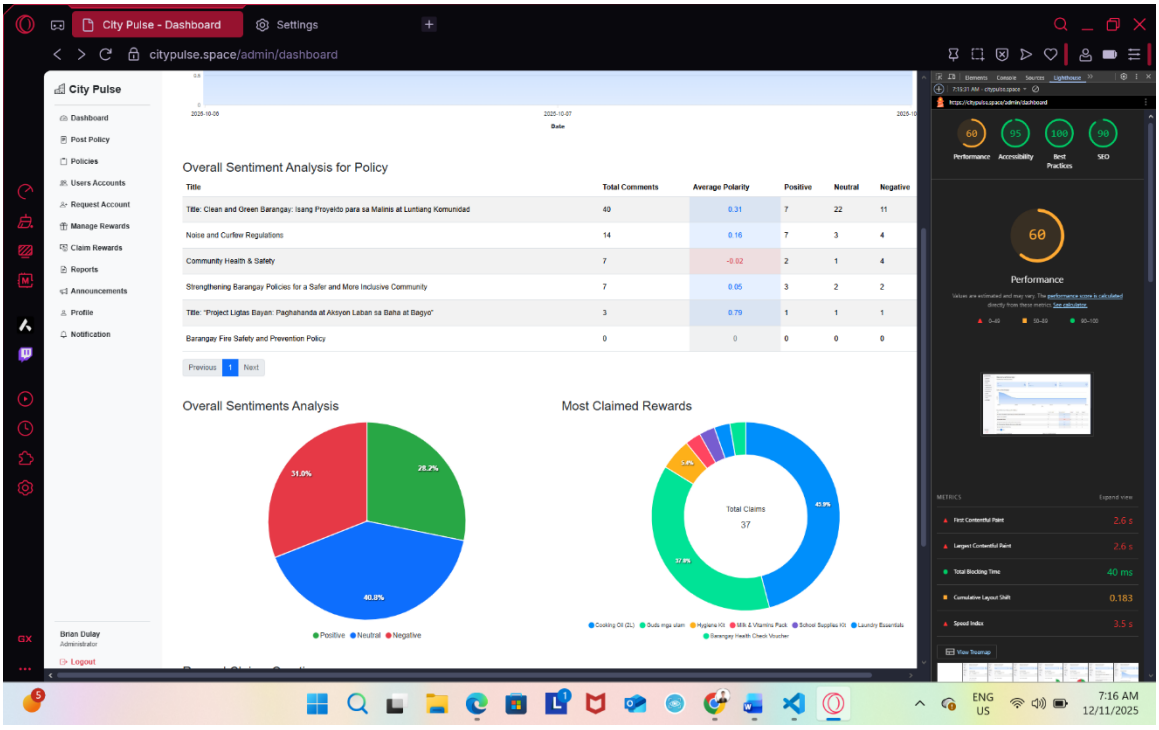
Microsoft Edge Lighthouse





Opera Gx Lighthouse





APPENDIX C

Survey

Questionnaires

Technology Acceptance Model Questionnaire

Dear Respondent,

Good day! We are 4th year students pursuing a BS in Information Technology specializing in Service Management Program conducting a research study and developing a system entitled, "**Local Government Web-Based Policy Recommendation System Based on Community Sentiment: Implementation of a Digital Reward System".** We seek to determine the acceptability of the technology developed by asking for your opinion using the indicators below. Please use the description for each score so that you will be guided accordingly. Rest assured that any information you provide will be treated with the utmost confidentiality in accordance with RA 10173 (Data Privacy Act) and will be used solely for research purposes. Thank you for your cooperation.
.

Sincerely,
Brian E. Dulay, Ken Zeldrix Geronimo, Kenneth Andrei Villegas
LSPU Los Baños, College of Computer Studies
09384430877

Please encircle the score associated with each category

1. Perceived Usefulness. Measures how well the system improves productivity, efficiency, and usefulness to the user.

Indicator	4 - Excellent	3 - Good	2 - Fair	1 - Poor
1. The system helps understand community sentiments on policies.	Effectively provides clear, valuable insights for decision-making.	Provides useful insights but occasionally unclear.	Provides limited insight; data sometimes confusing.	Does not help understand sentiments at all.
2. Improves the quality of local government decision-making.	Greatly enhances policy decisions and data-driven outcomes.	Helps improve some policy decisions.	Slightly helps but lacks sufficient data accuracy.	Does not influence decision quality.
3. Motivates community participation through rewards.	Strongly motivates consistent community engagement.	Encourages moderate participation.	Slight motivation, but participation remains low.	No motivation or engagement effect.
4. Makes policy recommendation tasks faster.	Significantly reduces time and effort.	Makes tasks somewhat quicker.	Minimal improvement in task efficiency.	No improvement; tasks remain slow.
5. Enhances overall effectiveness of community engagement.	Highly effective in improving engagement.	Generally effective but inconsistent.	Somewhat effective but weak impact.	Ineffective in community engagement.

2. Perceived Ease of Use . Measures simplicity, usability, and learning comfort.

Indicator	4 - Excellent	3 - Good	2 - Fair	1 - Poor
1. The interface is easy to navigate.	Interface is intuitive and clear.	Mostly easy with minor confusion.	Sometimes difficult to use.	Complicated and confusing interface.

2. System can be used without technical help.	Completely self-explanatory and simple.	Usable with minimal assistance.	Requires some guidance.	Difficult to use without constant help.
3. Feedback submission process is straightforward.	Very simple and clear submission process.	Mostly simple with few unclear steps.	A bit complicated or time-consuming.	Confusing and hard to complete.
4. Reward system features are easy to understand.	Fully understandable and transparent.	Mostly clear but slightly complex.	Requires extra explanation.	Hard to understand and confusing.
5. Learning the system is quick and effortless.	Easy to learn and master immediately.	Takes little time to learn.	Requires repeated trials to understand.	Very hard to learn or remember.

3. Behavioral Intention to Actual Use. Measures intention to continue use and integrate system into regular practice.

Indicator	4 - Excellent	3 - Good	2 - Fair	1 - Poor
1. Intention to continue using the system.	Strongly intends continued use.	Likely to continue using.	Unsure about continued use.	No intention to use again.
2. Recommending to others.	Actively recommends to others.	Sometimes recommends.	Rarely recommends.	Does not recommend.
3. Frequency of use for feedback or policy checking.	Frequently uses the system.	Occasionally uses.	Rarely uses.	Never uses.
4. Using digital rewards to engage.	Regularly uses and redeems rewards.	Sometimes participates in rewards.	Rarely engages with rewards.	Does not engage with reward system.
5. Preference over traditional feedback methods.	Fully prefers this system.	Prefers but still uses traditional methods.	Neutral between both.	Prefers traditional feedback only.

4. Attitude Toward Using the System. Measures user satisfaction, comfort, and emotional response.

Indicator	4 - Excellent	3 - Good	2 - Fair	1 - Poor
1. Enjoyment while using the system.	Greatly enjoys using it.	Enjoys moderately.	Rarely enjoys.	Does not enjoy at all.
2. Belief that it improves public participation.	Strongly believes it enhances participation.	Believes it helps somewhat.	Unsure about impact.	Does not believe it helps.

3. Reward system makes it engaging.	Highly engaging and satisfying.	Somewhat engaging.	Slightly engaging.	Not engaging at all.
4. Confidence and comfort in using the system.	Very confident and comfortable.	Mostly confident.	Slightly uneasy.	Not confident at all.
5. Overall positive perception of the system.	Very positive overall impression.	Generally positive impression.	Neutral or mixed feelings.	Negative impression.

ISO 25010 QUESTIONNAIRE

Dear IT Expert,

Good day! We are 4th year students conducting a research study and developing a system entitled, "**Local Government Web-Based Policy Recommendation System Based on Community Sentiment: Implementation of a Digital Reward System".** We are currently pursuing a BS in Information Technology specializing in Service Management Program. We now seek to determine if the said system adheres to ISO 25010 standards. To assess the product quality, we have devised a rubric. Your responses will help us evaluate various aspects of the system based on the said international standard.

Sincerely,"
Brian E. Dulay, Ken Zeldrix Geronimo, Kenneth Andrei Villegas
LSPU Los Baños, College of Computer Studies
09384430877

ISO 25010 QUESTIONNAIRE

Please encircle the score associated with each category
Functional Suitability

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
The system provides all required policy recommendation features.	Fully functional and complete.	Most features work with few missing.	Functions meet basic needs.	Some functions missing or incomplete.	Lacks essential features.
Outputs (reports, data) are accurate and reliable.	Consistently accurate and dependable.	Mostly accurate with minor errors.	Acceptable accuracy with occasional errors.	Frequent inaccuracies.	Unreliable outputs.
Reward system performs as intended.	Works perfectly and tracks all points correctly.	Minor tracking issues.	Generally works but with occasional errors.	Often fails to track correctly.	Fails to function as intended.
User feedback and sentiment analysis generate meaningful insights.	Always produces useful and relevant insights.	Mostly useful with few irrelevant outputs.	Somewhat useful.	Often irrelevant.	No useful insights generated.
System supports intended government and community functions.	Fully supports all functions seamlessly.	Supports most functions.	Supports basic functions only.	Limited support.	Does not support functional goals.

Reliability

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
The system runs without	Always stable.	Rare minor issues.	Occasionally unstable.	Frequent crashes.	Very unstable.

crashes or errors.					
Data is saved and retrieved correctly.	Always accurate.	Usually correct with minor delay.	Some inconsistencies.	Frequent data loss.	Major data failures.
System can recover after errors.	Quick full recovery.	Partial or slow recovery.	Requires restart to recover.	Difficult recovery process.	No recovery possible.
Uptime availability.	99–100% uptime.	95–98% uptime.	85–94% uptime.	70–84% uptime.	Below 70% uptime.
Consistent operation during heavy use.	Always stable under load.	Mostly stable.	Occasionally lags.	Frequently lags.	Crashes under load.

Performance Efficiency

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
System loads quickly.	Loads instantly (<2 sec).	Loads fast (2–4 sec).	Moderate speed.	Slow loading.	Very slow/unresponsive.
Handles multiple users efficiently.	No slowdowns.	Slight lag with many users.	Moderate lag.	Major lag with few users.	Unusable with multiple users.
Reward calculations are processed quickly.	Instant calculation.	Slight delay.	Moderate delay.	Long delay.	Often fails to compute.
Efficient use of server resources.	Very low resource usage.	Efficient overall.	Acceptable performance.	Inefficient and slow.	Very high resource usage.
Overall responsiveness.	Very responsive.	Mostly responsive.	Sometimes slow.	Frequently slow.	Not responsive at all.

Usability

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
Interface design is intuitive.	Very easy to use.	Mostly easy.	Acceptable usability.	Confusing at times.	Very hard to navigate.
Instructions and features are easy to understand.	Fully clear.	Mostly clear.	Understandable with effort.	Confusing explanations.	Unclear or missing instructions.

Feedback and forms are user-friendly.	Very easy to submit feedback.	Minor improvements needed.	Works but not smooth.	Some difficulty in use.	Hard to use.
User experience is satisfying.	Highly enjoyable and efficient.	Mostly positive.	Average satisfaction.	Sometimes frustrating.	Not enjoyable at all.
Accessibility for all users.	Fully accessible.	Mostly accessible.	Some accessibility issues.	Many accessibility issues.	Not accessible.

Security

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
User data is protected and private.	Fully protected.	Minor security gaps.	Basic security.	Weak protection.	No security at all.
Login and authentication are secure.	Strong multi-layer security.	Secure but with minor issues.	Basic password-based login.	Weak authentication.	No login protection.
Prevents unauthorized access.	Completely protected.	Rarely vulnerable.	Occasionally vulnerable.	Often compromised.	Always open to attacks.
Data transmission is encrypted.	Fully encrypted.	Mostly encrypted.	Partially encrypted.	Rarely encrypted.	Not encrypted.
Handles errors without exposing data.	Secure error handling.	Minor exposure risk.	Occasionally leaks minor info.	Frequently exposes data.	Always exposes data.

Compatibility

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
Works on different browsers.	Fully compatible.	Minor visual issues.	Works on most browsers.	Limited browser support.	Works only on one browser.
Works across various devices (PC, tablet, mobile).	Fully responsive.	Minor layout issues.	Works well on most devices.	Limited device compatibility.	Works only on one device type.
Integrates with external data or APIs.	Seamlessly integrated.	Minor configuration needed.	Acceptable integration.	Difficult integration.	No integration support.
Reward system syncs	Perfect synchronization.	Minor sync delays.	Sometimes delayed.	Frequently unsynced.	Does not sync.

with user data.					
Compatible with existing government databases or systems.	Fully compatible.	Mostly compatible.	Partially compatible.	Limited compatibility.	Not compatible.

Maintainability

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
Code is easy to modify and update.	Very well-structured and documented	Mostly organized with minor issues.	Acceptable structure.	Hard to understand.	Poorly written and undocumented.
System updates can be applied easily.	Smooth and automated updates.	Minor manual steps.	Somewhat easy with effort.	Requires complex steps.	Very difficult to update.
Error logs and debugging tools are effective.	Detailed and easy to trace.	Mostly detailed.	Sufficient for basic debugging.	Incomplete logs.	No usable logs or tools.
Modules are reusable for future development.	Highly reusable and modular.	Mostly reusable.	Some reusable elements.	Limited reusability.	Not reusable.
Maintenance requires minimal downtime.	Almost zero downtime.	Short downtime.	Acceptable downtime.	Frequent downtime.	Long, disruptive downtime.

Portability

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
Can be deployed on different platforms.	Fully platform-independent.	Works on most platforms.	Requires minor adjustments.	Works on limited platforms.	Only works on one platform.
Installation process is simple.	Quick and automated.	Mostly easy.	Moderate effort required.	Complicated process.	Very difficult to install.
Configuration is straightforward.	Easy to configure.	Mostly straightforward.	Somewhat complex.	Requires technical assistance.	Very hard to configure.
Data migration between environments is smooth.	Seamless and reliable migration.	Minor issues.	Occasionally successful.	Frequent issues.	Fails to migrate data.

Adapts easily to environment changes.	Fully adaptable.	Mostly adaptable.	Some adjustments needed.	Difficult to adapt.	Not adaptable at all.

APPENDIX D

Grammarly and

Turnitin Result

Document Details

Submission ID
trn:oid:::1:3452965752

Submission Date
Jan 5, 2026, 8:16 PM GMT+5:30

Download Date
Jan 5, 2026, 8:19 PM GMT+5:30

File Name
TURNITIN-TEST.pdf

File Size
2.4 MB

103 Pages

17,536 Words

106,360 Characters

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- 104 Not Cited or Quoted 8%
  Matches with neither in-text citation nor quotation marks
- 4 Missing Quotations 0%
  Matches that are still very similar to source material
- 4 Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

6% Internet sources
3% Publications
5% Submitted works (Student Papers)

turnitin Page 3 of 111 - Integrity Overview Submission ID trn:oid:::1:3452965752

Match Groups

104 Not Cited or Quoted 8%
Matches with neither in-text citation nor quotation marks

4 Missing Quotations 0%
Matches that are still very similar to source material

4 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

6% Internet sources
3% Publications
5% Submitted works (Student Papers)

General metrics

68,040	9,244	630	36 min 58 sec	1 hr 11 min
characters	words	sentences	reading time	speaking time

Score



This text scores better than 96% of all texts checked by Grammarly

Writing Issues

106	4	102
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

General metrics

45,717	6,058	606	24 min 13 sec	46 min 36 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 96% of all texts checked by Grammarly

Writing Issues

71	2	69
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

APPENDIX E

POLARITY & FORMULA

Polarity and Formula

Polarity $= \boldsymbol{p_pos} - \boldsymbol{p_neg}$

Formulas:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad Precision = \frac{TP}{TP + FP} \quad Recall = \frac{TP}{TP + FN}$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

APPENDIX F

Publishable Paper

APPENDIX G

Curriculum Vitae



Brian E. Dulay

Bachelor of Science in Information Technology
Major in SMP

CONTACT

briandulay3@gmail.com

09384430877

Elepano Strt. Brgy. 3, Calamba City, Laguna, 4027

PERSONAL DATA

Date of Birth	January 22, 2003
Citizenship	Filipino
Sex	Male
Marital Status	Single
Language	Filipino, English

PROFILE

A motivated IT student with a foundation in programming and system analysis. Eager to apply technical knowledge and problem-solving skill in real life environment. Committed to continuous learning.

EDUCATION

- HIGH SCHOOL
  Lecheria Integrated School
  2015-2019
- SENIOR HIGH SCHOOL
  St. John and Paul Educational Foundation
  2019-2021
- BACHELOR DEGREE
  Laguna State Polytechnic University
  2022-Present

SKILLS

- Proficient in programming languages. (Java, C++, and Python)
- Skilled in Microsoft Office Application (Word, PPT, Excel)
- Basic Knowledge in Web Development (HTML, CSS, PHP)
- Problem Solving Skill
- Excellent communication and teamwork



KEN ZELDRIX B. GERONIMO

IT TECHNICIAN

CONTACT

+123-456-7890

hello@reallygreatsite.com

123 Anywhere St., Any City

EDUCATION

Borcelle University
Bachelor International Management
2016 - 2019

Rimberio University
Bachelor International Management
2012 - 2016

SKILLS

Network setup (LAN/Wi-Fi)

Software installation and configuration

Basic programming (Python, HTML, CSS)

Database management (MySQL, MS Access)

Strong problem-solving and teamwork skills

LANGUAGE

Filipino

English

PROFILE

Motivated IT professional with hands-on experience in computer troubleshooting, network setup, and system maintenance. Skilled in providing technical support and ensuring smooth IT operations. Quick learner with strong problem-solving skills and a passion for technology.

EXPERIENCE

Borcello International Co.

IT Support Assistant | City Hall IT Department
June 2024 – October 2024

Assisted in troubleshooting hardware, software, and network issues.

Installed and configured computers and printers.

Provided technical support and maintained system updates.

Salford & Co.

January 2023 – May 2024

Built and repaired desktop and laptop systems for local clients.

Installed software applications and configured network connections.

Provided technical guidance and IT maintenance for small businesses and students.

REFERENCES

Olivia Wilson
Wardiere Inc. / CEO

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KENNETH ANDREI M. VILLEGAS

PROFESSIONAL SUMMARY

Motivated and detail-oriented IT student aspiring to become an IT Specialist. Possesses a strong foundation in computer systems, networking, and software troubleshooting, with hands-on experience in hardware maintenance and basic programming. Committed to continuous learning, innovation, and providing efficient technological solutions that support organizational goals and improve system performance.

EDUCATION

2022-2026 **Laguna State Polytechnic University - Los Baños Campus**
Brgy. Malinta, Los Baños, Laguna
COLLEGE

2018-2020 **Colegio de Los Baños**
Lopez Avenue, Brgy. Batong Malake Los Baños, Laguna
SENIOR HIGH SCHOOL

2014-2018 **Tuntungin-Putho Integrated National High School**
Brgy. Tuntungin Putho, Los Baños, Laguna
JUNIOR HIGH SCHOOL

2008-2014 **Tuntungin-Putho Elementary School**
Brgy. Putho, Los Baños, Laguna
ELEMENTARY

SKILLS

- Software Installation and Troubleshooting
- Cybersecurity and Data Protection
- Computer Hardware Installation and Maintenance
- Adaptability to New Technologies
- Problem-Solving and Critical Thinking
- Computer Hardware Installation and Maintenance

CONTACT

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